THEORY OF MATRICES - (MMT122J)

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Chapter 1

Matrix Algebra and Structural Properties

This unit introduces the fundamental algebraic structure of matrices over the real field $\mathbb{R}$ and complex field $\mathbb{C}$. It examines core operational properties, highlighting key structural differences from scalar algebra, particularly non-commutativity. The unit further covers transpose and conjugate transpose operations, reversal laws, structural classifications such as symmetric, skew-symmetric, Hermitian, and skew-Hermitian matrices, and matrix decompositions. Finally, it explores the adjoint of a square matrix and its essential relationship with matrix inversion.

1.1 Algebraic Operations on Matrices over Real and Complex Fields

Definition 1.1.1 (Matrix over $\mathbb{R}$ and $\mathbb{C}$)**.** Let $\mathbb{K}$ denote either the real field $\mathbb{R}$ or the complex field $\mathbb{C}$. A matrix A of size $m \times n$ over $\mathbb{K}$ is a rectangular array of scalars arranged in m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = (a_{ij})_{m\times n}, \quad a_{ij} \in \mathbb{K}.$$

The set of all $m \times n$ matrices with entries in $\mathbb{K}$ is denoted by $M_{m\times n}(\mathbb{K})$ or $M_n(\mathbb{K})$ when $m = n$.

1.1.1 Basic Matrix Operations

Definition 1.1.2 (Matrix Addition and Scalar Multiplication)**.** Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices in $M_{m\times n}(\mathbb{K})$, and let $\alpha \in \mathbb{K}$.

(i) **Matrix Addition:** The sum $A + B$ is an $m \times n$ matrix defined component-wise by:

$$(A + B)_{ij} = a_{ij} + b_{ij} \quad \text{for all } 1 \le i \le m,\ 1 \le j \le n.$$

Addition is defined *only* when A and B have identical dimensions.

(ii) **Scalar Multiplication:** The scalar multiple αA is an $m \times n$ matrix defined by:

$$(\alpha A)_{ij} = \alpha \cdot a_{ij} \quad \text{for all } 1 \le i \le m,\ 1 \le j \le n.$$

Theorem 1.1.3 (Properties of Addition and Scalar Multiplication)**.** *For any matrices $A, B, C \in M_{m\times n}(\mathbb{K})$ and scalars $\alpha, \beta \in \mathbb{K}$, the following properties hold:*

(i) Commutativity of Addition: $A + B = B + A$

(ii) Associativity of Addition: $(A + B) + C = A + (B + C)$

(iii) Additive Identity: $A + 0 = A$, *where* 0 *is the* $m \times n$ *zero matrix.*

(iv) Additive Inverse: $A + (-A) = 0$, *where* $-A = (-1)A$.

(v) Distributive Laws: $\alpha(A + B) = \alpha A + \alpha B$ *and* $(\alpha + \beta)A = \alpha A + \beta A$.

(vi) Compatibility of Scalar Multiplication: $(\alpha\beta)A = \alpha(\beta A)$.

1.1.2 Matrix Multiplication

Definition 1.1.4 (Matrix Multiplication)**.** Let $A = (a_{ij}) \in M_{m\times p}(\mathbb{K})$ and $B = (b_{jk}) \in M_{p\times n}(\mathbb{K})$. The product AB is the $m \times n$ matrix $C = (c_{ik}) \in M_{m\times n}(\mathbb{K})$ whose (i, k)-th entry is given by:

$$c_{ik} = (AB)_{ik} = \sum_{j=1}^{p} a_{ij}b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{ip}b_{pk}.$$

The product AB is defined if and only if the number of columns in A equals the number of rows in B.

1.1.3 Non-Commutativity of Matrix Multiplication

Unlike scalar multiplication in $\mathbb{R}$ or $\mathbb{C}$ where $ab = ba$, matrix multiplication is **generally non-commutative**. That is, for two matrices A and B:

$$AB \neq BA \quad \text{in general.}$$

Non-commutativity manifests in three distinct scenarios:

1. **Dimensional Mismatch:** AB is defined, but BA is not defined (e.g., $A \in M_{2\times 3}(\mathbb{K})$ and $B \in M_{3\times 4}(\mathbb{K})$).
2. **Dimension Discrepancy:** Both AB and BA exist, but they have different dimensions (e.g., $A \in M_{2\times 3}(\mathbb{K})$ and $B \in M_{3\times 2}(\mathbb{K})$ yield $AB \in M_{2\times 2}(\mathbb{K})$ and $BA \in M_{3\times 3}(\mathbb{K})$).
3. **Entry-wise Inequality:** $A, B \in M_n(\mathbb{K})$ (both are square matrices of the same order), so both AB and BA are $n \times n$ matrices, yet $AB \neq BA$.

Example 1.1.5 (Explicit Non-Commutativity over $\mathbb{R}$)**.** Consider the real matrices $A, B \in M_2(\mathbb{R})$:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Computing AB:

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1(0) + 2(1) & 1(1) + 2(0) \\ 3(0) + 4(1) & 3(1) + 4(0) \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}.$$

Computing BA:

$$BA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 0(1) + 1(3) & 0(2) + 1(4) \\ 1(1) + 0(3) & 1(2) + 0(4) \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}.$$

Clearly, $AB \neq BA$.

Example 1.1.6 (Non-Commutativity over $\mathbb{C}$)**.** Consider the complex matrices $A, B \in M_2(\mathbb{C})$:

$$A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Calculating AB:

$$AB = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}.$$

Calculating BA:

$$BA = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}.$$

Notice that $AB = -BA \neq BA$. Thus, A and B anti-commute.

1.1.4 Consequences of Non-Commutativity

The non-commutative nature of matrix multiplication leads to crucial algebraic differences between matrix algebra and scalar arithmetic:

1. Failure of the binomial theorem

In standard real or complex algebra, $(a+b)^2 = a^2 + 2ab + b^2$. However, for square matrices A and B:

$$(A+B)^2 = (A+B)(A+B) = A^2 + AB + BA + B^2.$$

Because $AB \neq BA$ in general:

$$(A+B)^2 \neq A^2 + 2AB + B^2 \quad \text{(unless } AB = BA\text{)}.$$

Similarly:

$$(A-B)(A+B) = A^2 + AB - BA - B^2 \neq A^2 - B^2 \quad \text{(unless } AB = BA\text{)}.$$

Example 1.1.7. Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Then $A + B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

$$(A+B)^2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}.$$

On the other hand:

$$A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad B^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad AB = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

So $A^2 + 2AB + B^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \neq (A+B)^2$.

2. Multiplication of two non-zero matrices can yeild a zero matrix

Example 1.1.8. Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \neq 0$.

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0.$$

3. Failure of Cancellation Law

If $AB = AC$, it does **not** generally follow that $B = C$, even if $A \neq 0$.

Example 1.1.9. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}$, and $C = \begin{bmatrix} 0 & 0 \\ 3 & 4 \end{bmatrix}$. Notice that $B \neq C$. However:

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

$$AC = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus $AB = AC = 0$, yet $B \neq C$.

Remark 1.1.10. The cancellation law $AB = AC \implies B = C$ holds if and only if A is non-singular (invertible).

1.1.5 Special Cases Where Matrices Commute

Although matrix multiplication is non-commutative in general, specific pairs of matrices commute ($AB = BA$):

(i) **Identity Matrix:** Any square matrix $A \in M_n(\mathbb{K})$ commutes with the identity matrix I_n:
$$AI_n = I_nA = A.$$

(ii) **Scalar Matrices:** Matrices of the form cI_n (where $c \in \mathbb{K}$) commute with all matrices in $M_n(\mathbb{K})$.

(iii) **Powers of the Same Matrix:** For any $k, m \in \mathbb{N}_0$, $A^kA^m = A^mA^k = A^{k+m}$.

(iv) **Matrix and Its Inverse:** If A is invertible, $AA^{-1} = A^{-1}A = I_n$.

(v) **Diagonal Matrices:** Two diagonal matrices of the same size always commute with each other.

1.2 Transpose of a Matrix and Its Properties

Definition 1.2.1 (Transpose of a Matrix). Let $A = (a_{ij})$ be an $m \times n$ matrix over $\mathbb{K}$. The **transpose** of A, denoted by A^T (or A^t), is the $n \times m$ matrix obtained by interchanging the rows and columns of A.

Formally, if $A^T = (b_{ji})_{n\times m}$, then its entries are defined by:

$$b_{ji} = a_{ij} \quad \text{for all } 1 \leq i \leq m,\ 1 \leq j \leq n.$$

Thus, the i-th row of A becomes the i-th column of A^T, and the j-th column of A becomes the j-th row of A^T.

1.2.1 Algebraic Properties of Transpose

Theorem 1.2.2 (Properties of Transposition). *Let A and B be matrices of appropriate sizes over $\mathbb{K}$, and let $\alpha \in \mathbb{K}$. Then:*

(i) ***Double Transpose Law:*** $(A^T)^T = A$.

(ii) ***Sum Rule:*** $(A + B)^T = A^T + B^T$ *(where A and B have the same dimensions).*

(iii) ***Scalar Multiplication Rule:*** $(\alpha A)^T = \alpha A^T$.

(iv) ***Reversal Law for Multiplication:*** $(AB)^T = B^T A^T$ *(provided the product AB is defined).*

Proof. We prove properties (i) and (iv):

(i) Let $A = (a_{ij})_{m\times n}$. Then $A^T = (b_{ji})_{n\times m}$ where $b_{ji} = a_{ij}$.
Taking the transpose again, $(A^T)^T = (c_{ij})_{m\times n}$ where $c_{ij} = b_{ji} = a_{ij}$.
Thus, $(A^T)^T = A$.

(iv) Let $A = (a_{ij})_{m\times p}$ and $B = (b_{jk})_{p\times n}$. Then AB is an $m \times n$ matrix.
The (i, k)-th entry of AB is given by:

$$(AB)_{ik} = \sum_{j=1}^{p} a_{ij} b_{jk}.$$

Therefore, the (k, i)-th entry of $(AB)^T$ is:

$$\left((AB)^T\right)_{ki} = (AB)_{ik} = \sum_{j=1}^{p} a_{ij} b_{jk}.$$

Now, consider $B^T = (d_{kj})_{n\times p}$ where $d_{kj} = b_{jk}$, and $A^T = (e_{ji})_{p\times m}$ where $e_{ji} = a_{ij}$.
The (k, i)-th entry of the product $B^T A^T$ is:

$$(B^T A^T)_{ki} = \sum_{j=1}^{p} d_{kj} e_{ji} = \sum_{j=1}^{p} b_{jk} a_{ij} = \sum_{j=1}^{p} a_{ij} b_{jk}.$$

Since $\left((AB)^T\right)_{ki} = (B^T A^T)_{ki}$ for all $1 \le k \le n$ and $1 \le i \le m$, it follows that $(AB)^T = B^T A^T$.

■

Theorem 1.2.3 (General Reversal Law for Transposition)**.** *Let $k \ge 2$ be an integer, and let $A_1, A_2, \ldots, A_k$ be a finite collection of conformable matrices over $\mathbb{K}$. Then the transpose of their product satisfies:*

$$(A_1 A_2 \cdots A_k)^T = A_k^T \cdots A_2^T A_1^T.$$

Proof. We prove the statement by mathematical induction on k, the number of matrices.

Base Case ($k = 2$): For two conformable matrices A_1 and A_2, the reversal law gives:

$$(A_1 A_2)^T = A_2^T A_1^T.$$

Thus, the statement holds for $k = 2$.

Inductive Hypothesis: Assume that the statement holds for $k = m$ matrices for some integer $m \ge 2$. That is, assume:

$$(A_1 A_2 \cdots A_m)^T = A_m^T \cdots A_2^T A_1^T.$$

Inductive Step ($k = m+1$): Let $A_1, A_2, \ldots, A_m, A_{m+1}$ be $m+1$ conformable matrices. Group the product of the first m matrices into a single matrix $B = A_1 A_2 \cdots A_m$.

Then, the product of all $m + 1$ matrices can be written as BA_{m+1}. Applying the base case ($k = 2$) to the product of B and A_{m+1}, we obtain:

$$(A_1 A_2 \cdots A_m A_{m+1})^T = (BA_{m+1})^T = A_{m+1}^T B^T.$$

By the inductive hypothesis, $B^T = (A_1A_2\cdots A_m)^T = A_m^T \cdots A_2^T A_1^T$. Substituting this expression for B^T gives:

$$(A_1A_2\cdots A_mA_{m+1})^T = A_{m+1}^T \left(A_m^T \cdots A_2^T A_1^T\right) = A_{m+1}^T A_m^T \cdots A_2^T A_1^T.$$

Hence, the statement holds for $k = m+1$.

By the principle of mathematical induction, the general reversal law $(A_1A_2\cdots A_k)^T = A_k^T \cdots A_2^T A_1^T$ holds for all integers $k \geq 2$. ■

Example 1.2.4. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}$. Then $A^T = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix}$ and $B^T = \begin{bmatrix} 2 & 0 & 1 \\ 1 & -1 & 3 \end{bmatrix}$.

Computing AB:

$$AB = \begin{bmatrix} 1(2)+2(0)+0(1) & 1(1)+2(-1)+0(3) \\ 3(2)+(-1)(0)+4(1) & 3(1)+(-1)(-1)+4(3) \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 10 & 16 \end{bmatrix}.$$

Thus, $(AB)^T = \begin{bmatrix} 2 & 10 \\ -1 & 16 \end{bmatrix}$.

Now computing B^TA^T:

$$B^TA^T = \begin{bmatrix} 2 & 0 & 1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 2(1)+0(2)+1(0) & 2(3)+0(-1)+1(4) \\ 1(1)+(-1)(2)+3(0) & 1(3)+(-1)(-1)+3(4) \end{bmatrix} = \begin{bmatrix} 2 & 10 \\ -1 & 16 \end{bmatrix}.$$

This confirms $(AB)^T = B^TA^T$.

1.3 Inverse of a Matrix and Its Properties

Definition 1.3.1 (Invertible Matrix and Inverse)**.** A square matrix $A \in M_n(\mathbb{K})$ is said to be **invertible** (or **non-singular**) if there exists a matrix $B \in M_n(\mathbb{K})$ such that:

$$AB = BA = I_n,$$

where I_n is the $n \times n$ identity matrix.

If such a matrix B exists, it is unique and is called the **inverse** of A, denoted by A^{-1}.

Theorem 1.3.2 (Uniqueness of Inverse)**.** *If a square matrix $A \in M_n(\mathbb{K})$ is invertible, its inverse is unique.*

Proof. Suppose B and C are both inverses of A. Then:

$$AB = BA = I_n \quad \text{and} \quad AC = CA = I_n.$$

Using associativity of matrix multiplication:

$$B = BI_n = B(AC) = (BA)C = I_nC = C.$$

Hence, $B = C$. ■

1.3.1 Properties of Inverse and Reversal Laws

Theorem 1.3.3 (Properties of Matrix Inverses). *Let $A, B \in M_n(\mathbb{K})$ be invertible matrices, and let $\alpha \in \mathbb{K} \setminus \{0\}$. Then:*

(i) ***Involution Law:*** *A^{-1} is invertible and $(A^{-1})^{-1} = A$.*

(ii) ***Scalar Multiplication Law:*** *αA is invertible and $(\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}$.*

(iii) ***Reversal Law for Inverse:*** *The product AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.*

(iv) ***Commutativity of Transpose and Inverse:*** *A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.*

Proof. We prove properties (iii) and (iv):

(iii) To show that $(AB)^{-1} = B^{-1}A^{-1}$, we verify that multiplying AB by $B^{-1}A^{-1}$ yields I_n from both sides:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n,$$

and

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n.$$

By the uniqueness of the matrix inverse, $(AB)^{-1} = B^{-1}A^{-1}$.

(iv) Using the reversal law of transpose and the definition of inverse:

$$A^T(A^{-1})^T = (A^{-1}A)^T = (I_n)^T = I_n,$$

and

$$(A^{-1})^TA^T = (AA^{-1})^T = (I_n)^T = I_n.$$

Thus, A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

■

Theorem 1.3.4 (General Reversal Law for Inverses). *Let $k \geq 2$ be an integer, and let $A_1, A_2, \ldots, A_k \in M_n(\mathbb{K})$ be invertible matrices. Then the product matrix $A_1A_2 \cdots A_k$ is invertible, and its inverse satisfies:*

$$(A_1A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1}A_1^{-1}.$$

Proof. We prove the theorem by mathematical induction on k, the number of invertible matrices.

Base Case ($k = 2$): For two invertible matrices $A_1, A_2 \in M_n(\mathbb{K})$, we verify that $(A_1A_2)(A_2^{-1}A_1^{-1}) = I_n$ and $(A_2^{-1}A_1^{-1})(A_1A_2) = I_n$:

$$(A_1A_2)(A_2^{-1}A_1^{-1}) = A_1(A_2A_2^{-1})A_1^{-1} = A_1I_nA_1^{-1} = A_1A_1^{-1} = I_n,$$

and

$$(A_2^{-1}A_1^{-1})(A_1A_2) = A_2^{-1}(A_1^{-1}A_1)A_2 = A_2^{-1}I_nA_2 = A_2^{-1}A_2 = I_n.$$

By uniqueness of the matrix inverse, $(A_1A_2)^{-1} = A_2^{-1}A_1^{-1}$. Thus, the base case holds.

Inductive Hypothesis: Assume that the statement holds for $k = m$ invertible matrices for some integer $m \geq 2$. That is, assume $A_1A_2 \cdots A_m$ is invertible and:

$$(A_1A_2 \cdots A_m)^{-1} = A_m^{-1} \cdots A_2^{-1}A_1^{-1}.$$

Inductive Step ($k = m+1$): Let $A_1, A_2, \ldots, A_m, A_{m+1} \in M_n(\mathbb{K})$ be $m+1$ invertible matrices. Group the product of the first m matrices as $B = A_1A_2 \cdots A_m$.

Since B is invertible by the inductive hypothesis and A_{m+1} is invertible by assumption, we apply the base case ($k = 2$) to the product BA_{m+1}:

$$(A_1A_2 \cdots A_mA_{m+1})^{-1} = (BA_{m+1})^{-1} = A_{m+1}^{-1}B^{-1}.$$

Substituting the inductive hypothesis $B^{-1} = (A_1A_2 \cdots A_m)^{-1} = A_m^{-1} \cdots A_2^{-1}A_1^{-1}$ yields:

$$(A_1A_2 \cdots A_mA_{m+1})^{-1} = A_{m+1}^{-1}\left(A_m^{-1} \cdots A_2^{-1}A_1^{-1}\right) = A_{m+1}^{-1}A_m^{-1} \cdots A_2^{-1}A_1^{-1}.$$

Thus, the statement holds for $k = m + 1$.

By the principle of mathematical induction, the general reversal law for inverses holds for all integers $k \geq 2$. ■

1.4 Adjoint of a Square Matrix and Matrix Inversion

The classical adjoint (also known as the **adjugate**) of a square matrix provides an explicit formula for matrix inversion and bridges matrix algebra with determinant theory.

1.4.1 Minors, Cofactors, and the Adjoint Matrix

Definition 1.4.1 (Minor and Cofactor)**.** Let $A = (a_{ij}) \in M_n(\mathbb{K})$ be a square matrix of order $n \geq 2$.

(i) The **minor** M_{ij} of the entry a_{ij} is the determinant of the $(n-1) \times (n-1)$ submatrix obtained by deleting the i-th row and j-th column of A.

(ii) The **cofactor** C_{ij} of the entry a_{ij} is defined as:

$$C_{ij} = (-1)^{i+j}M_{ij}.$$

The matrix $C = (C_{ij})_{n\times n}$ is called the **cofactor matrix** of A.

Definition 1.4.2 (Adjoint of a Matrix)**.** Let $A \in M_n(\mathbb{K})$ be a square matrix and let $C = (C_{ij})$ be its cofactor matrix. The **adjoint** (or **adjugate**) of A, denoted by $\operatorname{adj}(A)$, is the transpose of the cofactor matrix:

$$\operatorname{adj}(A) = C^T = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}.$$

Explicitly, the (i, j)-th entry of $\operatorname{adj}(A)$ is the cofactor C_{ji}.

1.4.2 Fundamental Theorem of Adjoint Matrix

The connection between A, $\operatorname{adj}(A)$, and $\det(A)$ is governed by Laplace's expansion theorem.

Theorem 1.4.3 (Fundamental Property of Adjoint)**.** *For any square matrix $A \in M_n(\mathbb{K})$:*

$$A \cdot \operatorname{adj}(A) = \operatorname{adj}(A) \cdot A = \det(A)\, I_n,$$

where I_n is the $n \times n$ identity matrix.

Proof. Let $A = (a_{ij})_{n\times n}$ and $\operatorname{adj}(A) = (b_{ij})_{n\times n}$, where $b_{ij} = C_{ji}$. Consider the (i, k)-th entry of the product $P = A \cdot \operatorname{adj}(A)$:

$$P_{ik} = \sum_{j=1}^{n} a_{ij}b_{jk} = \sum_{j=1}^{n} a_{ij}C_{kj}.$$

We analyze two cases for P_{ik}:

1. **Case 1 ($i = k$):** The diagonal entries are:

$$P_{ii} = \sum_{j=1}^{n} a_{ij} C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}.$$

   By Laplace expansion along the i-th row, this sum equals $\det(A)$.

2. **Case 2 ($i \neq k$):** The off-diagonal entries are:

$$P_{ik} = \sum_{j=1}^{n} a_{ij} C_{kj} \quad (i \neq k).$$

   This sum represents the determinant expansion of a matrix formed by replacing the k-th row of A with its i-th row. Because this auxiliary matrix has two identical rows (i-th and k-th rows), its determinant is zero. Thus, $P_{ik} = 0$ for $i \neq k$.

Combining both cases:

$$P_{ik} = \begin{cases} \det(A) & \text{if } i = k, \\ 0 & \text{if } i \neq k. \end{cases}$$

Hence, $A \cdot \operatorname{adj}(A) = \det(A) I_n$. A completely analogous argument expanding along columns shows that $\operatorname{adj}(A) \cdot A = \det(A) I_n$. ■

1.4.3 Connection with Matrix Inversion

Theorem 1.4.4 (Criterion for Invertibility and Adjoint Formula). *A square matrix $A \in M_n(\mathbb{K})$ is invertible (non-singular) if and only if $\det(A) \neq 0$. When $\det(A) \neq 0$, the inverse A^{-1} is uniquely given by:*

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Proof.

- **Sufficiency ($\det(A) \neq 0 \implies A$ is invertible):**
  Suppose $\det(A) \neq 0$. By the Fundamental Theorem of Adjoint:

$$A \cdot \operatorname{adj}(A) = \det(A) I_n \implies A \cdot \left(\frac{1}{\det(A)} \operatorname{adj}(A) \right) = I_n,$$

  and

$$\operatorname{adj}(A) \cdot A = \det(A) I_n \implies \left(\frac{1}{\det(A)} \operatorname{adj}(A) \right) \cdot A = I_n.$$

  Thus, $B = \frac{1}{\det(A)} \operatorname{adj}(A)$ satisfies $AB = BA = I_n$, proving A is invertible and $A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$.

- **Necessity (A is invertible $\implies \det(A) \neq 0$):**
  Suppose A is invertible. Then $AA^{-1} = I_n$. Taking determinants on both sides:

$$\det(AA^{-1}) = \det(I_n) \implies \det(A)\det(A^{-1}) = 1.$$

  Therefore, $\det(A) \neq 0$ and $\det(A^{-1}) = \frac{1}{\det(A)}$.

■

1.4.4 Properties of the Adjoint Matrix

Theorem 1.4.5 (Algebraic Properties of Adjoint)**.** *Let* $A, B \in M_n(\mathbb{K})$ *and* $\alpha \in \mathbb{K}$*. Then:*

(i) $\mathrm{adj}(I_n) = I_n$.

(ii) $\mathrm{adj}(A^T) = (\mathrm{adj}(A))^T$.

(iii) Reversal Law: $\mathrm{adj}(AB) = \mathrm{adj}(B)\,\mathrm{adj}(A)$.

(iv) $\mathrm{adj}(\alpha A) = \alpha^{n-1}\,\mathrm{adj}(A)$.

(v) Determinant of Adjoint: $\det(\mathrm{adj}(A)) = (\det(A))^{n-1}$.

(vi) Double Adjoint Identity: $\mathrm{adj}(\mathrm{adj}(A)) = (\det(A))^{n-2}A \quad (n \geq 2)$.

Proof. We prove properties (v) and (vi):

(v) From $A \cdot \mathrm{adj}(A) = \det(A)I_n$, take determinants on both sides:

$$\det(A \cdot \mathrm{adj}(A)) = \det(\det(A)I_n) \implies \det(A)\det(\mathrm{adj}(A)) = (\det(A))^n.$$

If $\det(A) \neq 0$, dividing by $\det(A)$ yields:

$$\det(\mathrm{adj}(A)) = (\det(A))^{n-1}.$$

(vi) Replace A with $\mathrm{adj}(A)$ in $M \cdot \mathrm{adj}(M) = \det(M)I_n$:

$$\mathrm{adj}(A) \cdot \mathrm{adj}(\mathrm{adj}(A)) = \det(\mathrm{adj}(A))I_n = (\det(A))^{n-1}I_n.$$

Multiplying on the left by A:

$$A \cdot \mathrm{adj}(A) \cdot \mathrm{adj}(\mathrm{adj}(A)) = A\left((\det(A))^{n-1}I_n\right).$$

Since $A \cdot \mathrm{adj}(A) = \det(A)I_n$:

$$\det(A)I_n \cdot \mathrm{adj}(\mathrm{adj}(A)) = (\det(A))^{n-1}A.$$

Assuming $\det(A) \neq 0$, dividing both sides by $\det(A)$ gives:

$$\mathrm{adj}(\mathrm{adj}(A)) = (\det(A))^{n-2}A.$$

■

Example 1.4.6 (Adjoint and Inverse of a 2×2 Matrix)**.** Find the adjoint and inverse of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ over $\mathbb{R}$.

Solution. 1. Compute cofactors:

$$C_{11} = (-1)^{1+1}(d) = d, \quad C_{12} = (-1)^{1+2}(c) = -c,$$
$$C_{21} = (-1)^{2+1}(b) = -b, \quad C_{22} = (-1)^{2+2}(a) = a.$$

2. The cofactor matrix is $C = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$. Taking the transpose gives:

$$\mathrm{adj}(A) = C^T = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Rule of thumb for 2×2 *matrices:* Swap diagonal elements and negate off-diagonal elements.

3. Determinant: $\det(A) = ad - bc$.
4. If $ad - bc \neq 0$, then:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

■

Example 1.4.7 (Adjoint and Inverse of a 3×3 Matrix)**.** Find $\mathrm{adj}(A)$ and A^{-1} for the real matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}.$$

Solution. Compute $\det(A)$ using expansion along Row 1:

$$\det(A) = 1 \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} - 2 \begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} + 3 \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix}$$

$$\det(A) = 1(0 - 24) - 2(0 - 20) + 3(0 - 5) = -24 + 40 - 15 = 1.$$

Since $\det(A) = 1 \neq 0$, A is invertible.

The cofactors are:

$$C_{11} = + \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} = -24, \quad C_{12} = - \begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} = 20, \quad C_{13} = + \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} = -5,$$

$$C_{21} = - \begin{vmatrix} 2 & 3 \\ 6 & 0 \end{vmatrix} = 18, \quad C_{22} = + \begin{vmatrix} 1 & 3 \\ 5 & 0 \end{vmatrix} = -15, \quad C_{23} = - \begin{vmatrix} 1 & 2 \\ 5 & 6 \end{vmatrix} = 4,$$

$$C_{31} = + \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = 5, \quad C_{32} = - \begin{vmatrix} 1 & 3 \\ 0 & 4 \end{vmatrix} = -4, \quad C_{33} = + \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1.$$

The cofactor matrix is:

$$C = \begin{bmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ 5 & -4 & 1 \end{bmatrix},$$

and so the adjoint is:

$$\mathrm{adj}(A) = C^T = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}.$$

Finally,

$$A^{-1} = \frac{1}{\det(A)} \mathrm{adj}(A) = \frac{1}{1} \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix} = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}.$$

■

Example 1.4.8 (Verification of Reversal Law for Inverses)**.** Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$. Compute A^{-1}, B^{-1}, and $(AB)^{-1}$, and verify that $(AB)^{-1} = B^{-1}A^{-1}$.

Solution. First, compute A^{-1} and B^{-1}:

$$A^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 1/2 & 0 \\ -1/2 & 1 \end{bmatrix}.$$

Now compute AB:

$$AB = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 1 & 1 \end{bmatrix}.$$

The inverse of AB is:

$$(AB)^{-1} = \frac{1}{4(1) - 2(1)} \begin{bmatrix} 1 & -2 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 1/2 & -1 \\ -1/2 & 2 \end{bmatrix}.$$

Computing $B^{-1}A^{-1}$:

$$B^{-1}A^{-1} = \begin{bmatrix} 1/2 & 0 \\ -1/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & -1 \\ -1/2 & 2 \end{bmatrix}.$$

This verifies that $(AB)^{-1} = B^{-1}A^{-1}$. ■

1.5 Conjugate Transpose and Its Algebraic Properties

When dealing with complex matrices $M_{m\times n}(\mathbb{C})$, taking the simple transpose is often insufficient because complex multiplication involves complex conjugation.

Definition 1.5.1 (Conjugate Matrix)**.** Let $A = (a_{ij})_{m\times n} \in M_{m\times n}(\mathbb{C})$. The **conjugate matrix** $\bar{A}$ is obtained by taking the complex conjugate of each entry of A:

$$\bar{A} = (\bar{a}_{ij})_{m\times n}, \quad \text{where if } a_{ij} = x + iy, \text{ then } \bar{a}_{ij} = x - iy.$$

Definition 1.5.2 (Conjugate Transpose)**.** Let $A \in M_{m\times n}(\mathbb{C})$. The **conjugate transpose** (also known as the **Hermitian transpose** or **adjoint matrix** in operator theory) of A, denoted by A^* (or A^H), is defined as the transpose of the complex conjugate of A:

$$A^* = (\bar{A})^T = \overline{(A^T)}.$$

Explicitly, if $A = (a_{ij})_{m\times n}$, then A^* is an $n \times m$ matrix whose (j, i)-th entry is:

$$(A^*)_{ji} = \bar{a}_{ij} \quad \text{for all } 1 \le i \le m,\ 1 \le j \le n.$$

1.5.1 Algebraic Properties of Conjugate Transpose

Theorem 1.5.3 (Properties of Conjugate Transposition)**.** *Let A, B be complex matrices of suitable sizes, and let $\alpha \in \mathbb{C}$. Then:*

(i) ***Involution Law:*** $(A^*)^* = A$.

(ii) ***Sum Rule:*** $(A + B)^* = A^* + B^*$.

(iii) ***Conjugate-Linearity:*** $(\alpha A)^* = \bar{\alpha} A^*$. *(Note the complex conjugation on the scalar α!)*

(iv) ***Reversal Law for Multiplication:*** $(AB)^* = B^*A^*$.

(v) ***Commutativity with Inversion:*** *If $A \in M_n(\mathbb{C})$ is invertible, then A^* is invertible and $(A^*)^{-1} = (A^{-1})^*$.*

Proof. We prove properties (iii), (iv), and (v):

(iii) Let $A = (a_{ij})$. The (j, i)-th entry of $(\alpha A)^*$ is:

$$((\alpha A)^*)_{ji} = \overline{(\alpha a_{ij})} = \bar{\alpha}\bar{a}_{ij} = \bar{\alpha}(A^*)_{ji} = (\bar{\alpha}A^*)_{ji}.$$

Hence, $(\alpha A)^* = \bar{\alpha}A^*$.

(iv) Using the properties of transposition and complex conjugation:

$$(AB)^* = \overline{(AB)^T} = \overline{B^T A^T} = \overline{B^T} \cdot \overline{A^T} = B^* A^*.$$

(v) Taking the conjugate transpose of $AA^{-1} = I_n$ and using the reversal law:

$$(AA^{-1})^* = (I_n)^* \implies (A^{-1})^* A^* = I_n.$$

Similarly, $(A^{-1}A)^* = I_n^* \implies A^*(A^{-1})^* = I_n$.
By uniqueness of inverse, $(A^*)^{-1} = (A^{-1})^*$.

■

Remark 1.5.4. If A is a real matrix ($A \in M_{m\times n}(\mathbb{R})$), then $\bar{A} = A$. Consequently, for real matrices, the conjugate transpose reduces directly to the transpose:

$$A^* = A^T \quad \text{for all } A \in M_{m\times n}(\mathbb{R}).$$

Example 1.5.5. Consider the complex matrix $A = \begin{bmatrix} 1+i & 2-3i \\ 4 & 5i \end{bmatrix}$.

1. The conjugate of A is $\bar{A}$:

$$\bar{A} = \begin{bmatrix} 1-i & 2+3i \\ 4 & -5i \end{bmatrix}.$$

2. The transpose of $\bar{A}$ to find A^*:

$$A^* = (\bar{A})^T = \begin{bmatrix} 1-i & 4 \\ 2+3i & -5i \end{bmatrix}.$$

3. Therefore $(\alpha A)^* = \bar{\alpha}A^*$ for $\alpha = i$:

$$\alpha A = i\begin{bmatrix} 1+i & 2-3i \\ 4 & 5i \end{bmatrix} = \begin{bmatrix} -1+i & 3+2i \\ 4i & -5 \end{bmatrix}.$$

Taking conjugate transpose:

$$(\alpha A)^* = \begin{bmatrix} -1-i & -4i \\ 3-2i & -5 \end{bmatrix}.$$

On the other hand, $\bar{\alpha} = -i$:

$$\bar{\alpha}A^* = -i\begin{bmatrix} 1-i & 4 \\ 2+3i & -5i \end{bmatrix} = \begin{bmatrix} -i(1-i) & -4i \\ -i(2+3i) & 5i^2 \end{bmatrix} = \begin{bmatrix} -1-i & -4i \\ 3-2i & -5 \end{bmatrix}.$$

Thus $(\alpha A)^* = \bar{\alpha}A^*$ holds precisely.

1.6 Properties of Symmetric, Skew-Symmetric, Hermitian, and Skew-Hermitian Matrices

In matrix theory, square matrices with specific symmetries under transposition or conjugate transposition play a central role in linear algebra, physics, and engineering.

1.6.1 Symmetric and Skew-Symmetric Matrices

Definition 1.6.1 (Symmetric and Skew-Symmetric Matrices)**.** Let $A = (a_{ij}) \in M_n(\mathbb{K})$ be a square matrix.

(i) A is called **symmetric** if $A^T = A$, which means:

$$a_{ij} = a_{ji} \quad \text{for all } 1 \leq i, j \leq n.$$

(ii) A is called **skew-symmetric** (or **anti-symmetric**) if $A^T = -A$, which means:

$$a_{ij} = -a_{ji} \quad \text{for all } 1 \leq i, j \leq n.$$

Theorem 1.6.2. *All main diagonal entries of a skew-symmetric matrix are zero.*

Proof. Let $A = (a_{ij}) \in M_n(\mathbb{K})$ be skew-symmetric. By definition, $a_{ij} = -a_{ji}$ for all i, j. For the diagonal entries ($i = j$), we have:

$$a_{ii} = -a_{ii} \implies 2a_{ii} = 0 \implies a_{ii} = 0.$$

Thus, every entry on the main diagonal of A must be zero. ■

Theorem 1.6.3. *Let $A, B \in M_n(\mathbb{K})$ be symmetric matrices. The product AB is symmetric if and only if A and B commute ($AB = BA$).*

Proof. Since A and B are symmetric, $A^T = A$ and $B^T = B$.

- ($\implies$) Suppose AB is symmetric. Then $(AB)^T = AB$. By the reversal law:

$$(AB)^T = B^T A^T = BA.$$

  Therefore, $AB = BA$.

- ($\impliedby$) Suppose $AB = BA$. Then:

$$(AB)^T = B^T A^T = BA = AB.$$

  Thus, AB is symmetric.

■

Theorem 1.6.4 (Symmetric Products AA^T and A^TA)**.** *For any rectangular matrix $A \in M_{m\times n}(\mathbb{K})$, the products AA^T (of order $m \times m$) and A^TA (of order $n \times n$) are always symmetric.*

Proof. Using the reversal law for transposition:

$$(AA^T)^T = (A^T)^T A^T = AA^T,$$

and

$$(A^TA)^T = A^T(A^T)^T = A^TA.$$

Hence, both AA^T and A^TA are symmetric. ■

Theorem 1.6.5 (Determinant of Odd-Order Skew-Symmetric Matrices)**.** *If $A \in M_n(\mathbb{K})$ is a skew-symmetric matrix and n is an odd integer, then $\det(A) = 0$.*

Proof. Since $A^T = -A$, taking the determinant on both sides yields:

$$\det(A^T) = \det(-A).$$

Using the property $\det(A^T) = \det(A)$ and $\det(cA) = c^n \det(A)$ for an $n \times n$ matrix:

$$\det(A) = (-1)^n \det(A).$$

Since n is odd, $(-1)^n = -1$, so:

$$\det(A) = -\det(A) \implies 2\det(A) = 0 \implies \det(A) = 0.$$

■

Theorem 1.6.6 (Decomposition into Symmetric and Skew-Symmetric Parts)**.** *Every square matrix $A \in M_n(\mathbb{K})$ can be uniquely expressed as the sum of a symmetric matrix S and a skew-symmetric matrix K.*

Proof. **Existence:** Define S and K as follows:

$$S = \frac{1}{2}(A + A^T) \quad \text{and} \quad K = \frac{1}{2}(A - A^T).$$

First, we verify their structural properties:

$$S^T = \left(\frac{1}{2}(A + A^T)\right)^T = \frac{1}{2}(A^T + (A^T)^T) = \frac{1}{2}(A^T + A) = S \quad (S \text{ is symmetric}),$$

$$K^T = \left(\frac{1}{2}(A - A^T)\right)^T = \frac{1}{2}(A^T - (A^T)^T) = \frac{1}{2}(A^T - A) = -K \quad (K \text{ is skew-symmetric}).$$

Adding S and K:

$$S + K = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = A.$$

Uniqueness: Suppose $A = S_1 + K_1$, where $S_1^T = S_1$ and $K_1^T = -K_1$. Taking the transpose:

$$A^T = (S_1 + K_1)^T = S_1^T + K_1^T = S_1 - K_1.$$

Adding A and A^T: $A + A^T = 2S_1 \implies S_1 = \frac{1}{2}(A + A^T) = S$.
Subtracting A^T from A: $A - A^T = 2K_1 \implies K_1 = \frac{1}{2}(A - A^T) = K$.
Thus, the decomposition is unique. ■

Example 1.6.7. Decompose $A = \begin{bmatrix} 2 & 5 \\ 1 & 4 \end{bmatrix}$ into symmetric and skew-symmetric parts:

$$A^T = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}.$$

$$S = \frac{1}{2}(A + A^T) = \frac{1}{2}\begin{bmatrix} 4 & 6 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix},$$

$$K = \frac{1}{2}(A - A^T) = \frac{1}{2}\begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}.$$

Clearly, $A = S + K$, with $S^T = S$ and $K^T = -K$.

1.6.2 Hermitian and Skew-Hermitian Matrices

Over the complex field $\mathbb{C}$, the complex conjugate transpose A^* plays the standard role of transpose.

Definition 1.6.8 (Hermitian and Skew-Hermitian Matrices)**.** Let $A = (a_{ij}) \in M_n(\mathbb{C})$ be a complex square matrix.

(i) A is called **Hermitian** if $A^* = A$, which means:

$$a_{ij} = \bar{a}_{ji} \quad \text{for all } 1 \leq i, j \leq n.$$

(ii) A is called **skew-Hermitian** if $A^* = -A$, which means:

$$a_{ij} = -\bar{a}_{ji} \quad \text{for all } 1 \leq i, j \leq n.$$

Theorem 1.6.9. *Let $A = (a_{ij}) \in M_n(\mathbb{C})$.*

(i) If A is Hermitian, all its main diagonal entries are purely real.

(ii) If A is skew-Hermitian, all its main diagonal entries are either zero or purely imaginary.

Proof. (i) For a Hermitian matrix A, $a_{ij} = \bar{a}_{ji}$. For diagonal entries ($i = j$):

$$a_{ii} = \bar{a}_{ii}.$$

A complex number equal to its own conjugate must have an imaginary part of zero. Thus, $a_{ii} \in \mathbb{R}$.

(ii) For a skew-Hermitian matrix A, $a_{ij} = -\bar{a}_{ji}$. For diagonal entries ($i = j$):

$$a_{ii} = -\bar{a}_{ii} \implies a_{ii} + \bar{a}_{ii} = 0 \implies 2\,\mathrm{Re}(a_{ii}) = 0 \implies \mathrm{Re}(a_{ii}) = 0.$$

Thus, a_{ii} is purely imaginary or zero.

■

Theorem 1.6.10. *Let $A \in M_n(\mathbb{C})$.*

(i) A is Hermitian if and only if iA is skew-Hermitian.

(ii) A is skew-Hermitian if and only if iA is Hermitian.

Proof. Using property $(\alpha M)^* = \bar{\alpha} M^*$:

(i) $(iA)^* = \bar{i}A^* = -iA^*$. If $A^* = A$ (Hermitian), then $(iA)^* = -iA = -(iA)$ (skew-Hermitian).

(ii) If $A^* = -A$ (skew-Hermitian), then $(iA)^* = -i(-A) = iA$ (Hermitian).

■

Theorem 1.6.11 (Hermitian Products AA^* and A^*A)**.** *For any rectangular complex matrix $A \in M_{m\times n}(\mathbb{C})$, AA^* and A^*A are always Hermitian.*

Proof. By the reversal law for conjugate transpose:

$$(AA^*)^* = (A^*)^*A^* = AA^*, \quad \text{and} \quad (A^*A)^* = A^*(A^*)^* = A^*A.$$

Hence, AA^* and A^*A are Hermitian. ■

Theorem 1.6.12 (Decomposition into Hermitian and Skew-Hermitian Parts)**.** *Every square complex matrix $A \in M_n(\mathbb{C})$ can be uniquely decomposed as:*

$$A = H + K,$$

where H is Hermitian ($H^* = H$) and K is skew-Hermitian ($K^* = -K$).

Proof. Construct $H = \frac{1}{2}(A + A^*)$ and $K = \frac{1}{2}(A - A^*)$. Checking conjugate transposes:

$$H^* = \left(\frac{1}{2}(A + A^*)\right)^* = \frac{1}{2}(A^* + A) = H \quad \text{(Hermitian)},$$

$$K^* = \left(\frac{1}{2}(A - A^*)\right)^* = \frac{1}{2}(A^* - A) = -K \quad \text{(skew-Hermitian)}.$$

Summing: $H + K = A$. Uniqueness follows identically to the symmetric case using A^*. ■

Example 1.6.13. Decompose $A = \begin{bmatrix} 2+i & 3 \\ 1-i & 4i \end{bmatrix}$ into Hermitian and skew-Hermitian components:

$$A^* = (\bar{A})^T = \begin{bmatrix} 2-i & 1+i \\ 3 & -4i \end{bmatrix}.$$

$$H = \frac{1}{2}(A + A^*) = \frac{1}{2}\begin{bmatrix} 4 & 4+i \\ 4-i & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2+\frac{1}{2}i \\ 2-\frac{1}{2}i & 0 \end{bmatrix},$$

$$K = \frac{1}{2}(A - A^*) = \frac{1}{2}\begin{bmatrix} 2i & 2-i \\ -2-i & 8i \end{bmatrix} = \begin{bmatrix} i & 1-\frac{1}{2}i \\ -1-\frac{1}{2}i & 4i \end{bmatrix}.$$

Notice $H^* = H$ (diagonal entries are real: $2, 0$) and $K^* = -K$ (diagonal entries are purely imaginary: $i, 4i$).

Theorem 1.6.14 (Cartesian Decomposition of a Matrix)**.** *Every square complex matrix $A \in M_n(\mathbb{C})$ can be uniquely expressed in the form:*

$$A = P + iQ,$$

where both P and Q are Hermitian matrices (i.e., $P^* = P$ and $Q^* = Q$).

Proof. **Existence**

Let $A \in M_n(\mathbb{C})$ be any square complex matrix, and let A^* be its conjugate transpose. Define two matrices P and Q as:

$$P = \frac{1}{2}(A + A^*) \quad \text{and} \quad Q = \frac{1}{2i}(A - A^*).$$

We first verify that both P and Q are Hermitian:
We have

$$P^* = \left(\frac{1}{2}(A + A^*)\right)^* = \frac{1}{2}(A^* + (A^*)^*) = \frac{1}{2}(A^* + A) = P.$$

Thus, P is Hermitian ($P^* = P$).
Also,

$$Q^* = \left(\frac{1}{2i}(A - A^*)\right)^* = -\frac{1}{2i}(A^* - (A^*)^*) = -\frac{1}{2i}(A^* - A) = \frac{1}{2i}(A - A^*) = Q.$$

Thus, Q is also Hermitian ($Q^* = Q$).

Therefore,

$$P + iQ = \frac{1}{2}(A + A^*) + i\left(\frac{1}{2i}(A - A^*)\right) = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*) = A.$$

This proves that the decomposition $A = P + iQ$ exists.

Uniqueness

Suppose there exist two Hermitian matrices R and S (so $R^* = R$ and $S^* = S$) such that:

$$A = R + iS.$$

Taking the conjugate transpose of both sides:

$$A^* = (R + iS)^* = R^* + (iS)^* = R^* - iS^* = R - iS.$$

We now have a system of two linear matrix equations:

$$A = R + iS \tag{1}$$

$$A^* = R - iS \tag{2}$$

Adding equations (1) and (2):

$$A + A^* = 2R \implies R = \frac{1}{2}(A + A^*) = P.$$

Subtracting equation (2) from equation (1):

$$A - A^* = 2iS \implies S = \frac{1}{2i}(A - A^*) = Q.$$

Since $R = P$ and $S = Q$, the decomposition $A = P + iQ$ is unique. ■

Remark 1.6.15 (Connection to Hermitian and Skew-Hermitian Decomposition). If we set $K = iQ$, then since Q is Hermitian ($Q^* = Q$), we have:

$$K^* = (iQ)^* = -iQ^* = -iQ = -K.$$

Thus, $K = iQ$ is a **skew-Hermitian** matrix. Hence, the Cartesian decomposition $A = P + iQ$ (where P, Q are Hermitian) is mathematically equivalent to writing $A = P + K$ (where P is Hermitian and K is skew-Hermitian).

Example 1.6.16. Decompose the matrix $A = \begin{bmatrix} 2+3i & 1-i \\ 3+5i & 4 \end{bmatrix}$ into the form $P + iQ$, where P and Q are Hermitian.

Solution: First, compute the conjugate transpose $A^* = (\bar{A})^T$:

$$\bar{A} = \begin{bmatrix} 2-3i & 1+i \\ 3-5i & 4 \end{bmatrix} \implies A^* = \begin{bmatrix} 2-3i & 3-5i \\ 1+i & 4 \end{bmatrix}.$$

1. Calculate $P = \frac{1}{2}(A + A^*)$:

$$A + A^* = \begin{bmatrix} (2+3i) + (2-3i) & (1-i) + (3-5i) \\ (3+5i) + (1+i) & 4+4 \end{bmatrix} = \begin{bmatrix} 4 & 4-6i \\ 4+6i & 8 \end{bmatrix}.$$

$$P = \frac{1}{2}\begin{bmatrix} 4 & 4-6i \\ 4+6i & 8 \end{bmatrix} = \begin{bmatrix} 2 & 2-3i \\ 2+3i & 4 \end{bmatrix}.$$

Notice $P^* = P$ (diagonal entries are real: $2, 4$).

2. Calculate $Q = \frac{1}{2i}(A - A^*)$:

$$A - A^* = \begin{bmatrix} (2+3i)-(2-3i) & (1-i)-(3-5i) \\ (3+5i)-(1+i) & 4-4 \end{bmatrix} = \begin{bmatrix} 6i & -2+4i \\ 2+4i & 0 \end{bmatrix}.$$

Dividing by $2i$ (note that $\frac{1}{2i} = -\frac{i}{2}$):

$$Q = \frac{1}{2i}\begin{bmatrix} 6i & -2+4i \\ 2+4i & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2+i \\ 2-i & 0 \end{bmatrix}.$$

Thus, $Q^* = Q$.

3. Verification:

$$P + iQ = \begin{bmatrix} 2 & 2-3i \\ 2+3i & 4 \end{bmatrix} + i\begin{bmatrix} 3 & 2+i \\ 2-i & 0 \end{bmatrix} = \begin{bmatrix} 2+3i & (2-3i)+(2i-1) \\ (2+3i)+(2i+1) & 4+0 \end{bmatrix}$$
$$= \begin{bmatrix} 2+3i & 1-i \\ 3+5i & 4 \end{bmatrix} = A.$$

This completes the decomposition.

Summary Comparison

Matrix Type	Defining Condition	Diagonal Entries	Real Analogue
Symmetric	$A^T = A$	Unrestricted	$a_{ij} = a_{ji}$
Skew-Symmetric	$A^T = -A$	All Zero ($a_{ii} = 0$)	$a_{ij} = -a_{ji}$
Hermitian	$A^* = A$	Purely Real ($\text{Im}(a_{ii}) = 0$)	Symmetric
Skew-Hermitian	$A^* = -A$	Purely Imaginary or Zero ($\text{Re}(a_{ii}) = 0$)	Skew-Symmetric

Exercise 1.6.17. Solve the following exercises:

1. Construct two 2×2 real matrices A and B such that $AB = 0$ (the zero matrix) but $BA \neq 0$. What does this imply about zero divisors in matrix algebra?

2. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$. Compute $(A+B)^2$ and $A^2 + 2AB + B^2$. Show that $(A+B)^2 \neq A^2 + 2AB + B^2$, and state the necessary and sufficient condition under which equality holds.

3. Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ be matrices over $\mathbb{C}$. Compute AB and BA, and show that A and B anti-commute (i.e., $AB = -BA$).

4. Verify the reversal law for transposition $(AB)^T = B^TA^T$ for the matrices:
$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ -1 & 1 \end{bmatrix}.$$

5. Using the reversal law for two matrices, prove by induction that for any k conformable matrices $A_1, A_2, \ldots, A_k$:
$$(A_1A_2\cdots A_k)^T = A_k^T \cdots A_2^TA_1^T.$$

6. For $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$, find A^{-1} and explicitly verify that $AA^{-1} = A^{-1}A = I_2$.

7. Given $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, calculate $(AB)^{-1}$ and $B^{-1}A^{-1}$, and confirm that $(AB)^{-1} = B^{-1}A^{-1}$.

8. Show that if $A \in M_n(\mathbb{K})$ is an invertible matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$. Verify this result for $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$.

9. Let $A = \begin{bmatrix} 2+i & 3-2i \\ 4i & 1+5i \end{bmatrix} \in M_2(\mathbb{C})$. Find the conjugate transpose A^*, and verify that $(2iA)^* = -2iA^*$.

10. Let $A \in M_{m\times p}(\mathbb{C})$ and $B \in M_{p\times n}(\mathbb{C})$. Prove formally from definition that $(AB)^* = B^*A^*$.

11. Express the real matrix $A = \begin{bmatrix} 3 & 8 & 2 \\ 2 & 5 & 6 \\ 4 & 0 & 1 \end{bmatrix}$ uniquely in the form $A = S + K$, where S is symmetric ($S^T = S$) and K is skew-symmetric ($K^T = -K$).

12. Prove that if A is an $n \times n$ skew-symmetric matrix and n is an odd integer, then $\det(A) = 0$. Using this property, state $\det(A)$ without expansion for:
$$A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}.$$

13. Prove that all main diagonal entries of a Hermitian matrix are real numbers, whereas all main diagonal entries of a skew-Hermitian matrix are either zero or purely imaginary.

14. Show that if H is a Hermitian matrix, then $K = iH$ is a skew-Hermitian matrix. Conversely, show that if K is skew-Hermitian, then $H = -iK$ is Hermitian.

15. Decompose the complex matrix $A = \begin{bmatrix} 1+2i & 3-i \\ 2+4i & 5i \end{bmatrix}$ into the form $A = P + iQ$, where P is a Hermitian matrix and Q is a skew-Hermitian matrix.

16. Find the cofactor matrix C and the adjoint matrix $\mathrm{adj}(A)$ for:
$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}.$$

17. For $A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}$, compute $\mathrm{adj}(A)$ and verify the fundamental identity $A \cdot \mathrm{adj}(A) = \det(A)I_3$.

18. Let A be a 3×3 non-singular matrix with $\det(A) = 4$.
    (a) Prove that $\det(\mathrm{adj}(A)) = (\det(A))^2$.
    (b) Find the numerical value of $\det(\mathrm{adj}(A))$.

19. Prove that for any non-singular matrix $A \in M_n(\mathbb{K})$ ($n \geq 2$):
$$\mathrm{adj}(\mathrm{adj}(A)) = (\det(A))^{n-2}A.$$

20. Using the adjoint method ($A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$), determine the inverse of the matrix:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}.$$

Chapter 2

Elementary Operations and Systems of Linear Equations

This unit focuses on the fundamental theory of elementary operations and their matrix representations to analyze matrix structure and solve linear systems. It covers the reduction of matrices to row echelon and normal forms (PAQ), establishing the equivalence of non-singularity with row reducibility to identity and expressibility as products of elementary matrices. Furthermore, it details the systematic investigation of homogeneous ($AX = 0$) and non-homogeneous ($AX = B$) linear systems. Finally, it provides complete criteria for consistency and geometric/algebraic characterizations of their solution sets.

2.1 Elementary Operations, Elementary Matrices, and Matrix Normal Forms

Elementary matrix operations provide the fundamental computational engine for solving linear systems, finding matrix inverses, and reducing matrices to canonical forms.

2.1.1 Elementary Operations and Elementary Matrices

Definition 2.1.1 (Elementary Row and Column Operations)**.** An **elementary operation** on an $m \times n$ matrix A over $\mathbb{K}$ is an operation of any of the following three types:

Type I: **Interchange:** Interchanging two rows (or columns).

$$R_i \leftrightarrow R_j \quad (\text{or } C_i \leftrightarrow C_j).$$

Type II: **Scaling:** Multiplying a row (or column) by a non-zero scalar $c \in \mathbb{K} \setminus \{0\}$.

$$R_i \to cR_i \quad (\text{or } C_i \to cC_i).$$

Type III: **Addition/Replacement:** Adding a scalar multiple $c \in \mathbb{K}$ of one row (or column) to another row (or column).

$$R_i \to R_i + cR_j \quad (i \neq j) \quad (\text{or } C_i \to C_i + cC_j).$$

Definition 2.1.2 (Elementary Matrix)**.** An **elementary matrix** of order n is a matrix obtained by performing a single elementary row (or column) operation on the identity matrix I_n.

Representation of Elementary Operations via Matrix Multiplication

Proof. Let $A \in M_{m\times n}(\mathbb{K})$ be an $m \times n$ matrix with rows $R_1, R_2, \dots, R_m$, written as a row-block matrix:

$$A = \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_m \end{bmatrix},$$

where each $R_k \in \mathbb{K}^{1\times n}$ is a $1 \times n$ row vector.

Let $I_m = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_m \end{bmatrix}$ denote the $m \times m$ identity matrix, where $e_k = \begin{bmatrix} 0 & \cdots & 1 & \cdots & 0 \end{bmatrix}$ is the k-th standard row basis vector of $\mathbb{K}^{1\times m}$. Note that for any $1 \times m$ vector $v = \sum_{k=1}^{m} v_k e_k$, we have $vA = \sum_{k=1}^{m} v_k R_k$.

Part (i): Representation of Elementary Row Operations

Let e be an elementary row operation, and let $E = e(I_m)$ be the corresponding $m \times m$ elementary matrix whose k-th row is $r_k = e(e_k)$. The product EA has k-th row given by $(EA)_{k\cdot} = r_k A$.

We consider the three types of elementary row operations separately:

Type I: Interchange ($R_i \leftrightarrow R_j$):
Applying $R_i \leftrightarrow R_j$ to I_m yields the elementary matrix E with rows:

$$r_k = e_k \quad \text{for } k \neq i, j, \quad r_i = e_j, \quad \text{and} \quad r_j = e_i.$$

Multiplying E by A gives the rows of EA:

$$(EA)_{k\cdot} = e_k A = R_k \quad (k \neq i, j), \quad (EA)_{i\cdot} = e_j A = R_j, \quad \text{and} \quad (EA)_{j\cdot} = e_i A = R_i.$$

Thus, $EA = \begin{bmatrix} \vdots \\ R_j \\ \vdots \\ R_i \\ \vdots \end{bmatrix}$, which is precisely A after interchanging R_i and R_j.

Type II: Scaling ($R_i \to cR_i$, $c \neq 0$):
Applying $R_i \to cR_i$ to I_m yields the elementary matrix E with rows:

$$r_k = e_k \quad \text{for } k \neq i, \quad \text{and} \quad r_i = ce_i.$$

Multiplying E by A gives the rows of EA:

$$(EA)_{k\cdot} = e_k A = R_k \quad (k \neq i), \quad \text{and} \quad (EA)_{i\cdot} = (ce_i)A = c(e_i A) = cR_i.$$

Thus, EA is precisely the matrix obtained by multiplying R_i of A by c.

Type III: Addition/Replacement ($R_i \to R_i + cR_j$, $i \neq j$):
Applying $R_i \to R_i + cR_j$ to I_m yields the elementary matrix E with rows:

$$r_k = e_k \quad \text{for } k \neq i, \quad \text{and} \quad r_i = e_i + ce_j.$$

Multiplying E by A gives the rows of EA:

$$(EA)_{k\cdot} = e_k A = R_k \quad (k \neq i), \quad \text{and} \quad (EA)_{i\cdot} = (e_i + ce_j)A = e_i A + c(e_j A) = R_i + cR_j.$$

Thus, EA is precisely the matrix obtained by adding cR_j to R_i in A.

This establishes that performing any elementary row operation on A is equivalent to left-multiplying A by the corresponding elementary matrix E.

Part (ii): Representation of Elementary Column Operations

Performing an elementary column operation f on A is algebraically equivalent to performing the corresponding elementary row operation e on A^T, followed by taking the transpose of the result.

By Part (i), applying the row operation e to A^T is equivalent to left-multiplying A^T by the $n \times n$ elementary matrix $E = e(I_n)$:

$$\text{Result of column operation } f \text{ on } A = \left(e(A^T)\right)^T = (E \cdot A^T)^T.$$

Using the reversal law for matrix transposition:

$$(E \cdot A^T)^T = (A^T)^T \cdot E^T = A \cdot E^T.$$

Define $F = E^T$. Since $E = e(I_n)$ is the elementary matrix for the row operation e, taking its transpose $F = E^T = (e(I_n))^T = f(I_n)$ yields the $n \times n$ elementary matrix corresponding to the column operation f.

Hence, performing an elementary column operation on A is equivalent to right-multiplying A by the corresponding elementary matrix F. ∎

Theorem 2.1.3 (Invertibility of Elementary Matrices). *Every elementary matrix is non-singular (invertible), and its inverse is also an elementary matrix of the same type corresponding to the inverse elementary operation.*

Proof. Let I_n denote the $n \times n$ identity matrix, and let e be any elementary row operation on I_n so that the corresponding elementary matrix is given by $E = e(I_n)$.

Every elementary row operation e possesses a unique inverse elementary row operation e^{-1} of the exact same type, defined as follows:

(i) **Type I ($R_i \leftrightarrow R_j$):** The inverse operation e^{-1} is $R_i \leftrightarrow R_j$ (itself).

(ii) **Type II ($R_i \to cR_i$, $c \neq 0$):** The inverse operation e^{-1} is $R_i \to c^{-1}R_i$ ($c^{-1} \neq 0$).

(iii) **Type III ($R_i \to R_i + cR_j$, $i \neq j$):** The inverse operation e^{-1} is $R_i \to R_i - cR_j$.

Let $E' = e^{-1}(I_n)$ be the elementary matrix formed by applying the inverse operation e^{-1} to I_n. By definition, E' is an elementary matrix of the same type as E.

Recall that applying an elementary row operation to any matrix A is algebraically equivalent to left-multiplying A by the corresponding elementary matrix, i.e., $e(A) = e(I_n)A = EA$.

Applying this property to E' and E:

$$E \cdot E' = e(E') = e\left(e^{-1}(I_n)\right) = (e \circ e^{-1})(I_n) = I_n,$$

and

$$E' \cdot E = e^{-1}(E) = e^{-1}\left(e(I_n)\right) = (e^{-1} \circ e)(I_n) = I_n.$$

Since $E \cdot E' = E' \cdot E = I_n$, E is non-singular and its inverse is given by:

$$E^{-1} = E' = e^{-1}(I_n),$$

which is precisely the elementary matrix corresponding to the inverse elementary operation e^{-1}. ∎

Remark 2.1.4 (Matrix Representation of Row Operations). Performing an elementary row operation e on an $m \times n$ matrix A is algebraically equivalent to **left-multiplying** (pre-multiplying) A by an $m \times m$ elementary matrix E.

The elementary matrix E is obtained by applying the exact same row operation e to the identity matrix I_m:

$$A \xrightarrow{\text{row operation } e} e(A) = E \cdot A, \quad \text{where } E = e(I_m).$$

Key takeaway: To transform rows, pre-multiply by the elementary matrix; to transform columns, post-multiply by the elementary matrix.

Example 2.1.5 (Representing a Row Operation via Pre-Multiplication)**.** Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ be a 2×3 matrix. Represent the elementary row operation $R_2 \to R_2 - 3R_1$ by pre-multiplying A by an elementary matrix E, and verify the result.

Solution. First, applying the operation $R_2 \to R_2 - 3R_1$ directly to A yields:

$$A \xrightarrow{R_2 \to R_2 - 3R_1} \begin{bmatrix} 1 & 2 & 3 \\ 4 - 3(1) & 5 - 3(2) & 6 - 3(3) \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -1 & -3 \end{bmatrix}.$$

Next, construct the 2×2 elementary matrix E by applying $R_2 \to R_2 - 3R_1$ to the identity matrix I_2:

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 3R_1} E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}.$$

Pre-multiplying A by E gives:

$$\begin{aligned} E \cdot A &= \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 0(4) & 1(2) + 0(5) & 1(3) + 0(6) \\ -3(1) + 1(4) & -3(2) + 1(5) & -3(3) + 1(6) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 & 3 \\ 1 & -1 & -3 \end{bmatrix}. \end{aligned}$$

The result of pre-multiplying by E is identical to the direct row operation, verifying that $e(A) = E \cdot A$. ■

Example 2.1.6 (Row Interchange Operation $R_1 \leftrightarrow R_2$)**.** Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ be a 2×3 matrix.

Interchanging the first and second rows ($R_1 \leftrightarrow R_2$) directly yields:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}.$$

To represent this operation via matrix multiplication, we apply $R_1 \leftrightarrow R_2$ to the identity matrix I_2:

$$E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Left-multiplying A by E gives:

$$EA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 0(1) + 1(4) & 0(2) + 1(5) & 0(3) + 1(6) \\ 1(1) + 0(4) & 1(2) + 0(5) & 1(3) + 0(6) \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}.$$

This verifies that left-multiplying A by E executes the row interchange operation $R_1 \leftrightarrow R_2$.

Example 2.1.7 (Row Addition Operation $R_2 \to R_2 - 3R_1$)**.** Let $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \\ 1 & 0 \end{bmatrix}$ be a 3×2 matrix.

Applying the row operation $R_2 \to R_2 - 3R_1$ directly to A gives:

$$A \xrightarrow{R_2 \to R_2 - 3R_1} \begin{bmatrix} 2 & 1 \\ 3 - 3(2) & 4 - 3(1) \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -3 & 1 \\ 1 & 0 \end{bmatrix}.$$

The corresponding 3×3 elementary matrix E is constructed by performing $R_2 \to R_2 - 3R_1$ on I_3:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Computing the product EA:

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 4 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1(2) + 0 + 0 & 1(1) + 0 + 0 \\ -3(2) + 1(3) + 0 & -3(1) + 1(4) + 0 \\ 0 + 0 + 1(1) & 0 + 0 + 1(0) \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -3 & 1 \\ 1 & 0 \end{bmatrix}.$$

Hence, left-multiplication by E performs the row addition $R_2 \to R_2 - 3R_1$.

Example 2.1.8 (Column Scaling Operation $C_2 \to 3C_2$)**.** Let $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$ be a 2×3 matrix.

Scaling the second column by a factor of 3 ($C_2 \to 3C_2$) directly gives:

$$A \xrightarrow{C_2 \to 3C_2} \begin{bmatrix} 1 & 3(3) & 5 \\ 2 & 3(4) & 6 \end{bmatrix} = \begin{bmatrix} 1 & 9 & 5 \\ 2 & 12 & 6 \end{bmatrix}.$$

To represent this column operation, we apply $C_2 \to 3C_2$ to I_3 to obtain the elementary matrix F:

$$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Right-multiplying A by F yields:

$$AF = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1(1) + 0 + 0 & 0 + 3(3) + 0 & 0 + 0 + 5(1) \\ 2(1) + 0 + 0 & 0 + 4(3) + 0 & 0 + 0 + 6(1) \end{bmatrix} = \begin{bmatrix} 1 & 9 & 5 \\ 2 & 12 & 6 \end{bmatrix}.$$

Thus, right-multiplication by F performs the column scaling $C_2 \to 3C_2$.

Example 2.1.9 (Column Addition Operation $C_2 \to C_2 - 2C_1$)**.** Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ be a 2×2 matrix.

Applying the column operation $C_2 \to C_2 - 2C_1$ directly to A gives:

$$A \xrightarrow{C_2 \to C_2 - 2C_1} \begin{bmatrix} 1 & 2 - 2(1) \\ 3 & 4 - 2(3) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix}.$$

The corresponding elementary matrix F obtained by applying $C_2 \to C_2 - 2C_1$ to I_2 is:

$$F = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Right-multiplying A by F verifies the result:

$$AF = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1(1) + 2(0) & 1(-2) + 2(1) \\ 3(1) + 4(0) & 3(-2) + 4(1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & -2 \end{bmatrix}.$$

Example 2.1.10 (Inverse of Scaling Matrix (Type II))**.** Consider the 3×3 elementary matrix E corresponding to the row scaling operation $R_2 \to 4R_2$:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The inverse operation of multiplying R_2 by 4 is multiplying R_2 by $\frac{1}{4}$ ($R_2 \to \frac{1}{4}R_2$). Applying this inverse operation to I_3 yields:

$$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Direct calculation verifies that E^{-1} is indeed the two-sided inverse of E:

$$EE^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3,$$

and

$$E^{-1}E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

This demonstrates that E is invertible and E^{-1} is an elementary matrix of the same type.

Example 2.1.11 (Inverse of Replacement Matrix (Type III))**.** Consider the 3×3 elementary matrix E corresponding to the operation $R_3 \to R_3 - 3R_1$:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}.$$

The inverse operation of subtracting $3R_1$ from R_3 is adding $3R_1$ to R_3 ($R_3 \to R_3 + 3R_1$). Performing this inverse operation on I_3 gives:

$$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}.$$

We verify that $EE^{-1} = E^{-1}E = I_3$:

$$EE^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3(1) + 0 + 1(3) & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3,$$

and

$$E^{-1}E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3(1) + 0 + 1(-3) & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

Example 2.1.12 (Elementary Matrices and Their Inverses)**.** Starting with the 3×3 identity matrix $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, we construct elementary matrices and their inverses for each type of row operation:

- **Type I ($R_1 \leftrightarrow R_3$):** Interchanging the first and third rows gives $E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. Since interchanging twice restores the matrix, $E_1^{-1} = E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.
- **Type II ($R_2 \to 5R_2$):** Scaling the second row by 5 gives $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The inverse scaling operation $R_2 \to \frac{1}{5}R_2$ yields $E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
- **Type III ($R_3 \to R_3 - 2R_1$):** Subtracting twice the first row from the third row gives $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$. The inverse operation $R_3 \to R_3 + 2R_1$ yields $E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$.

2.1.2 Row Echelon Form (REF) and Reduced Row Echelon Form (RREF)

Definition 2.1.13 (Row Echelon Form)**.** An $m \times n$ matrix A is in **Row Echelon Form (REF)** if it satisfies:

(i) All zero rows (rows with all zero entries) are placed at the bottom of the matrix.

(ii) The first non-zero entry in a non-zero row (called the **leading entry** or **pivot**) is strictly to the right of the leading entry of the row above it.

Definition 2.1.14 (Reduced Row Echelon Form)**.** A matrix A is in **Reduced Row Echelon Form (RREF)** if it is in Row Echelon Form and satisfies two additional conditions:

(iii) Every leading entry (pivot) is 1.

(iv) Each column containing a leading 1 has zeros in all its other entries.

Theorem 2.1.15 (Uniqueness of RREF)**.** *Every matrix A is row-equivalent to a **unique** matrix in Reduced Row Echelon Form (RREF). (Note: The REF of a matrix is not unique, but its RREF is unique).*

Proof. Let A be an $m \times n$ matrix over $\mathbb{K}$. To prove uniqueness, suppose A is row-equivalent to two matrices $R = (r_{ij})$ and $S = (s_{ij})$, both of which are in Reduced Row Echelon Form (RREF).

Since elementary row operations preserve the solution set of a system of linear equations, the homogeneous systems $RX = 0$ and $SX = 0$ have the exact same solution set $V = \{X \in \mathbb{K}^n \mid RX = 0\} = \{X \in \mathbb{K}^n \mid SX = 0\}$.

We establish that $R = S$ by demonstrating that R and S share identical pivot column positions, and subsequently showing that all their corresponding entries coincide.

Identity of Pivot Column Positions

Let $c_1 < c_2 < \cdots < c_k$ be the indices of the pivot columns of R, and let $d_1 < d_2 < \cdots < d_l$ be the indices of the pivot columns of S.

Suppose for contradiction that the pivot column positions of R and S are not identical. Let j be the smallest column index where R and S differ in pivot status. Without loss of generality, assume column j is a pivot column in R (so $j = c_p$ for some $1 \leq p \leq k$), but column j is a non-pivot (free) column in S.

Since column j is a free column in S, we can construct a non-zero solution vector $X = (x_1, x_2, \ldots, x_n)^T \in \mathbb{K}^n$ to $SX = 0$ by setting the free variable $x_j = 1$ and setting all other free variables with index greater than j to zero ($x_f = 0$ for $f > j$).

By definition of RREF, for $SX = 0$, the pivot variables x_{d_i} (for $d_i < j$) are uniquely determined by x_j, while $x_m = 0$ for all $m > j$. Since $X \in V$, X must also satisfy $RX = 0$.

Now consider the p-th row equation of $RX = 0$. Because R is in RREF with pivot at $(p, c_p) = (p, j)$:

- $r_{p,j} = 1$.
- $r_{p,c_i} = 0$ for all other pivot columns $c_i \neq j$.
- $r_{p,f} = 0$ for all columns $f > j$ (since $x_f = 0$).

Evaluating the p-th row of $RX = 0$ at X gives:

$$(RX)_p = r_{p,1}x_1 + \cdots + r_{p,j}x_j + \cdots + r_{p,n}x_n = 0 + 1(1) + 0 = 1 \neq 0.$$

This contradicts $RX = 0$. Hence, column j must also be a pivot column in S.

By symmetry, R and S must have the exact same number of pivot columns ($k = l$) at the exact same positions:

$$c_i = d_i \quad \text{for all } i = 1, 2, \ldots, k.$$

Equality of Matrix Entries

Since R and S have identical pivot columns $c_1 < c_2 < \cdots < c_k$, both matrices have leading 1s at positions (i, c_i) and zero entries everywhere else in column c_i for $1 \leq i \leq k$.

Now consider any non-pivot (free) column $j \notin \{c_1, c_2, \ldots, c_k\}$. In $RX = 0$, construct a basic solution $X^{(j)} = (x_1, x_2, \ldots, x_n)^T$ corresponding to the free variable $x_j = 1$ and all other free variables set to zero. The i-th row equation of $RX = 0$ yields:

$$x_{c_i} + r_{i,j}x_j = 0 \implies x_{c_i} = -r_{i,j} \quad \text{for } 1 \leq i \leq k.$$

Since $SX = 0$ has the same solution set, $X^{(j)}$ must also satisfy $SX = 0$. Substituting the components of $X^{(j)}$ into the i-th row equation of $SX = 0$ gives:

$$x_{c_i} + s_{i,j}x_j = 0 \implies -r_{i,j} + s_{i,j}(1) = 0 \implies s_{i,j} = r_{i,j}.$$

Since $s_{i,j} = r_{i,j}$ holds for every row i and every non-pivot column j, and all pivot columns are identical, it follows that:

$$R = S.$$

Thus, the Reduced Row Echelon Form of A is unique. ■

Example 2.1.16. Reduce $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 7 \\ -1 & 2 & 1 \end{bmatrix}$ to REF and RREF.

Solution. Eliminate entries below first pivot ($a_{11} = 1$):**

$$R_2 \to R_2 - 2R_1, \quad R_3 \to R_3 + R_1 \implies \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 3 \\ 0 & 1 & 3 \end{bmatrix}.$$

Create second pivot:**

$$R_2 \to \frac{1}{3}R_2 \implies \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix}.$$

$$R_3 \to R_3 - R_2 \implies \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \quad \textbf{(Row Echelon Form - REF)}.$$

Create third pivot and clear entries above pivots (RREF):**

$$R_3 \to \frac{1}{2}R_3 \implies \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Clear column 3: $R_2 \to R_2 - R_3$, $R_1 \to R_1 - 2R_3$:

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Clear column 2: $R_1 \to R_1 + R_2$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \quad \textbf{(Reduced Row Echelon Form - RREF)}.$$

■

2.1.3 Reduction of a Matrix to Normal Form PAQ

Definition 2.1.17 (Normal Form / Rank Canonical Form)**.** An $m \times n$ matrix N is said to be in **Normal Form** (or **Rank Canonical Form**) if it has the block form:

$$N = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}_{m \times n},$$

where I_r is the $r \times r$ identity matrix ($r \geq 0$), and 0 represents zero submatrices of appropriate dimensions. The integer r is the **rank** of the matrix.

Theorem 2.1.18 (Reduction to Normal Form PAQ)**.** *For any $m \times n$ matrix A of rank r, there exist non-singular matrices $P \in M_m(\mathbb{K})$ (representing a sequence of elementary row operations) and $Q \in M_n(\mathbb{K})$ (representing a sequence of elementary column operations) such that:*

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}.$$

Proof. Let $A \in M_{m \times n}(\mathbb{K})$. 1. By applying a finite sequence of elementary row operations represented by elementary matrices $E_1, E_2, \ldots, E_k$, A can be reduced to its RREF R:

$$PA = R, \quad \text{where } P = E_k \cdots E_2 E_1 \text{ is non-singular.}$$

2. In R, the r leading 1s are in r distinct pivot columns. By applying column operations (column interchanges $C_i \leftrightarrow C_j$ to move pivot columns to the left, followed by column additions $C_j \to C_j - cC_i$ to clear entries to the right of each pivot 1), represented by non-singular elementary matrices $F_1, F_2, \ldots, F_p$:

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}, \quad \text{where } Q = F_1 F_2 \cdots F_p \text{ is non-singular.}$$

■

Simultaneous Algorithm to Compute P and Q

To compute P and Q simultaneously for $A_{m\times n}$: 1. Write the matrix array $\begin{bmatrix} A \\ I_n \end{bmatrix}$ or setup $A = I_m A I_n$. 2. Perform elementary **row operations** on A and simultaneously on I_m. 3. Perform elementary **column operations** on A and simultaneously on I_n. 4. When A is transformed into $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$, I_m becomes P and I_n becomes Q.

Example 2.1.19 (Reduction to Normal Form PAQ)**.** Find non-singular matrices P and Q such that PAQ is in normal form for:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 1 & 3 & 8 \end{bmatrix}.$$

Solution: Write $A = I_3 A I_3$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 1 & 3 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Step 1: Row Operations on A and I_3 (left matrix):

- $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 2 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- $R_3 \to R_3 - 2R_2$:

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Step 2: Column Operations on A and I_3 (right matrix):

- $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - 2C_1$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- $C_3 \to C_3 - 3C_2$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Result: The normal form is $PAQ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$, where:

$$P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{and } \operatorname{rank}(A) = 2.$$

2.2 Representation of non-singular matrices as products of elementary matrices

The following theorem connects invertibility to row operations.

Theorem 2.2.1. *Every non-singular matrix A can be expressed as a product of elementary matrices.*

Proof. Let A be an $n \times n$ non-singular matrix. By definition, a non-singular matrix has full rank (n), which means its reduced row echelon form is the identity matrix, I_n. This implies that there exists a finite sequence of elementary row operations that transforms A into I_n. Let the elementary matrices corresponding to these operations be $E_1, E_2, \ldots, E_k$. Applying these in order to A gives:

$$(E_k E_{k-1} \cdots E_2 E_1)A = I_n$$

This equation shows that the product of the elementary matrices is the inverse of A:

$$E_k E_{k-1} \cdots E_2 E_1 = A^{-1}$$

Since A is invertible, so is its inverse A^{-1}. We can take the inverse of both sides of this equation to solve for A:

$$A = (A^{-1})^{-1} = (E_k E_{k-1} \cdots E_2 E_1)^{-1}$$

Using the reversal law for the inverse of a product, we get:

$$A = E_1^{-1} E_2^{-1} \cdots E_{k-1}^{-1} E_k^{-1}$$

The inverse of any elementary matrix is also an elementary matrix. For example, the inverse of a row addition is a row subtraction, and the inverse of scaling by c is scaling by $1/c$. Therefore, we have successfully expressed A as a product of elementary matrices. ∎

Example 2.2.2 (Decomposition into Elementary Matrices)**.** Express $C = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ as a product of elementary matrices.

Solution. We reduce C to the identity matrix. The only required operation is $R_1 \to R_1 - 2R_2$. The corresponding elementary matrix is $E = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$. Applying this operation gives $EC = I$. Therefore, $C = E^{-1}$. The inverse of the operation $R_1 \to R_1 - 2R_2$ is $R_1 \to R_1 + 2R_2$. The inverse elementary matrix is:

$$E^{-1} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

Thus, C is itself an elementary matrix. In this simple case, the "product" is just one matrix. ∎

Example 2.2.3. Express the non-singular matrix A as a product of elementary matrices.

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Solution. The process involves two main parts: first, we find the sequence of elementary matrices that reduces A to the identity matrix I. Second, we use the inverses of these elementary matrices to construct A.

Reduce A to the Identity Matrix We apply row operations sequentially, identifying the elementary matrix at each step.

Start with $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$. The first operation is $R_3 \to R_3 - R_1$. The elementary matrix for this is $E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$.

$$E_1A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 1 \end{pmatrix}$$

The second operation is $R_3 \to R_3 + 2R_2$. The elementary matrix is $E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$.

$$E_2(E_1A) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

The third operation is $R_3 \to \frac{1}{3}R_3$. The elementary matrix is $E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/3 \end{pmatrix}$.

$$E_3(E_2E_1A) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

The fourth operation is $R_2 \to R_2 - R_3$. The elementary matrix is $E_4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$.

$$E_4(E_3E_2E_1A) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The final operation is $R_1 \to R_1 - 2R_2$. The elementary matrix is $E_5 = \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

$$E_5(E_4E_3E_2E_1A) = \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$$

The complete reduction is given by the equation $E_5E_4E_3E_2E_1A = I$. Solving for A gives:

$$A = (E_5E_4E_3E_2E_1)^{-1} = E_1^{-1}E_2^{-1}E_3^{-1}E_4^{-1}E_5^{-1}$$

Find the Inverse Elementary Matrices We find the inverse of each elementary matrix by finding the inverse of its corresponding operation.

- E_1^{-1} (inverse of $R_3 \to R_3 - R_1$) is $R_3 \to R_3 + R_1$: $E_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$.

- E_2^{-1} (inverse of $R_3 \to R_3 + 2R_2$) is $R_3 \to R_3 - 2R_2$: $E_2^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}$.
- E_3^{-1} (inverse of $R_3 \to \frac{1}{3}R_3$) is $R_3 \to 3R_3$: $E_3^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.
- E_4^{-1} (inverse of $R_2 \to R_2 - R_3$) is $R_2 \to R_2 + R_3$: $E_4^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.
- E_5^{-1} (inverse of $R_1 \to R_1 - 2R_2$) is $R_1 \to R_1 + 2R_2$: $E_5^{-1} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Conclusion The matrix A can be written as the following product of elementary matrices, in the reverse order of their inverses:

$$A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} E_5^{-1}$$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

This product, when multiplied out, will yield the original matrix A. ■

2.3 Systems of Linear Equations

A general system of m linear equations in n unknowns $(x_1, x_2, \ldots, x_n)$ can be written as:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

where the coefficients a_{ij} and the constants b_i are scalars.

A system of m linear equations in n variables can be compactly represented in matrix form as:

$$A\mathbf{x} = \mathbf{b}$$

where A is an $m \times n$ matrix of coefficients, $\mathbf{x}$ is an $n \times 1$ column vector of variables, and $\mathbf{b}$ is an $m \times 1$ column vector of constants. We categorize these systems into two types based on the vector $\mathbf{b}$.

2.3.1 The Non-Homogeneous System: $A\mathbf{x} = \mathbf{b}$

Definition 2.3.1 (Non-Homogeneous System)**.** A system of linear equations $A\mathbf{x} = \mathbf{b}$ is called **non-homogeneous** if the vector of constants $\mathbf{b}$ is not the zero vector (i.e., $\mathbf{b} \neq \mathbf{0}$).

The primary question for a non-homogeneous system is whether a solution exists at all. A system that has at least one solution is called **consistent**; otherwise, it is **inconsistent**. The rank of the matrix provides a definitive test for consistency.

Theorem 2.3.2 (Rouché–Capelli Theorem for Consistency). *A system of linear equations $A\mathbf{x} = \mathbf{b}$ is consistent if and only if the rank of the coefficient matrix A equals the rank of the augmented matrix $[A \mid \mathbf{b}]$:*

$$\operatorname{rank}(A) = \operatorname{rank}([A \mid \mathbf{b}]).$$

Proof. Let $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$ be represented in terms of its column vectors.

By matrix-vector multiplication, $A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$. Thus, the system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ can be expressed as a linear combination of the columns of A, which means $\mathbf{b} \in \operatorname{col}(A)$.

We prove the two directions of the equivalence:

Direct Direction:
Suppose $A\mathbf{x} = \mathbf{b}$ is consistent. Then $\mathbf{b} \in \operatorname{col}(A)$. Adjoining $\mathbf{b}$ to the set of columns of A does not introduce a new linearly independent vector, so the spanned space is unchanged:

$$\operatorname{col}([A \mid \mathbf{b}]) = \operatorname{col}(A).$$

Taking the dimension of both column spaces yields:

$$\operatorname{rank}([A \mid \mathbf{b}]) = \dim(\operatorname{col}([A \mid \mathbf{b}])) = \dim(\operatorname{col}(A)) = \operatorname{rank}(A).$$

Converse Direction:
Suppose $\operatorname{rank}(A) = \operatorname{rank}([A \mid \mathbf{b}])$. The column space $\operatorname{col}(A)$ is inherently a subspace of $\operatorname{col}([A \mid \mathbf{b}])$. Since a subspace having the same dimension as its enclosing space must be equal to the space itself:

$$\operatorname{col}(A) = \operatorname{col}([A \mid \mathbf{b}]).$$

Because $\mathbf{b} \in \operatorname{col}([A \mid \mathbf{b}])$, it follows that $\mathbf{b} \in \operatorname{col}(A)$. Hence, $\mathbf{b}$ is a linear combination of the columns of A, guaranteeing the existence of a solution $\mathbf{x}$.

■

Once consistency is established, the rank can further tell us about the number of solutions.

Theorem 2.3.3 (Nature of Solutions). *If a non-homogeneous system $A\mathbf{x} = \mathbf{b}$ is consistent and rank$(A) = r$, where n is the number of variables (columns), then:*

1. *The system has a **unique solution** if $r = n$.*
2. *The system has **infinitely many solutions** if $r < n$. The number of free parameters in the solution will be $n - r$.*

Proof. Since the system $A\mathbf{x} = \mathbf{b}$ is consistent, there exists at least one particular solution $\mathbf{x}_p$ such that $A\mathbf{x}_p = \mathbf{b}$.

If $\mathbf{x}$ is any other solution to $A\mathbf{x} = \mathbf{b}$, define $\mathbf{x}_h = \mathbf{x} - \mathbf{x}_p$. Applying A gives:

$$A\mathbf{x}_h = A(\mathbf{x} - \mathbf{x}_p) = A\mathbf{x} - A\mathbf{x}_p = \mathbf{b} - \mathbf{b} = \mathbf{0}.$$

Thus, every solution $\mathbf{x}$ to the non-homogeneous system can be expressed in the form:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h,$$

where $\mathbf{x}_h$ is a solution to the associated homogeneous system $A\mathbf{x}_h = \mathbf{0}$. By the Rank-Nullity Theorem, the dimension of the solution space (null space) of $A\mathbf{x}_h = \mathbf{0}$ is:

$$\dim(\operatorname{null}(A)) = n - r.$$

We now analyze the two cases based on the value of r:

1. **Unique Solution ($r = n$):**
When $r = n$, the dimension of the null space is $n - n = 0$. Consequently, the null space contains only the zero vector, forcing $\mathbf{x}_h = \mathbf{0}$ as the unique homogeneous solution. Therefore, the only solution to $A\mathbf{x} = \mathbf{b}$ is:
$$\mathbf{x} = \mathbf{x}_p + \mathbf{0} = \mathbf{x}_p,$$
which is unique.

2. **Infinitely Many Solutions ($r < n$):**
When $r < n$, the dimension of the null space is $k = n - r > 0$. Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ be a basis for this k-dimensional null space. Every homogeneous solution $\mathbf{x}_h$ can be written uniquely as a linear combination of these basis vectors:
$$\mathbf{x}_h = t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_k\mathbf{x}_k \quad \text{for scalars } t_1, t_2, \dots, t_k \in \mathbb{K}.$$
Substituting $\mathbf{x}_h$ back gives the general solution:
$$\mathbf{x} = \mathbf{x}_p + t_1\mathbf{x}_1 + t_2\mathbf{x}_2 + \cdots + t_{n-r}\mathbf{x}_{n-r}.$$
Since the parameters $t_1, t_2, \dots, t_{n-r} \in \mathbb{K}$ can be chosen arbitrarily from an infinite field $\mathbb{K}$, there are infinitely many solutions parameterized by exactly $n - r$ free variables.

■

Example 2.3.4 (Consistency and Unique Solution)**.** Determine if the following system is consistent, and if so, find its solution.

$$\begin{aligned} x + y + z &= 3 \\ x + 2y + 2z &= 5 \\ 2x + 3y + 4z &= 9 \end{aligned}$$

Solution. We form the augmented matrix $[A \mid \mathbf{b}]$ and reduce it to row echelon form.

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 2 & 2 & 5 \\ 2 & 3 & 4 & 9 \end{array}\right) \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{array}\right) \xrightarrow{R_3 \to R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

From the echelon form, we see that the rank of the coefficient matrix A is 3, and the rank of the augmented matrix $[A \mid \mathbf{b}]$ is also 3. Since $\text{rank}(A) = \text{rank}([A \mid \mathbf{b}])$, the system is consistent. Furthermore, since the rank (3) is equal to the number of variables (3), the solution is unique. By back substitution, the last row gives $z = 1$. The second row gives $y + z = 2 \implies y + 1 = 2 \implies y = 1$. The first row gives $x + y + z = 3 \implies x + 1 + 1 = 3 \implies x = 1$. The unique solution is $\mathbf{x} = (1, 1, 1)^T$. ■

Example 2.3.5 (Infinite Solutions)**.** Find the complete solution set for the system: $x - y + z = 2$, $x + y + 2z = 0$, $3x - y + 4z = 4$.

Solution. We reduce the augmented matrix for the system.

$$\left(\begin{array}{ccc|c} 1 & -1 & 1 & 2 \\ 1 & 1 & 2 & 0 \\ 3 & -1 & 4 & 4 \end{array}\right) \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - R_1} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 2 \\ 0 & 2 & 1 & -2 \\ 0 & 2 & 1 & -2 \end{array}\right) \xrightarrow{R_3 \to R_3 - R_2} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 2 \\ 0 & 2 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The rank of both the coefficient matrix and the augmented matrix is 2. Since this is less than the number of variables ($r = 2 < n = 3$), the system is consistent and has infinitely many solutions. The variable z is a free variable; let $z = t$. The second row gives $2y + z = -2 \implies 2y = -2 - t \implies y = -1 - \frac{1}{2}t$. The first row gives $x - y + z = 2 \implies x = 2 + y - z = 2 + (-1 - \frac{1}{2}t) - t = 1 - \frac{3}{2}t$. The solution set is $\mathbf{x} = (1 - \frac{3}{2}t, -1 - \frac{1}{2}t, t)^T$ for any $t \in \mathbb{R}$. ■

Example 2.3.6 (Inconsistent System)**.** Show that the system $x + 2y - z = 1$, $2x + 4y - 2z = 3$ has no solution.

Solution. We form the augmented matrix and row-reduce.

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 4 & -2 & 3 \end{array}\right) \xrightarrow{R_2 \to R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

From the echelon form, the rank of the coefficient matrix A is 1, while the rank of the augmented matrix $[A \mid \mathbf{b}]$ is 2. Since rank$(A) \neq$ rank$([A \mid \mathbf{b}])$, the system is inconsistent. The last row corresponds to the contradictory equation $0x + 0y + 0z = 1$. Therefore, no solution exists. ■

2.3.2 The Homogeneous System: $A\mathbf{x} = \mathbf{0}$

Definition 2.3.7 (Homogeneous System)**.** A system of linear equations $A\mathbf{x} = \mathbf{b}$ is called **homogeneous** if the vector of constants is the zero vector, i.e., $A\mathbf{x} = \mathbf{0}$.

A homogeneous system is *always* consistent because $\mathbf{x} = \mathbf{0}$ (the **trivial solution**) is always a solution. The central question is whether non-zero solutions exist, which are called **non-trivial solutions**.

Theorem 2.3.8 (Existence of Non-Trivial Solutions)**.** *The homogeneous system $A\mathbf{x} = \mathbf{0}$, where A is an $m \times n$ matrix, has a non-trivial solution if and only if the rank of A is strictly less than the number of variables n:*

$$\operatorname{rank}(A) < n.$$

Proof. Let $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ denote the column vectors of A. Matrix-vector multiplication allows the system $A\mathbf{x} = \mathbf{0}$ to be written as a linear combination of columns:

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{0}.$$

A non-trivial solution $\mathbf{x} \neq \mathbf{0}$ exists if and only if there exist scalars $x_1, \ldots, x_n$, not all zero, satisfying this equation, which is the exact condition for the columns of A to be linearly dependent.

Since the rank of A is the maximum number of linearly independent columns, n column vectors are linearly dependent if and only if $\operatorname{rank}(A) < n$. ■

The set of all solutions to a homogeneous system forms a subspace of $\mathbb{K}^n$, known as the **null space** (or solution space) of the matrix A.

Theorem 2.3.9 (Dimension of the Solution Space)**.** *For an $m \times n$ matrix A with $\operatorname{rank}(A) = r$, the number of linearly independent solutions to the homogeneous system $A\mathbf{x} = \mathbf{0}$ (the dimension of its null space) is $n - r$.*

Proof. Row-reducing A to its Reduced Row Echelon Form (RREF) preserves the solution set. The RREF matrix contains r pivot columns and $n - r$ non-pivot (free) columns.

Assigning a value of 1 to one free variable at a time while setting all other free variables to 0 produces $n - r$ fundamental solution vectors. These $n - r$ vectors are linearly independent by construction (owing to the positions of the free variables) and span the entire solution space. Thus, the dimension of the solution space is $n - r$. ■

Corollary 2.3.10 (Maximum Number of Linearly Independent Solutions for $A\mathbf{x} = \mathbf{b}$)**.** *Let $A\mathbf{x} = \mathbf{b}$ be a consistent non-homogeneous system ($b \neq \mathbf{0}$) with $A \in M_{m\times n}(\mathbb{K})$ and $\operatorname{rank}(A) = r$. The maximum number of linearly independent solution vectors to $A\mathbf{x} = \mathbf{b}$ is $n - r + 1$.*

Proof. Fix a particular solution $\mathbf{x}_p \neq \mathbf{0}$ to $A\mathbf{x}_p = \mathbf{b}$. By the previous theorem, the associated homogeneous system $A\mathbf{x} = \mathbf{0}$ has a basis of $n - r$ linearly independent solutions $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n-r}\}$.

Construct the following set of $n - r + 1$ solution vectors to $A\mathbf{x} = \mathbf{b}$:

$$S = \{\mathbf{x}_p,\ \mathbf{x}_p + \mathbf{x}_1,\ \mathbf{x}_p + \mathbf{x}_2,\ \dots,\ \mathbf{x}_p + \mathbf{x}_{n-r}\}.$$

Each vector in S is a valid solution because $A(\mathbf{x}_p + \mathbf{x}_i) = A\mathbf{x}_p + A\mathbf{x}_i = \mathbf{b} + \mathbf{0} = \mathbf{b}$.

To prove that S is linearly independent, consider the combination:

$$c_0\mathbf{x}_p + c_1(\mathbf{x}_p + \mathbf{x}_1) + \cdots + c_{n-r}(\mathbf{x}_p + \mathbf{x}_{n-r}) = \mathbf{0}.$$

Regrouping terms yields:

$$(c_0 + c_1 + \cdots + c_{n-r})\mathbf{x}_p + c_1\mathbf{x}_1 + \cdots + c_{n-r}\mathbf{x}_{n-r} = \mathbf{0}.$$

Applying A to both sides gives $(c_0 + c_1 + \cdots + c_{n-r})\mathbf{b} = \mathbf{0}$. Since $\mathbf{b} \neq \mathbf{0}$, we must have $c_0 + c_1 + \cdots + c_{n-r} = 0$.

The equation then reduces to $c_1\mathbf{x}_1 + \cdots + c_{n-r}\mathbf{x}_{n-r} = \mathbf{0}$. Since $\{\mathbf{x}_1, \dots, \mathbf{x}_{n-r}\}$ is linearly independent, $c_1 = c_2 = \cdots = c_{n-r} = 0$, which in turn forces $c_0 = 0$.

Thus, S consists of $n - r + 1$ linearly independent solution vectors. ∎

Example 2.3.11 (Homogeneous System with Non-Trivial Solutions)**.** Find the solution space for the homogeneous system $A\mathbf{x} = \mathbf{0}$, where $A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 2 \\ 3 & -1 & 4 \end{pmatrix}$.

Solution. We reduce the coefficient matrix A to its echelon form.

$$\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 2 \\ 3 & -1 & 4 \end{pmatrix} \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - R_1} \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 1 \end{pmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The rank of A is 2. Since the rank is less than the number of variables ($r = 2 < n = 3$), there are non-trivial solutions. The number of linearly independent solutions is $n - r = 3 - 2 = 1$. Let the free variable be $z = t$. The second row gives $2y + z = 0 \implies y = -\frac{1}{2}t$. The first row gives $x - y + z = 0 \implies x = y - z = -\frac{1}{2}t - t = -\frac{3}{2}t$. The solution space is the set of all vectors of the form $\mathbf{x} = (-\frac{3}{2}t, -\frac{1}{2}t, t)^T$, which is spanned by the basis vector $(-3, -1, 2)^T$. ∎

Example 2.3.12 (Homogeneous System with Only the Trivial Solution)**.** Solve the system $x + y = 0$, $y + z = 0$, $x + z = 0$.

Solution. The coefficient matrix is $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$. We find its rank.

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} \xrightarrow{R_3 \to R_3 + R_2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

The rank of the matrix is 3. Since the rank is equal to the number of variables ($r = n = 3$), the homogeneous system has only the trivial solution. Thus, the only solution is $\mathbf{x} = (0, 0, 0)^T$. ∎

Exercise 2.3.13. 1. Write down the 3×3 elementary matrices corresponding to the following elementary row operations:

(a) Interchanging the first and third rows ($R_1 \leftrightarrow R_3$).

(b) Multiplying the second row by -4 ($R_2 \to -4R_2$).

(c) Adding 3 times the second row to the first row ($R_1 \to R_1 + 3R_2$).

2. Prove that every elementary matrix is non-singular and that its inverse is also an elementary matrix of the same type. Write down the inverses of the elementary matrices obtained in Problem 1.

3. Reduce the following matrix A to Row Echelon Form (REF) using elementary row operations and determine its rank:

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -1 & 7 \\ -1 & -2 & 3 & -1 \end{bmatrix}.$$

4. Reduce the following matrix A to its Reduced Row Echelon Form (RREF):

$$A = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 0 & 4 & 8 \end{bmatrix}.$$

5. Find non-singular matrices P and Q such that PAQ is in the normal canonical form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ for:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 1 & 3 & 8 \end{bmatrix}.$$

State the rank r of matrix A.

6. Find the normal form PAQ for the matrix:

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}.$$

7. Using elementary row operations on the augmented matrix $[A \mid I_3]$, compute A^{-1} for:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}.$$

8. Prove that a square matrix $A \in M_n(\mathbb{K})$ is non-singular if and only if A is row-equivalent to the identity matrix I_n.

9. Express the non-singular matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$ as a product of elementary matrices.

10. Express $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ as a product of elementary matrices.

11. Find the complete solution set and a basis for the solution space of the homogeneous system $AX = 0$, where:
$$x_1 + 2x_2 - 3x_3 + x_4 = 0,$$
$$2x_1 + 5x_2 - 8x_3 + 6x_4 = 0,$$
$$3x_1 + 4x_2 - 5x_3 - 4x_4 = 0.$$

12. Determine the values of k for which the following homogeneous system has non-trivial solutions, and find the corresponding solutions:
$$(1-k)x_1 + 2x_2 + 3x_3 = 0,$$
$$2x_1 + (4-k)x_2 + 6x_3 = 0,$$
$$3x_1 + 6x_2 + (9-k)x_3 = 0.$$

13. Show that a homogeneous system $AX = 0$ with m equations and n variables has a non-trivial solution if $n > m$. What is the dimension of the solution space in terms of n and $r = \operatorname{rank}(A)$?

14. Test the consistency of the following non-homogeneous system using rank criteria ($\operatorname{rank}(A) = \operatorname{rank}([A \mid B])$), and solve if consistent:
$$x_1 + x_2 + x_3 = 6,$$
$$x_1 + 2x_2 + 3x_3 = 10,$$
$$x_1 + 2x_2 + 4x_3 = 11.$$

15. Investigate for what values of λ and μ the system:
$$x + y + z = 6,$$
$$x + 2y + 3z = 10,$$
$$x + 2y + \lambda z = \mu,$$
has: (i) no solution, (ii) a unique solution, (iii) infinitely many solutions.

16. Solve the following system completely, or show that it is inconsistent:
$$x_1 - 2x_2 + x_3 + x_4 = 2,$$
$$3x_1 + 0x_2 + 2x_3 + 2x_4 = 7,$$
$$2x_1 - 4x_2 + 2x_3 + 2x_4 = 4,$$
$$4x_1 - 2x_2 + 3x_3 + 3x_4 = 9.$$

17. Let X_p be a particular solution of $AX = B$, and let $V_0 = \{X \in \mathbb{K}^n \mid AX = 0\}$ be the solution space of the corresponding homogeneous system. Prove that the complete solution set of $AX = B$ is given by $S = \{X_p + X_h \mid X_h \in V_0\}$.

18. Solve the following system of linear equations using Gauss-Jordan elimination:
$$2x_1 + x_2 + 5x_3 = 4,$$
$$x_1 + 3x_2 + 2x_3 = 7,$$
$$3x_1 + 4x_2 + 7x_3 = 11.$$

19. Given $A \in M_n(\mathbb{K})$, suppose there exist non-singular matrices P and Q such that $PAQ = I_n$. Show that A is invertible and express A^{-1} explicitly in terms of P and Q.

20. Determine whether $B = \begin{bmatrix} 1 \\ 5 \\ 3 \end{bmatrix}$ lies in the column space of $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 0 \end{bmatrix}$ by testing the consistency of $AX = B$.

Chapter 3

LINEAR COMBINATIONS, RANK AND TRACE

This unit explores vector concepts in $\mathbb{R}^n$ and $\mathbb{C}^n$, focusing on linear combinations, linear span, linear independence, and the structural concept of a basis. It establishes matrix rank via row-reduction, interpreting rank as the maximum number of linearly independent rows or columns in a matrix. The unit examines the fundamental relationship between matrix rank r, variable count n, and the dimension $n - r$ of the solution space for homogeneous systems $AX = 0$. Finally, it details the trace operator, covering its algebraic properties and cyclic invariance, including the identity $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.

3.1 Linear Combinations and Span

Let $\mathbb{K}$ denote either the real field $\mathbb{R}$ or the complex field $\mathbb{C}$. The space $\mathbb{K}^n$ consists of all n-tuples of scalars:

$$\mathbb{K}^n = \left\{ v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \;\middle|\; v_1, v_2, \dots, v_n \in \mathbb{K} \right\}.$$

Under standard vector addition and scalar multiplication, $\mathbb{K}^n$ forms the prototypical vector space of dimension n.

3.1.1 Linear Combinations

Definition 3.1.1 (Linear Combination). Let $v_1, v_2, \dots, v_k \in \mathbb{K}^n$ be a finite set of vectors. A vector $v \in \mathbb{K}^n$ is said to be a **linear combination** of $v_1, v_2, \dots, v_k$ if there exist scalars $c_1, c_2, \dots, c_k \in \mathbb{K}$ such that:

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = \sum_{i=1}^{k} c_i v_i.$$

The scalars $c_1, c_2, \dots, c_k$ are called the **coefficients** of the linear combination.

Remark 3.1.2 (Matrix Formulation of Linear Combinations). Determining whether v is a linear combination of $\{v_1, v_2, \dots, v_k\}$ is equivalent to solving a linear system. Let $A = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix}$ be the $n \times k$ matrix whose columns are the vectors v_i, and let $C = \begin{bmatrix} c_1 & c_2 & \cdots & c_k \end{bmatrix}^T$. Then:

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = v \iff AC = v.$$

Thus, v is a linear combination of $v_1, \dots, v_k$ if and only if the system $AC = v$ is consistent.

Example 3.1.3 (Expressing a Vector as a Linear Combination)**.** Consider vectors in $\mathbb{R}^3$:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 & 1 & 3 \end{bmatrix}^T, \quad \text{and} \quad v = \begin{bmatrix} 2 & \\ 5 & 8 \end{bmatrix}^T.$$

To express v as a linear combination $c_1v_1 + c_2v_2 = v$, we set up the matrix system:

$$c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} \implies \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}.$$

Row reducing the augmented matrix $[A \mid v]$:

$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 5 \\ 1 & 3 & 8 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1,\, R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 3 & 6 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{bmatrix}.$$

The third row gives $0c_1 + 0c_2 = 3$, which is impossible. Thus, v cannot be written as a linear combination of v_1 and v_2.

3.1.2 Linear Span

Definition 3.1.4 (Linear Span)**.** Let $S = \{v_1, v_2, \ldots, v_k\} \subseteq \mathbb{K}^n$ be a non-empty subset of vectors. The **linear span** (or simply **span**) of S, denoted by $\operatorname{span}(S)$ or $\operatorname{span}\{v_1, \ldots, v_k\}$, is the set of all possible linear combinations of vectors in S:

$$\operatorname{span}(S) = \left\{ \sum_{i=1}^{k} c_i v_i \;\middle|\; c_1, c_2, \ldots, c_k \in \mathbb{K} \right\}.$$

By convention, the span of the empty set is defined as $\operatorname{span}(\emptyset) = \{0\}$.

Definition 3.1.5 (Spanning Set)**.** A set of vectors $S = \{v_1, v_2, \ldots, v_k\}$ is said to **span** $\mathbb{K}^n$ if $\operatorname{span}(S) = \mathbb{K}^n$. That is, every vector $b \in \mathbb{K}^n$ can be expressed as a linear combination of vectors in S.

Example 3.1.6 ($\mathbb{R}^n$ as the Span of the Standard Basis)**.** Show that $\mathbb{R}^n$ is the linear span of the standard basis vectors $\mathcal{E} = \{e_1, e_2, \ldots, e_n\}$.

Solution. Let $x = \begin{bmatrix} x_1 & x_2 & \ldots & x_n \end{bmatrix}^T \in \mathbb{R}^n$ be an arbitrary vector, where $x_1, x_2, \ldots, x_n \in \mathbb{R}$.

Using the standard basis vectors $e_1 = \begin{bmatrix} 1 & 0 & \ldots & 0 \end{bmatrix}^T, e_2 = \begin{bmatrix} 0 & 1 & \ldots & 0 \end{bmatrix}^T, \ldots, e_n = \begin{bmatrix} 0 & 0 & \ldots & 1 \end{bmatrix}^T$, we write:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \cdots + x_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} = x_1e_1 + x_2e_2 + \cdots + x_ne_n.$$

Since $x_1, x_2, \ldots, x_n$ serve as the scalar coefficients in $\mathbb{R}$, every vector $x \in \mathbb{R}^n$ is a linear combination of $\{e_1, e_2, \ldots, e_n\}$.

Therefore, $\mathbb{R}^n = \operatorname{span}\{e_1, e_2, \ldots, e_n\}$. ■

Example 3.1.7 ($\mathbb{R}^3$ as the Span of a Non-Standard Set of Vectors)**.** Show that $\mathbb{R}^3 = \mathrm{span}(S)$, where $S = \{v_1, v_2, v_3\} \subset \mathbb{R}^3$ is given by:

$$v_1 = \begin{bmatrix}1\\1\\0\end{bmatrix}, \quad v_2 = \begin{bmatrix}0\\1\\1\end{bmatrix}, \quad v_3 = \begin{bmatrix}1\\0\\1\end{bmatrix}.$$

Solution. Let $b = \begin{bmatrix}x & y & z\end{bmatrix}^T \in \mathbb{R}^3$ be an arbitrary vector. We must show that there always exist real scalars $c_1, c_2, c_3 \in \mathbb{R}$ such that:

$$c_1v_1 + c_2v_2 + c_3v_3 = b.$$

Writing this vector equation as a linear system $AC = b$:

$$\begin{bmatrix}1 & 0 & 1\\1 & 1 & 0\\0 & 1 & 1\end{bmatrix}\begin{bmatrix}c_1\\c_2\\c_3\end{bmatrix} = \begin{bmatrix}x\\y\\z\end{bmatrix}.$$

To verify that a solution $C = \begin{bmatrix}c_1 & c_2 & c_3\end{bmatrix}^T$ exists for any choice of x, y, z, we compute the determinant of the coefficient matrix A:

$$\det(A) = 1(1(1) - 0(1)) - 0 + 1(1(1) - 1(0)) = 1 + 1 = 2 \neq 0.$$

Since $\det(A) \neq 0$, A is non-singular and invertible. Thus, for any vector $b \in \mathbb{R}^3$, a unique coefficient vector $C = A^{-1}b$ exists.

Solving the system explicitly via row reduction on the augmented matrix $[A \mid b]$:

$$\begin{bmatrix}1 & 0 & 1 & x\\1 & 1 & 0 & y\\0 & 1 & 1 & z\end{bmatrix} \xrightarrow{R_2 \to R_2 - R_1} \begin{bmatrix}1 & 0 & 1 & x\\0 & 1 & -1 & y-x\\0 & 1 & 1 & z\end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix}1 & 0 & 1 & x\\0 & 1 & -1 & y-x\\0 & 0 & 2 & z-y+x\end{bmatrix}.$$

Back-substitution yields explicit formulas for the scalars in terms of x, y, z:

$$c_3 = \frac{x-y+z}{2}, \quad c_2 = \frac{-x+y+z}{2}, \quad c_1 = \frac{x+y-z}{2}.$$

Because every vector $\begin{bmatrix}x & y & z\end{bmatrix}^T \in \mathbb{R}^3$ can be expressed as a linear combination of v_1, v_2, v_3 using these scalars, it follows that $\mathbb{R}^3 = \mathrm{span}\{v_1, v_2, v_3\}$. ■

Example 3.1.8 ($\mathbb{C}^2$ as a Linear Span over $\mathbb{C}$)**.** Show that $\mathbb{C}^2 = \mathrm{span}(S)$ over the complex field $\mathbb{C}$, where $S = \{v_1, v_2\} \subset \mathbb{C}^2$ is given by:

$$v_1 = \begin{bmatrix}1+i\\0\end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix}i\\1\end{bmatrix}.$$

Solution. Let $w = \begin{bmatrix}z_1\\z_2\end{bmatrix} \in \mathbb{C}^2$ be an arbitrary complex vector, where $z_1, z_2 \in \mathbb{C}$. We seek complex scalars $c_1, c_2 \in \mathbb{C}$ such that:

$$c_1\begin{bmatrix}1+i\\0\end{bmatrix} + c_2\begin{bmatrix}i\\1\end{bmatrix} = \begin{bmatrix}z_1\\z_2\end{bmatrix}.$$

This yields the matrix equation:

$$\begin{bmatrix}1+i & i\\0 & 1\end{bmatrix}\begin{bmatrix}c_1\\c_2\end{bmatrix} = \begin{bmatrix}z_1\\z_2\end{bmatrix}.$$

The system expands to:

$$(1+i)c_1 + ic_2 = z_1,$$
$$c_2 = z_2.$$

Substituting $c_2 = z_2$ into the first equation:

$$(1+i)c_1 + iz_2 = z_1 \implies (1+i)c_1 = z_1 - iz_2.$$

Dividing by $1+i$:

$$c_1 = \frac{z_1 - iz_2}{1+i} = \frac{(z_1 - iz_2)(1-i)}{(1+i)(1-i)} = \frac{(1-i)z_1 - (1+i)z_2}{2}.$$

Since c_1 and c_2 are well-defined complex scalars for any $z_1, z_2 \in \mathbb{C}$, every vector $w \in \mathbb{C}^2$ belongs to $\operatorname{span}(S)$.

Hence, $\mathbb{C}^2 = \operatorname{span}\{v_1, v_2\}$. ■

3.1.3 Linear Dependence and Independence

Definition 3.1.9 (Linear Dependence and Independence)**.** Let $S = \{v_1, v_2, \ldots, v_k\} \subset \mathbb{K}^n$.

(i) S is **linearly dependent** if there exist scalars $c_1, c_2, \ldots, c_k \in \mathbb{K}$, **not all zero**, such that:

$$c_1v_1 + c_2v_2 + \cdots + c_kv_k = 0.$$

An equation of this form with at least one non-zero coefficient $c_i \neq 0$ is called a **non-trivial linear relation**.

(ii) S is **linearly independent** if the only scalar solution to the vector equation:

$$c_1v_1 + c_2v_2 + \cdots + c_kv_k = 0$$

is the **trivial solution** $c_1 = c_2 = \cdots = c_k = 0$.

Before examining large sets of vectors using matrix reduction techniques, it is essential to understand the structural properties of small-order sets (containing $0, 1$, or 2 vectors) and how to apply the formal definition of linear independence directly.

Theorem 3.1.10 (Single-Element Sets)**.** *Let $S = \{v\}$ be a set containing a single vector $v \in \mathbb{K}^n$.*

*(i) If $v = 0$, then $S = \{0\}$ is **linearly dependent**.*

*(ii) If $v \neq 0$, then $S = \{v\}$ is **linearly independent**.*

Proof. (i) Consider the vector equation $c \cdot 0 = 0$. Any non-zero scalar $c \in \mathbb{K} \setminus \{0\}$ satisfies this equation. Since a non-trivial scalar solution exists, $\{0\}$ is linearly dependent.

(ii) Consider $cv = 0$ with $v \neq 0$. In a field $\mathbb{K}$, the product of a scalar c and a non-zero vector v equals zero if and only if $c = 0$. Thus, $c = 0$ is the unique solution, proving $\{v\}$ is linearly independent.

■

Theorem 3.1.11 (Two-Element Sets)**.** *A set $S = \{v_1, v_2\} \subset \mathbb{K}^n$ containing two vectors is **linearly dependent** if and only if one vector is a scalar multiple of the other (i.e., $v_1 = cv_2$ or $v_2 = cv_1$ for some scalar $c \in \mathbb{K}$).*

Proof. Suppose $\{v_1, v_2\}$ is linearly dependent. By definition, there exist scalars $c_1, c_2 \in \mathbb{K}$, not both zero, such that $c_1v_1 + c_2v_2 = 0$. If $c_1 \neq 0$, then $v_1 = \left(-\frac{c_2}{c_1}\right) v_2$, so v_1 is a scalar multiple of v_2. Conversely, if $v_1 = cv_2$ for some $c \in \mathbb{K}$, then $1 \cdot v_1 - c \cdot v_2 = 0$. Since the coefficient of v_1 is $1 \neq 0$, this is a non-trivial linear relation, proving $\{v_1, v_2\}$ is linearly dependent. ■

Corollary 3.1.12 (Geometric Interpretation in $\mathbb{R}^2$ and $\mathbb{R}^3$)**.** *Two non-zero vectors $v_1, v_2 \in \mathbb{R}^n$ are linearly dependent if and only if they lie on the same line passing through the origin (collinear). They are linearly independent if and only if they point in non-parallel directions.*

Theorem 3.1.13 (Sets Containing the Zero Vector)**.** *Any set $S = \{v_1, v_2, \ldots, v_k\} \subset \mathbb{K}^n$ containing the zero vector ($0 \in S$) is **linearly dependent**.*

Proof. Without loss of generality, assume $v_1 = 0$. Consider the linear combination:

$$1 \cdot v_1 + 0 \cdot v_2 + 0 \cdot v_3 + \cdots + 0 \cdot v_k = 1 \cdot 0 + 0 = 0.$$

Because the coefficient of v_1 is $1 \neq 0$, this provides a non-trivial linear combination that equals zero. Hence, S is linearly dependent. ■

Theorem 3.1.14 (Subset and Superset Inheritance Properties)**.** *Let S be a finite set of vectors in $\mathbb{K}^n$.*

(i) Any non-empty subset of a linearly independent set is linearly independent.

(ii) Any superset of a linearly dependent set is linearly dependent.

To test whether a set $\{v_1, v_2, \ldots, v_k\}$ is linearly independent using the formal definition, set up $c_1v_1 + c_2v_2 + \cdots + c_kv_k = 0$ and solve for the scalars $c_1, c_2, \ldots, c_k$. If $c_1 = c_2 = \cdots = c_k = 0$ is the unique solution, the set is **linearly independent**. If a non-zero scalar solution exists, the set is **linearly dependent**.

Example 3.1.15 (Linearly Independent Set in $\mathbb{R}^3$)**.** Using the definition, determine whether the set $S = \{v_1, v_2, v_3\} \subset \mathbb{R}^3$ is linearly independent, where:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Solution. We set up the vector equation $c_1v_1 + c_2v_2 + c_3v_3 = 0$:

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Equating corresponding components yields a system of three homogeneous linear equations:

$$c_1 + c_2 = 0, \tag{1}$$
$$c_2 + c_3 = 0, \tag{2}$$
$$c_1 + c_3 = 0. \tag{3}$$

From equation (1), $c_1 = -c_2$. From equation (2), $c_3 = -c_2$. Substituting both into equation (3):

$$(-c_2) + (-c_2) = 0 \implies -2c_2 = 0 \implies c_2 = 0.$$

Substituting $c_2 = 0$ back gives $c_1 = 0$ and $c_3 = 0$.

Since $c_1 = c_2 = c_3 = 0$ is the unique solution to $c_1v_1 + c_2v_2 + c_3v_3 = 0$, the set $\{v_1, v_2, v_3\}$ is linearly independent. ■

Example 3.1.16 (Linearly Dependent Set in $\mathbb{R}^3$)**.** Using the definition, test whether the set $S = \{v_1, v_2, v_3\} \subset \mathbb{R}^3$ is linearly independent, where:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Solution. Set up $c_1v_1 + c_2v_2 + c_3v_3 = 0$:

$$c_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Notice that $v_2 = 2v_1$. We can construct a non-trivial linear combination by choosing $c_1 = -2$, $c_2 = 1$, and $c_3 = 0$:

$$(-2) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + (1) \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} + (0) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2+2+0 \\ -4+4+0 \\ -6+6+0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since we found scalars $c_1 = -2 \neq 0$ and $c_2 = 1 \neq 0$ that satisfy the vector equation, a non-trivial linear relation exists.

Therefore, the set $\{v_1, v_2, v_3\}$ is linearly dependent. ■

Example 3.1.17 (Linear Independence over the Complex Field $\mathbb{C}^2$)**.** Using the definition, test whether $S = \{v_1, v_2\} \subset \mathbb{C}^2$ is linearly independent over $\mathbb{C}$, where:

$$v_1 = \begin{bmatrix} 1 \\ i \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} i \\ 1 \end{bmatrix}.$$

Solution. Set $c_1v_1 + c_2v_2 = 0$ for complex scalars $c_1, c_2 \in \mathbb{C}$:

$$c_1 \begin{bmatrix} 1 \\ i \end{bmatrix} + c_2 \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This gives the system of complex linear equations:

$$c_1 + ic_2 = 0, \tag{1}$$
$$ic_1 + c_2 = 0. \tag{2}$$

From equation (1), $c_1 = -ic_2$. Substituting c_1 into equation (2):

$$i(-ic_2) + c_2 = 0 \implies (-i^2)c_2 + c_2 = 0.$$

Since $i^2 = -1$, $-i^2 = 1$:

$$c_2 + c_2 = 0 \implies 2c_2 = 0 \implies c_2 = 0.$$

Substituting $c_2 = 0$ into $c_1 = -ic_2$ yields $c_1 = 0$.

Since $c_1 = c_2 = 0$ is the unique solution in $\mathbb{C}$, the set $\{v_1, v_2\}$ is linearly independent over $\mathbb{C}$. ■

Example 3.1.18 (Testing a Two-Vector Set in $\mathbb{R}^2$)**.** Using the definition, determine if the set $S = \{v_1, v_2\} \subset \mathbb{R}^2$ is linearly independent, where:

$$v_1 = \begin{bmatrix} 3 \\ -2 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} -6 \\ 4 \end{bmatrix}.$$

Solution. We set up the vector equation $c_1v_1 + c_2v_2 = 0$:

$$c_1 \begin{bmatrix} 3 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} -6 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This corresponds to the system:

$$\begin{aligned} 3c_1 - 6c_2 &= 0, \\ -2c_1 + 4c_2 &= 0. \end{aligned}$$

Both equations simplify to $c_1 = 2c_2$. Choosing $c_2 = 1$ gives $c_1 = 2$.

Since $2v_1 + 1v_2 = 0$ with $c_1 = 2 \neq 0$ and $c_2 = 1 \neq 0$, a non-trivial linear combination equals zero. Thus, the set $\{v_1, v_2\}$ is linearly dependent. ∎

Theorem 3.1.19 (Matrix Test for Linear Independence). *Let $v_1, v_2, \ldots, v_k \in \mathbb{K}^n$, and let $A = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix}_{n \times k}$. The set $\{v_1, v_2, \ldots, v_k\}$ is linearly independent if and only if the homogeneous system $AC = 0$ has only the trivial solution $C = 0$, which occurs if and only if $\operatorname{rank}(A) = k$ (the number of vectors).*

Proof. Let $C = \begin{bmatrix} c_1 & c_2 & \cdots & c_k \end{bmatrix}^T \in \mathbb{K}^k$ be a column vector of scalars. By the definition of matrix-vector multiplication in terms of columns, the product AC expands directly as a linear combination of the column vectors of A:

$$AC = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{bmatrix} = c_1v_1 + c_2v_2 + \cdots + c_kv_k.$$

Equivalence of Linear Independence and Trivial Solution

By definition, the set of vectors $\{v_1, v_2, \ldots, v_k\}$ is linearly independent if and only if the vector equation:

$$c_1v_1 + c_2v_2 + \cdots + c_kv_k = 0$$

has $c_1 = c_2 = \cdots = c_k = 0$ as its unique solution.

Since $AC = c_1v_1 + c_2v_2 + \cdots + c_kv_k$, the vector equation $c_1v_1 + \cdots + c_kv_k = 0$ is identical to the homogeneous matrix equation $AC = 0$. Thus, $\{v_1, v_2, \ldots, v_k\}$ is linearly independent if and only if $AC = 0$ has only the trivial solution $C = 0$.

Equivalence of Trivial Solution and Full Column Rank

The set of all solutions to $AC = 0$ forms the null space $\operatorname{null}(A) = \{C \in \mathbb{K}^k \mid AC = 0\}$. By the Rank-Nullity Theorem applied to the $n \times k$ matrix A:

$$\operatorname{rank}(A) + \operatorname{nullity}(A) = k,$$

where k is the number of columns (vectors), $\operatorname{rank}(A)$ is the column rank, and $\operatorname{nullity}(A) = \dim(\operatorname{null}(A))$.

The homogeneous system $AC = 0$ has only the trivial solution $C = 0$ if and only if its null space consists solely of the zero vector, which means $\operatorname{nullity}(A) = 0$. Substituting $\operatorname{nullity}(A) = 0$ into the Rank-Nullity equation gives:

$$\operatorname{rank}(A) + 0 = k \iff \operatorname{rank}(A) = k.$$

Equivalently, when A is reduced to Row Echelon Form, $AC = 0$ has only the trivial solution if and only if there are no free variables, which requires every column of A to contain a pivot position. Since the number of pivot positions equals $\operatorname{rank}(A)$, this occurs if and only if $\operatorname{rank}(A) = k$. ∎

Theorem 3.1.20 (Size Criterion for Linear Dependence). *Any set of k vectors in $\mathbb{K}^n$ with $k > n$ is automatically linearly dependent.*

Proof. Let $A = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix}$ be the $n \times k$ matrix whose columns are the given vectors. The homogeneous system $AC = 0$ consists of n equations in k unknowns. Since $k > n$, the number of unknowns exceeds the number of equations, ensuring the existence of at least $k - \operatorname{rank}(A) \geq k - n > 0$ free variables. This guarantees non-trivial solutions to $AC = 0$, making the set linearly dependent. ■

Example 3.1.21 (Testing Linear Independence in $\mathbb{R}^3$). Determine whether the set $S = \{v_1, v_2, v_3\} \subset \mathbb{R}^3$ is linearly independent, where:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 & 3 & 4 \end{bmatrix}^T.$$

We set up $c_1v_1 + c_2v_2 + c_3v_3 = 0$ and reduce the matrix $A = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 7 & 4 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1,\, R_3 \to R_3 - 3R_1} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Since $\operatorname{rank}(A) = 2 < 3$, there is one free variable (c_3). Choosing $c_3 = -1$ gives $c_2 = 1$ and $c_1 = -1$. Thus, $-v_1 + v_2 - v_3 = 0$ is a non-trivial linear relation, proving S is **linearly dependent**.

3.2 Rank of a Matrix and Solution Space of Linear Systems

3.2.1 Rank via Row Reduction and Linear Independence

The rank of a matrix is a fundamental numerical property that measures its structural complexity and operational capacity as a linear mapping.

Definition 3.2.1 (Row Space, Column Space, and Rank). Let $A \in M_{m\times n}(\mathbb{K})$ be an $m \times n$ matrix over a field $\mathbb{K}$.

(i) The **row space** of A, denoted by $\operatorname{row}(A)$, is the subspace of $\mathbb{K}^n$ spanned by the rows of A. Its dimension is called the **row rank** of A.

(ii) The **column space** of A, denoted by $\operatorname{col}(A)$, is the subspace of $\mathbb{K}^m$ spanned by the columns of A. Its dimension is called the **column rank** of A.

(iii) The **rank** of A, denoted by $\operatorname{rank}(A)$ or r, is the common dimension of its row space and column space:
$$\operatorname{rank}(A) = \dim(\operatorname{row}(A)) = \dim(\operatorname{col}(A)).$$

Thus, the rank r is equal to the maximum number of linearly independent rows, or equivalently, the maximum number of linearly independent columns in A.

Theorem 3.2.2 (Preservation of Rank under Elementary Row Operations). *Elementary row operations do not alter the row space or the rank of a matrix.*

Proof. Let A be an $m \times n$ matrix with rows $R_1, R_2, \ldots, R_m$, so $\operatorname{row}(A) = \operatorname{span}\{R_1, R_2, \ldots, R_m\}$. We analyze the effect of each type of elementary row operation on the row space:

Type I: **Interchange ($R_i \leftrightarrow R_j$):** Interchanging two rows merely reorders the vectors in the spanning set $\{R_1, \dots, R_m\}$. The spanned subspace remains identical.

Type II: **Scaling ($R_i \to cR_i$, $c \neq 0$):** Let $R_i' = cR_i$. Any vector in $\mathrm{span}\{R_1, \dots, cR_i, \dots, R_m\}$ is a linear combination of the new rows and thus belongs to $\mathrm{row}(A)$. Conversely, since $c \neq 0$, $R_i = \frac{1}{c}R_i'$, so any vector in $\mathrm{row}(A)$ is a linear combination of the new rows. Hence, the row space is unchanged.

Type III: **Addition ($R_i \to R_i + cR_j$, $i \neq j$):** Let $R_i' = R_i + cR_j$. Since $R_i' \in \mathrm{row}(A)$, the new row space is contained in $\mathrm{row}(A)$. Conversely, $R_i = R_i' - cR_j$, so the original row R_i is in the new row space. Thus, both row spaces are identical.

Since elementary row operations preserve the row space, they preserve its dimension. Hence, $\mathrm{rank}(A)$ is invariant under row reduction. ■

Corollary 3.2.3 (Computation of Rank via Row Echelon Form)**.** *When a matrix A is reduced to its Row Echelon Form (REF) or Reduced Row Echelon Form (RREF), the non-zero rows form a basis for* $\mathrm{row}(A)$*. Consequently, the rank r of A equals the number of non-zero rows (or equivalently, the number of pivot positions) in its echelon form.*

Example 3.2.4 (Computing Rank via Row Reduction for a 3×4 Matrix)**.** Find the rank of the matrix A by reducing it to Row Echelon Form:

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 0 & 5 \end{bmatrix}.$$

Solution. We apply elementary row operations to introduce zeros below the leading entry of the first row:

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 0 & 5 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -4 \\ 0 & 0 & 3 & -4 \end{bmatrix}$$
$$\xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The matrix is now in Row Echelon Form. Counting the non-zero rows gives $r = 2$.

Therefore, the rank of A is 2, which indicates that A has 2 linearly independent rows and 2 linearly independent columns. ■

Example 3.2.5 (Computing Rank for a 3×3 Matrix)**.** Find the rank of the matrix A by reducing it to Row Echelon Form:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 7 & 11 \end{bmatrix}.$$

Solution. Applying elementary row operations:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 7 & 11 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$
$$\xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

The Row Echelon Form contains 2 non-zero rows. Hence, $\mathrm{rank}(A) = 2$. ■

3.2.2 Relationship Between Rank r, Variables n, and Solution Space of $AX = 0$

Consider a homogeneous system of m linear equations in n variables:

$$AX = 0, \quad \text{where } A \in M_{m\times n}(\mathbb{K}) \text{ and } X = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}^T \in \mathbb{K}^n.$$

The set of all solutions forms a subspace of $\mathbb{K}^n$, called the **solution space** (or **null space**) of A:

$$V_0 = \{X \in \mathbb{K}^n \mid AX = 0\}.$$

Theorem 3.2.6 (Dimension of the Solution Space / Rank-Nullity Theorem). *Let $A \in M_{m\times n}(\mathbb{K})$ be a matrix with n columns (variables), and let $r = \operatorname{rank}(A)$. Then the dimension of the solution space V_0 of the homogeneous system $AX = 0$ is given by:*

$$\dim(V_0) = n - r.$$

Proof. Let R be the Reduced Row Echelon Form (RREF) of A. Since row operations do not alter the solution set, $AX = 0 \iff RX = 0$.

Since $\operatorname{rank}(A) = r$, the RREF matrix R contains exactly r pivot columns and r leading 1s.

(i) The r variables corresponding to the pivot columns are called **pivot variables** (or bound variables). Each pivot variable is uniquely expressed in terms of the remaining variables.

(ii) The remaining $n-r$ variables corresponding to non-pivot columns are called **free variables**.

Assigning independent parameters $t_1, t_2, \dots, t_{n-r} \in \mathbb{K}$ to the $n - r$ free variables allows every solution $X \in V_0$ to be expressed uniquely as a linear combination:

$$X = t_1X_1 + t_2X_2 + \cdots + t_{n-r}X_{n-r},$$

where $X_1, X_2, \dots, X_{n-r}$ are linearly independent fundamental solution vectors obtained by setting one free variable to 1 and all other free variables to 0.

Because $\{X_1, X_2, \dots, X_{n-r}\}$ is linearly independent and spans V_0, it forms a basis for V_0. Consequently:

$$\dim(V_0) = n - r = (\text{Total Number of Variables}) - (\text{Rank of Matrix}).$$

∎

Corollary 3.2.7 (Characterization of Solutions). *For an $m \times n$ homogeneous system $AX = 0$ with rank r:*

(i) If $r = n$, then $\dim(V_0) = n - n = 0$. The system has only the trivial solution $X = 0$.

(ii) If $r < n$, then $\dim(V_0) = n - r > 0$. The system possesses infinitely many non-trivial solutions parameterized by $n - r$ free variables.

Example 3.2.8 (Rank, Variable Count, and Basis for $AX = 0$). Determine the rank r of the coefficient matrix A, state the number of variables n, compute the dimension $n - r$ of the solution space of $AX = 0$, and construct a basis for its solution space, where:

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 3 \\ 3 & 6 & 2 & 7 \end{bmatrix}.$$

Solution. The matrix A has $m = 3$ rows and $n = 4$ columns (variables x_1, x_2, x_3, x_4).

We reduce A to Reduced Row Echelon Form (RREF) using elementary row operations:

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 3 \\ 3 & 6 & 2 & 7 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 5 & -5 \\ 0 & 0 & 5 & -5 \end{bmatrix}$$
$$\xrightarrow{R_2 \to \frac{1}{5}R_2} \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 5 & -5 \end{bmatrix}$$
$$\xrightarrow{R_3 \to R_3 - 5R_2} \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$
$$\xrightarrow{R_1 \to R_1 + R_2} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

From the RREF matrix:

- The rank $r = \operatorname{rank}(A) = 2$ (since there are 2 non-zero rows / pivot positions).
- The total number of variables is $n = 4$.
- The dimension of the solution space is $\dim(V_0) = n - r = 4 - 2 = 2$.

Columns 1 and 3 contain pivots, so x_1 and x_3 are pivot variables. Columns 2 and 4 lack pivots, so x_2 and x_4 are free variables.

Writing out the system equations from the RREF:

$$x_1 + 2x_2 + 3x_4 = 0 \implies x_1 = -2x_2 - 3x_4,$$
$$x_3 - x_4 = 0 \implies x_3 = x_4.$$

Expressing an arbitrary solution $X \in V_0$ in vector form:

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2x_2 - 3x_4 \\ x_2 \\ x_4 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Setting $(x_2 = 1, x_4 = 0)$ and $(x_2 = 0, x_4 = 1)$ yields the fundamental basis vectors:

$$X_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad X_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Thus, $\{X_1, X_2\}$ forms a basis for the solution space V_0, confirming that $\dim(V_0) = 2 = n - r$. ■

3.2.3 Additional Theorems and Properties of Matrix Rank

In addition to the row-reduction characterization, matrix rank possesses several important algebraic properties regarding minors, matrix multiplication, and non-singular transformations.

Definition 3.2.9 (Minor-Based Definition of Rank)**.** The **rank** r of a matrix $A \in M_{m\times n}(\mathbb{K})$ is the order of the largest non-singular (non-zero determinant) square submatrix that can be formed from A. Equivalently, r is the order of the highest-order non-vanishing minor of A.

Theorem 3.2.10 (Sylvester's Rank Inequality)**.** *If $A \in M_{m\times p}(\mathbb{K})$ and $B \in M_{p\times n}(\mathbb{K})$ are conformable matrices, then the rank of their product AB is bounded by the rank of each individual factor matrix:*

$$\operatorname{rank}(AB) \leq \min(\operatorname{rank}(A), \operatorname{rank}(B)).$$

Proof. We prove the inequality by establishing two separate bounds: $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$ and $\operatorname{rank}(AB) \leq \operatorname{rank}(B)$.

Bound by the Rank of the First Factor: $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$

The j-th column of the product matrix AB is given by Ab_j, where b_j is the j-th column of B. Because Ab_j is a linear combination of the columns of A, every column of AB lies in the column space of A:

$$\operatorname{col}(AB) \subseteq \operatorname{col}(A).$$

Since the dimension of a subspace cannot exceed the dimension of the containing space:

$$\dim(\operatorname{col}(AB)) \leq \dim(\operatorname{col}(A)).$$

By the definition of rank as column rank, this yields $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$.

Bound by the Rank of the Second Factor: $\operatorname{rank}(AB) \leq \operatorname{rank}(B)$

Similarly, the i-th row of the product matrix AB is given by a_iB, where a_i is the i-th row of A. Because a_iB is a linear combination of the rows of B, every row of AB lies in the row space of B:

$$\operatorname{row}(AB) \subseteq \operatorname{row}(B).$$

Comparing dimensions gives:

$$\dim(\operatorname{row}(AB)) \leq \dim(\operatorname{row}(B)).$$

By the definition of rank as row rank, this yields $\operatorname{rank}(AB) \leq \operatorname{rank}(B)$.

Combining both bounds, $\operatorname{rank}(AB)$ is less than or equal to both $\operatorname{rank}(A)$ and $\operatorname{rank}(B)$, so:

$$\operatorname{rank}(AB) \leq \min(\operatorname{rank}(A), \operatorname{rank}(B)).$$

∎

Theorem 3.2.11 (Invariance of Rank under Non-Singular Multiplication)**.** *Let $A \in M_{m\times n}(\mathbb{K})$. If $P \in M_m(\mathbb{K})$ and $Q \in M_n(\mathbb{K})$ are non-singular matrices of appropriate dimensions, then:*

$$\operatorname{rank}(PA) = \operatorname{rank}(AQ) = \operatorname{rank}(A).$$

Proof. We prove $\operatorname{rank}(AQ) = \operatorname{rank}(A)$.

First, applying Sylvester's rank inequality to the product AQ:

$$\operatorname{rank}(AQ) \leq \operatorname{rank}(A).$$

To establish the reverse inequality, notice that since Q is non-singular, its inverse Q^{-1} exists. We write $A = (AQ)Q^{-1}$ and apply Sylvester's rank inequality again:

$$\operatorname{rank}(A) = \operatorname{rank}((AQ)Q^{-1}) \leq \operatorname{rank}(AQ).$$

Combining both inequalities $\operatorname{rank}(AQ) \leq \operatorname{rank}(A)$ and $\operatorname{rank}(A) \leq \operatorname{rank}(AQ)$ forces:

$$\operatorname{rank}(AQ) = \operatorname{rank}(A).$$

A completely analogous argument using $A = P^{-1}(PA)$ shows that $\operatorname{rank}(PA) = \operatorname{rank}(A)$. ∎

Example 3.2.12 (Verification of Sylvester's Rank Inequality). Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$. Verify that $\operatorname{rank}(AB) \le \min(\operatorname{rank}(A), \operatorname{rank}(B))$.

Solution. By inspection, matrix A has 1 non-zero row, so $\operatorname{rank}(A) = 1$. Matrix B has 1 non-zero row, so $\operatorname{rank}(B) = 1$. Thus, the minimum of their ranks is:

$$\min(\operatorname{rank}(A), \operatorname{rank}(B)) = \min(1, 1) = 1.$$

Computing the matrix product AB:

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The rank of the zero matrix is $\operatorname{rank}(AB) = 0$.

Comparing ranks confirms the inequality:

$$0 \le 1 \implies \operatorname{rank}(AB) \le \min(\operatorname{rank}(A), \operatorname{rank}(B)).$$

■

3.3 Trace of a Matrix and Its Properties

The trace of a matrix is a fundamental scalar invariant in linear algebra. It plays a vital role in operator theory, differential equations, quantum mechanics, and representation theory.

Definition 3.3.1 (Trace of a Matrix). Let $A = (a_{ij})_{n\times n} \in M_n(\mathbb{K})$ be a square matrix over a field $\mathbb{K}$. The **trace** of A, denoted by $\operatorname{Tr}(A)$ (or $\operatorname{tr}(A)$), is the sum of all elements on its main diagonal:

$$\operatorname{Tr}(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}.$$

The trace is defined *only* for square matrices.

Theorem 3.3.2 (Basic Properties of the Trace Operator). *Let $A, B \in M_n(\mathbb{K})$ be square matrices, and let $c \in \mathbb{K}$ be a scalar. Then:*

(i) ***Additive Linearity:*** $\operatorname{Tr}(A + B) = \operatorname{Tr}(A) + \operatorname{Tr}(B)$.

(ii) ***Scalar Homogeneity:*** $\operatorname{Tr}(cA) = c\operatorname{Tr}(A)$.

(iii) ***Transpose Invariance:*** $\operatorname{Tr}(A^T) = \operatorname{Tr}(A)$.

(iv) ***Conjugate Transpose Invariance:*** $\operatorname{Tr}(A^*) = \overline{\operatorname{Tr}(A)}$ *for* $A \in M_n(\mathbb{C})$.

(v) ***Identity Matrix Trace:*** $\operatorname{Tr}(I_n) = n$.

Proof. Let $A = (a_{ij})$ and $B = (b_{ij})$.

(i) The diagonal entries of $A + B$ are $(A + B)_{ii} = a_{ii} + b_{ii}$. Summing gives:

$$\operatorname{Tr}(A + B) = \sum_{i=1}^{n}(a_{ii} + b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} = \operatorname{Tr}(A) + \operatorname{Tr}(B).$$

(ii) The diagonal entries of cA are $(cA)_{ii} = ca_{ii}$. Summing gives:

$$\operatorname{Tr}(cA) = \sum_{i=1}^{n} ca_{ii} = c\sum_{i=1}^{n} a_{ii} = c\operatorname{Tr}(A).$$

(iii) Taking the transpose interchanges rows and columns, but leaves main diagonal entries a_{ii} unchanged. Hence, $\mathrm{Tr}(A^T) = \mathrm{Tr}(A)$.

(iv) Taking conjugate transpose replaces diagonal entries a_{ii} with $\bar{a}_{ii}$. Hence, $\mathrm{Tr}(A^*) = \sum_{i=1}^{n} \bar{a}_{ii} = \overline{\sum_{i=1}^{n} a_{ii}} = \overline{\mathrm{Tr}(A)}$.

(v) Since $(I_n)_{ii} = 1$ for all $1 \le i \le n$, $\mathrm{Tr}(I_n) = \sum_{i=1}^{n} 1 = n$.

■

3.3.1 The Fundamental Product Identity $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$

A central property of the trace operator is that it allows matrix factors to commute under the trace function, even though matrix multiplication itself is generally non-commutative ($AB \neq BA$).

Theorem 3.3.3 (Commutativity of Trace Under Matrix Multiplication). *Let $A \in M_{m\times n}(\mathbb{K})$ and $B \in M_{n\times m}(\mathbb{K})$ be rectangular matrices such that both product matrices $AB \in M_m(\mathbb{K})$ and $BA \in M_n(\mathbb{K})$ are square. Then:*

$$\mathrm{Tr}(AB) = \mathrm{Tr}(BA).$$

Proof. Let $A = (a_{ik})_{m\times n}$ and $B = (b_{ki})_{n\times m}$.

By definition of matrix multiplication, the (i,i)-th diagonal entry of the $m \times m$ matrix AB is given by:

$$(AB)_{ii} = \sum_{k=1}^{n} a_{ik} b_{ki} \quad \text{for } 1 \le i \le m.$$

Summing over all diagonal entries of AB:

$$\mathrm{Tr}(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m} \sum_{k=1}^{n} a_{ik} b_{ki}.$$

Similarly, the (k,k)-th diagonal entry of the $n \times n$ matrix BA is given by:

$$(BA)_{kk} = \sum_{i=1}^{m} b_{ki} a_{ik} \quad \text{for } 1 \le k \le n.$$

Summing over all diagonal entries of BA:

$$\mathrm{Tr}(BA) = \sum_{k=1}^{n} (BA)_{kk} = \sum_{k=1}^{n} \sum_{i=1}^{m} b_{ki} a_{ik}.$$

Because addition in the scalar field $\mathbb{K}$ is commutative and associative, we swap the finite summations and scalar terms:

$$\mathrm{Tr}(BA) = \sum_{k=1}^{n} \sum_{i=1}^{m} b_{ki} a_{ik} = \sum_{i=1}^{m} \sum_{k=1}^{n} a_{ik} b_{ki} = \mathrm{Tr}(AB).$$

Hence, $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$. ■

Corollary 3.3.4 (Cyclic Permutation Property of Trace). *For any conformable matrices A, B, C:*

$$\mathrm{Tr}(ABC) = \mathrm{Tr}(BCA) = \mathrm{Tr}(CAB).$$

Proof. Treat BC as a single matrix $M = BC$. Applying $\mathrm{Tr}(AM) = \mathrm{Tr}(MA)$ gives:

$$\mathrm{Tr}(A(BC)) = \mathrm{Tr}((BC)A) \implies \mathrm{Tr}(ABC) = \mathrm{Tr}(BCA).$$

Re-grouping AB as a single matrix $N = AB$ gives:

$$\mathrm{Tr}((AB)C) = \mathrm{Tr}(C(AB)) \implies \mathrm{Tr}(ABC) = \mathrm{Tr}(CAB).$$

■

Remark 3.3.5. The cyclic property allows cyclic shifts, but arbitrary permutations are **invalid**. In general:

$$\mathrm{Tr}(ABC) \neq \mathrm{Tr}(BAC).$$

3.3.2 Similarity Invariance of the Trace

Theorem 3.3.6 (Similarity Invariance). *Similar matrices have equal traces. That is, if $A, B \in M_n(\mathbb{K})$ and $B = P^{-1}AP$ for some non-singular matrix $P \in M_n(\mathbb{K})$, then:*

$$\mathrm{Tr}(B) = \mathrm{Tr}(A).$$

Proof. Using the property $\mathrm{Tr}(XY) = \mathrm{Tr}(YX)$ with $X = P^{-1}A$ and $Y = P$:

$$\mathrm{Tr}(B) = \mathrm{Tr}(P^{-1}AP) = \mathrm{Tr}((P^{-1}A)P) = \mathrm{Tr}(P(P^{-1}A)) = \mathrm{Tr}((PP^{-1})A) = \mathrm{Tr}(I_nA) = \mathrm{Tr}(A).$$

Thus, the trace is invariant under change of basis. ■

Example 3.3.7 (Verifying $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ for Rectangular Matrices). Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$ (2×3) and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}$ (3×2). Compute AB and BA, and verify that $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.

Solution. First, compute the 2×2 product matrix AB:

$$AB = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1(2) + 2(0) + 0(1) & 1(1) + 2(-1) + 0(3) \\ 3(2) + (-1)(0) + 4(1) & 3(1) + (-1)(-1) + 4(3) \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 10 & 16 \end{bmatrix}.$$

The trace of AB is:

$$\mathrm{Tr}(AB) = 2 + 16 = 18.$$

Next, compute the 3×3 product matrix BA:

$$BA = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix} = \begin{bmatrix} 2(1) + 1(3) & 2(2) + 1(-1) & 2(0) + 1(4) \\ 0(1) + (-1)(3) & 0(2) + (-1)(-1) & 0(0) + (-1)(4) \\ 1(1) + 3(3) & 1(2) + 3(-1) & 1(0) + 3(4) \end{bmatrix} = \begin{bmatrix} 5 & 3 & 4 \\ -3 & 1 & -4 \\ 10 & -1 & 12 \end{bmatrix}.$$

The trace of BA is:

$$\mathrm{Tr}(BA) = 5 + 1 + 12 = 18.$$

Thus, $\mathrm{Tr}(AB) = \mathrm{Tr}(BA) = 18$, despite AB and BA having different matrix dimensions (2×2 versus 3×3). ■

Example 3.3.8 (Impossibility of $AB - BA = I_n$). Prove that there exist no $n \times n$ matrices $A, B \in M_n(\mathbb{K})$ over a field $\mathbb{K}$ of characteristic zero ($\mathbb{R}$ or $\mathbb{C}$) satisfying the commutation relation:

$$AB - BA = I_n.$$

Solution. Suppose for contradiction that such matrices $A, B \in M_n(\mathbb{K})$ exist, so that $AB - BA = I_n$.

Applying the trace operator to both sides of the equation:

$$\operatorname{Tr}(AB - BA) = \operatorname{Tr}(I_n).$$

By additive linearity of the trace operator:

$$\operatorname{Tr}(AB) - \operatorname{Tr}(BA) = \operatorname{Tr}(I_n).$$

Using the identity $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$ on the left-hand side, and $\operatorname{Tr}(I_n) = n$ on the right-hand side:

$$\operatorname{Tr}(AB) - \operatorname{Tr}(AB) = n \implies 0 = n.$$

For matrices of dimension $n \geq 1$ over $\mathbb{R}$ or $\mathbb{C}$, $n \neq 0$. This contradiction establishes that no matrices A, B can satisfy $AB - BA = I_n$. ■

Example 3.3.9 (Trace Under Similarity Transformations). Given $A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, find $B = P^{-1}AP$ and verify that $\operatorname{Tr}(B) = \operatorname{Tr}(A)$.

Solution. First, compute $\operatorname{Tr}(A) = 3 + 4 = 7$.

Next, find P^{-1}. Since $\det(P) = (1)(2) - (1)(1) = 1$:

$$P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}.$$

Now compute AP:

$$AP = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 6 & 10 \end{bmatrix}.$$

Now compute $B = P^{-1}(AP)$:

$$B = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 5 \\ 6 & 10 \end{bmatrix} = \begin{bmatrix} 2(4) - 1(6) & 2(5) - 1(10) \\ -1(4) + 1(6) & -1(5) + 1(10) \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 2 & 5 \end{bmatrix}.$$

The trace of B is:

$$\operatorname{Tr}(B) = 2 + 5 = 7.$$

Thus, $\operatorname{Tr}(B) = \operatorname{Tr}(A) = 7$, verifying similarity invariance. ■

Exercise 3.3.10.

1. Determine whether $v = \begin{bmatrix} 3 \\ 8 \\ 11 \end{bmatrix}$ can be expressed as a linear combination of $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$ in $\mathbb{R}^3$.

2. Show that the set $S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$ spans $\mathbb{R}^3$.

3. Show that $\mathcal{B} = \left\{ v_1 = \begin{bmatrix} 1+i \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$ forms a basis for $\mathbb{C}^2$ over $\mathbb{C}$.

4. Find the coordinates of $x = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$ relative to the ordered basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ for $\mathbb{R}^3$.

5. Using the definition directly, test whether $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \right\} \subset \mathbb{R}^3$ is linearly independent.

6. Using the matrix rank test, determine if $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 3 \end{bmatrix} \right\} \subset \mathbb{R}^4$ is linearly independent.

7. Prove that any set of k vectors in $\mathbb{K}^n$ with $k > n$ is automatically linearly dependent.

8. Determine all real values of k for which the set $\left\{ \begin{bmatrix} 1 \\ 1 \\ k \end{bmatrix}, \begin{bmatrix} 1 \\ k \\ 1 \end{bmatrix}, \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} \right\} \subset \mathbb{R}^3$ is linearly independent.

9. Reduce the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 2 \\ 3 & 6 & 0 & 5 \end{bmatrix}$ to Row Echelon Form (REF), state its rank r, and specify the maximum number of linearly independent row vectors.

10. Compute the rank of $A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 2 & 3 & 7 & 11 \\ 1 & 3 & 8 & 10 \end{bmatrix}$ by reducing it to Reduced Row Echelon Form (RREF).

11. Prove that elementary row operations preserve the row space and therefore do not alter the rank of a matrix.

12. Prove that for any rectangular matrix $A \in M_{m\times n}(\mathbb{K})$, $\operatorname{rank}(A) \leq \min(m, n)$.

13. For the coefficient matrix $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 3 \\ 3 & 6 & 2 & 7 \end{bmatrix}$:
    (a) Find its rank r and the total variable count n.
    (b) Determine the dimension $n - r$ of the solution space for $AX = 0$.
    (c) Construct an explicit basis for the solution space.

14. State the Rank-Nullity Theorem for a matrix $A \in M_{m\times n}(\mathbb{K})$. If A is a 4×6 matrix with $\operatorname{rank}(A) = 3$, find the dimension of the solution space of $AX = 0$.

15. Find the general solution and a basis for the null space (solution space) of:

$$A = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 0 & 4 & 8 \end{bmatrix}.$$

16. Prove that if an $m \times n$ matrix A has $\operatorname{rank}(A) = n$, then the homogeneous system $AX = 0$ possesses only the trivial solution $X = 0$.

17. Compute the trace of $A = \begin{bmatrix} 3 & -1 & 4 \\ 2 & 5 & -2 \\ 1 & 0 & 6 \end{bmatrix}$, and verify that $\operatorname{Tr}(A^T) = \operatorname{Tr}(A)$.

18. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$ (2 × 3) and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}$ (3 × 2), calculate AB and BA, and verify the fundamental product identity $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$.

19. Prove that there exist no $n \times n$ matrices $A, B \in M_n(\mathbb{K})$ over $\mathbb{R}$ or $\mathbb{C}$ satisfying the commutator identity $AB - BA = I_n$.

20. Prove the similarity invariance of trace: if $A, B \in M_n(\mathbb{K})$ and $B = P^{-1}AP$ for a nonsingular matrix $P \in M_n(\mathbb{K})$, then $\operatorname{Tr}(B) = \operatorname{Tr}(A)$.

Chapter 4

MATRIX POLYNOMIALS AND INNER PRODUCT SPACES

This unit explores matrix polynomials, algebraic matrix functions, and the spectral properties of square matrices. It introduces characteristic and minimal polynomials, establishing eigenvalues and eigenvectors as intrinsic invariants of linear transformations. The unit examines the Cayley–Hamilton theorem alongside its practical computational applications in matrix inversion and power calculations. Finally, it investigates inner product structures on $\mathbb{R}^n$ and $\mathbb{C}^n$, establishing norms, orthogonality, orthonormal sets, and geometric interpretations of length and angles.

4.1 Matrix Polynomials and Algebraic Functions of Matrices

In scalar algebra, polynomials form the fundamental class of functions. By substituting a square matrix $A \in M_n(\mathbb{K})$ in place of a scalar variable x, we extend polynomial algebra to matrix spaces. This leads to matrix polynomials, annihilating polynomials, and formal algebraic matrix functions.

Definition 4.1.1 (Matrix Polynomial). Let $p(x) = a_k x^k + a_{k-1}x^{k-1} + \cdots + a_1 x + a_0$ be a polynomial of degree k over a field $\mathbb{K}$ (where $a_k \neq 0$). For a square matrix $A \in M_n(\mathbb{K})$, the **matrix polynomial** $p(A)$ is the $n \times n$ matrix defined by:

$$p(A) = a_k A^k + a_{k-1}A^{k-1} + \cdots + a_1 A + a_0 I_n,$$

where $A^0 = I_n$ is the $n \times n$ identity matrix, and A^j denotes the j-th matrix power ($A^j = \underbrace{A \cdot A \cdots A}_{j \text{ times}}$).

In scalar algebra, a_0 is a scalar. In matrix algebra, addition is defined only between matrices of identical dimensions. Therefore, the scalar constant a_0 in $p(x)$ **must** be multiplied by the identity matrix I_n when evaluating $p(A)$.

Theorem 4.1.2 (Commutativity and Homomorphism of Matrix Polynomials). *Let $p(x), q(x) \in \mathbb{K}[x]$ be polynomials, $A \in M_n(\mathbb{K})$, and $c \in \mathbb{K}$.*

(i) ***Additive Homomorphism:*** $(p + q)(A) = p(A) + q(A)$.

(ii) ***Scalar Homomorphism:*** $(c \cdot p)(A) = c \cdot p(A)$.

(iii) ***Multiplicative Homomorphism:*** $(p \cdot q)(A) = p(A)q(A)$.

(iv) ***Commutativity:*** *Matrix polynomials in the* same *matrix A always commute under multiplication:*

$$p(A)q(A) = q(A)p(A).$$

Proof. Let $p(x) = \sum_{i=0}^{k} a_i x^i$ and $q(x) = \sum_{j=0}^{m} b_j x^j$ be polynomials in $\mathbb{K}[x]$. Without loss of generality, by padding with zero coefficients if necessary, we may assume both polynomials are expressed up to degree k.

We prove each part systematically:

Part (i): Additive Homomorphism

The sum polynomial is $(p+q)(x) = \sum_{i=0}^{k}(a_i + b_i)x^i$. Evaluating at A:

$$\begin{aligned}
(p+q)(A) &= \sum_{i=0}^{k}(a_i + b_i)A^i \\
&= \sum_{i=0}^{k}(a_i A^i + b_i A^i) \\
&= \sum_{i=0}^{k} a_i A^i + \sum_{i=0}^{k} b_i A^i \\
&= p(A) + q(A).
\end{aligned}$$

Part (ii): Scalar Homomorphism

The scaled polynomial is $(c \cdot p)(x) = \sum_{i=0}^{k}(ca_i)x^i$. Evaluating at A:

$$\begin{aligned}
(c \cdot p)(A) &= \sum_{i=0}^{k}(ca_i)A^i \\
&= c\sum_{i=0}^{k} a_i A^i \\
&= c \cdot p(A).
\end{aligned}$$

Part (iii): Multiplicative Homomorphism

By definition of polynomial multiplication, the product polynomial is $(p \cdot q)(x) = \sum_{l=0}^{k+m} c_l x^l$, where $c_l = \sum_{i+j=l} a_i b_j$. Evaluating at A and using $A^{i+j} = A^i A^j$:

$$\begin{aligned}
(p \cdot q)(A) &= \sum_{l=0}^{k+m}\left(\sum_{i+j=l} a_i b_j\right) A^{i+j} \\
&= \sum_{i=0}^{k}\sum_{j=0}^{m} a_i b_j A^i A^j \\
&= \left(\sum_{i=0}^{k} a_i A^i\right)\left(\sum_{j=0}^{m} b_j A^j\right) \\
&= p(A)q(A).
\end{aligned}$$

Part (iv): Commutativity

In scalar polynomial arithmetic, polynomial multiplication is commutative, so $(p \cdot q)(x) = (q \cdot p)(x)$ in $\mathbb{K}[x]$.

Applying the multiplicative homomorphism from Part (iii) to both sides:

$$\begin{aligned} p(A)q(A) &= (p \cdot q)(A) \\ &= (q \cdot p)(A) \quad \text{(since } p(x)q(x) = q(x)p(x) \text{ in } \mathbb{K}[x]) \\ &= q(A)p(A). \end{aligned}$$

Thus, $p(A)q(A) = q(A)p(A)$, proving that matrix polynomials in the same matrix A always commute. ∎

Theorem 4.1.3 (Spectral Mapping Theorem for Matrix Polynomials). *Let $A \in M_n(\mathbb{K})$. If $\lambda \in \mathbb{K}$ is an eigenvalue of A with corresponding eigenvector $x \in \mathbb{K}^n \setminus \{0\}$, then $p(\lambda)$ is an eigenvalue of the matrix polynomial $p(A)$ with the exact same eigenvector x:*

$$p(A)x = p(\lambda)x.$$

Proof. Since $Ax = \lambda x$, applying A repeatedly yields $A^i x = \lambda^i x$ for all integers $i \geq 0$.

Let $p(x) = a_k x^k + \cdots + a_1 x + a_0$. Substituting A and applying $p(A)$ to x:

$$\begin{aligned} p(A)x &= (a_k A^k + \cdots + a_1 A + a_0 I_n)x \\ &= a_k A^k x + \cdots + a_1 Ax + a_0 x \\ &= a_k \lambda^k x + \cdots + a_1 \lambda x + a_0 x \\ &= (a_k \lambda^k + \cdots + a_1 \lambda + a_0)x = p(\lambda)x. \end{aligned}$$

Thus, x is an eigenvector of $p(A)$ with corresponding eigenvalue $p(\lambda)$. ∎

4.1.1 Annihilating Polynomials and Special Matrix Classes

Definition 4.1.4 (Annihilating Polynomial). A polynomial $p(x) \in \mathbb{K}[x]$ is called an **annihilating polynomial** (or **zeroing polynomial**) of a square matrix $A \in M_n(\mathbb{K})$ if:

$$p(A) = 0_{n \times n}.$$

Many prominent classes of matrices are defined by simple polynomial algebraic relations:

1. **Idempotent Matrices (Projections):** A matrix A satisfying $A^2 = A$. Its annihilating polynomial is $p(x) = x^2 - x = x(x-1)$.
2. **Involutory Matrices:** A matrix A satisfying $A^2 = I_n$. Its annihilating polynomial is $p(x) = x^2 - 1 = (x-1)(x+1)$.
3. **Nilpotent Matrices:** A matrix A for which there exists an integer $k \geq 1$ such that $A^k = 0$. Its annihilating polynomial is $p(x) = x^k$.
4. **Nilpotent Index:** The smallest positive integer k such that $A^k = 0$ is called the **index of nilpotency**.

4.1.2 Algebraic Matrix Functions and Power Series

Beyond finite polynomials, algebraic functions of matrices can be defined via infinite power series and rational expressions, provided convergence criteria or termination conditions are met.

Theorem 4.1.5 (Inverse of $(I_n - A)$ for a Nilpotent Matrix)**.** *Let $A \in M_n(\mathbb{K})$ be a nilpotent matrix with index k (such that $A^k = 0$). Then the matrix $(I_n - A)$ is invertible, and its inverse is given by the finite polynomial sum:*

$$(I_n - A)^{-1} = I_n + A + A^2 + \cdots + A^{k-1}.$$

Proof. Let $B = I_n + A + A^2 + \cdots + A^{k-1}$. Multiplying $(I_n - A)$ by B expands as:

$$\begin{aligned}(I_n - A)B &= I_n(I_n + A + A^2 + \cdots + A^{k-1}) - A(I_n + A + A^2 + \cdots + A^{k-1}) \\ &= (I_n + A + A^2 + \cdots + A^{k-1}) - (A + A^2 + A^3 + \cdots + A^k).\end{aligned}$$

Notice that all intermediate terms cancel in pairs, leaving only:

$$(I_n - A)B = I_n - A^k.$$

Since A is nilpotent with $A^k = 0$, this simplifies to $(I_n - A)B = I_n$.

By identical expansion, $B(I_n - A) = I_n$. Therefore, $(I_n - A)$ is invertible and $(I_n - A)^{-1} = B$. ∎

Definition 4.1.6 (Matrix Exponential Function)**.** For any square complex matrix $A \in M_n(\mathbb{C})$, the **matrix exponential** e^A (or $\exp(A)$) is defined by the convergent matrix power series:

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k = I_n + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \ldots$$

If A is nilpotent with $A^k = 0$, the series terminates after k terms, yielding a finite matrix polynomial.

Example 4.1.7 (Evaluating a Matrix Polynomial)**.** Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \in M_2(\mathbb{R})$ and $p(x) = x^2 - 4x + 3$. Evaluate $p(A)$.

Solution. We compute the matrix power A^2:

$$A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}.$$

Now substitute A into $p(A) = A^2 - 4A + 3I_2$:

$$\begin{aligned}p(A) &= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} - 4\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ 0 & 12 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1-4+3 & 8-8+0 \\ 0-0+0 & 9-12+3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.\end{aligned}$$

Thus, $p(A) = 0_{2\times 2}$, showing that $p(x) = x^2 - 4x + 3$ is an annihilating polynomial for A. ∎

Example 4.1.8 (Algebraic Matrix Functions for Nilpotent Matrices)**.** Let $A = \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$. Show that A is nilpotent, and compute $(I_3 - A)^{-1}$ and e^A as matrix polynomials.

Solution. We compute powers of A:

$$A^2 = \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$A^3 = A^2 A = \begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Since $A^3 = 0$, A is nilpotent with index $k = 3$.

Using the algebraic series formula for $(I_3 - A)^{-1}$:

$$(I_3 - A)^{-1} = I_3 + A + A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

For the matrix exponential e^A, the infinite series terminates at $k = 3$:

$$\begin{aligned} e^A = I_3 + A + \frac{1}{2}A^2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

■

Example 4.1.9 (Eigenvalues of Matrix Polynomials)**.** Let $A \in M_n(\mathbb{R})$ be a 2×2 matrix with eigenvalues $\lambda_1 = 2$ and $\lambda_2 = -1$. Find the eigenvalues of the matrix polynomial $p(A) = A^3 - 2A + 4I_2$.

Solution. By the Spectral Mapping Theorem for Matrix Polynomials, if λ is an eigenvalue of A, then $p(\lambda)$ is an eigenvalue of $p(A)$, where $p(x) = x^3 - 2x + 4$.

Evaluating $p(x)$ at each eigenvalue of A:

- For $\lambda_1 = 2$:
$$p(\lambda_1) = p(2) = 2^3 - 2(2) + 4 = 8 - 4 + 4 = 8.$$
- For $\lambda_2 = -1$:
$$p(\lambda_2) = p(-1) = (-1)^3 - 2(-1) + 4 = -1 + 2 + 4 = 5.$$

Thus, the eigenvalues of $p(A)$ are 8 and 5. ■

4.2 Characteristic and Minimal Polynomials of a Matrix

To every square matrix $A \in M_n(\mathbb{K})$, we can associate two special polynomials: the **characteristic polynomial**, which encodes the matrix's determinantal structure and eigenvalues, and the **minimal polynomial**, which represents the simplest algebraic equation satisfied by the matrix.

Definition 4.2.1 (Characteristic Matrix and Polynomial)**.** Let $A \in M_n(\mathbb{K})$ be an $n \times n$ matrix over a field $\mathbb{K}$, and let λ be an indeterminate scalar.

(i) The matrix $\lambda I_n - A$ is called the **characteristic matrix** of A.

(ii) The determinant of the characteristic matrix, denoted by $p_A(\lambda)$ or $\chi_A(\lambda)$, is a polynomial in λ of degree n called the **characteristic polynomial** of A:

$$p_A(\lambda) = \det(\lambda I_n - A).$$

Theorem 4.2.2 (Structure of the Characteristic Polynomial). *For any $n \times n$ matrix $A \in M_n(\mathbb{K})$, the characteristic polynomial $p_A(\lambda)$ has the general expansion:*

$$p_A(\lambda) = \lambda^n - \mathrm{Tr}(A)\lambda^{n-1} + c_{n-2}\lambda^{n-2} - \cdots + (-1)^n \det(A).$$

In particular:

(i) $p_A(\lambda)$ is a monic polynomial of degree n (its leading coefficient is 1).

(ii) The coefficient of λ^{n-1} is $-\mathrm{Tr}(A)$.

(iii) The constant term $p_A(0)$ is $(-1)^n \det(A)$.

(iv) The roots of $p_A(\lambda) = 0$ in $\mathbb{K}$ (or an algebraic extension of $\mathbb{K}$) are precisely the eigenvalues of A.

Proof. By definition, the characteristic polynomial of $A = (a_{ij})_{n\times n} \in M_n(\mathbb{K})$ is given by $p_A(\lambda) = \det(\lambda I_n - A)$, which is the determinant of the matrix:

$$\lambda I_n - A = \begin{bmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{bmatrix}.$$

By the Leibniz formula for determinants, $\det(\lambda I_n - A)$ is expressed as the sum:

$$p_A(\lambda) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} (\lambda I_n - A)_{i,\sigma(i)},$$

where S_n is the symmetric group of permutations on $\{1, 2, \ldots, n\}$.

We prove each part of the theorem explicitly:

(i) For the identity permutation $\sigma = \text{id}$, the corresponding term in the Leibniz sum is the product of the main diagonal entries:

$$\prod_{i=1}^{n} (\lambda - a_{ii}) = (\lambda - a_{11})(\lambda - a_{22}) \cdots (\lambda - a_{nn}) = \lambda^n - \left(\sum_{i=1}^{n} a_{ii}\right) \lambda^{n-1} + \ldots$$

For any non-identity permutation $\sigma \neq \text{id}$, $\sigma(i) \neq i$ for at least two indices i. Thus, the product contains at most $n-2$ diagonal entries of the form $(\lambda - a_{ii})$, meaning its degree in λ is at most $n-2$.

Therefore, the leading term of $p_A(\lambda)$ comes exclusively from the identity permutation term, giving $1 \cdot \lambda^n = \lambda^n$. Hence, $p_A(\lambda)$ is a monic polynomial of degree n.

(ii) From the Leibniz expansion, non-identity permutations ($\sigma \neq \text{id}$) contribute terms of degree at most $n-2$ in λ. Consequently, the term involving λ^{n-1} arises solely from expanding the main diagonal product:

$$\prod_{i=1}^{n} (\lambda - a_{ii}) = \lambda^n - (a_{11} + a_{22} + \cdots + a_{nn})\lambda^{n-1} + \text{lower-degree terms}.$$

Since $\mathrm{Tr}(A) = \sum_{i=1}^{n} a_{ii}$, the coefficient of λ^{n-1} in $p_A(\lambda)$ is exactly $-\mathrm{Tr}(A)$.

(iii) The constant term of $p_A(\lambda)$ is equal to its value at $\lambda = 0$:

$$p_A(0) = \det(0 \cdot I_n - A) = \det(-A).$$

By the multilinearity of the determinant function, factoring out -1 from each of the n rows yields:

$$\det(-A) = (-1)^n \det(A).$$

Thus, the constant term $p_A(0)$ is $(-1)^n \det(A)$.

(iv) Let $\lambda_0 \in \mathbb{K}$ (or an algebraic extension of $\mathbb{K}$). By definition, λ_0 is an eigenvalue of A if and only if there exists a non-zero eigenvector $v \in \mathbb{K}^n \setminus \{0\}$ satisfying:

$$\begin{aligned} Av = \lambda_0 v &\iff \lambda_0 v - Av = 0 \\ &\iff (\lambda_0 I_n - A)v = 0. \end{aligned}$$

The homogeneous linear system $(\lambda_0 I_n - A)v = 0$ has a non-trivial solution $v \neq 0$ if and only if the coefficient matrix $(\lambda_0 I_n - A)$ is singular (non-invertible), which holds if and only if its determinant vanishes:

$$\det(\lambda_0 I_n - A) = 0 \iff p_A(\lambda_0) = 0.$$

Therefore, λ_0 is a root of $p_A(\lambda) = 0$ if and only if λ_0 is an eigenvalue of A. ■

Theorem 4.2.3 (Similarity Invariance of the Characteristic Polynomial)**.** *Similar matrices have identical characteristic polynomials.*

Proof. Let $A, B \in M_n(\mathbb{K})$ be similar matrices, so $B = P^{-1}AP$ for some non-singular $P \in M_n(\mathbb{K})$. Using determinant multiplicative properties:

$$\begin{aligned} p_B(\lambda) &= \det(\lambda I_n - B) = \det(\lambda I_n - P^{-1}AP) \\ &= \det(P^{-1}(\lambda I_n)P - P^{-1}AP) = \det(P^{-1}(\lambda I_n - A)P) \\ &= \det(P^{-1})\det(\lambda I_n - A)\det(P) = \det(\lambda I_n - A) = p_A(\lambda). \end{aligned}$$

Thus, $p_B(\lambda) = p_A(\lambda)$, showing that the characteristic polynomial is a similarity invariant. ■

4.2.1 The Minimal Polynomial

While many polynomials $p(x)$ satisfy $p(A) = 0$, there exists a unique polynomial of lowest degree that annihilates A.

Definition 4.2.4 (Minimal Polynomial)**.** The **minimal polynomial** of a square matrix $A \in M_n(\mathbb{K})$, denoted by $m_A(x)$ (or $\mu_A(x)$), is the unique monic polynomial of minimum degree over $\mathbb{K}$ such that:

$$m_A(A) = 0_{n\times n}.$$

Theorem 4.2.5 (Divisibility Property of the Minimal Polynomial)**.** *Let $A \in M_n(\mathbb{K})$, and let $m_A(x)$ be its minimal polynomial. A polynomial $p(x) \in \mathbb{K}[x]$ satisfies $p(A) = 0$ if and only if $m_A(x)$ divides $p(x)$ (i.e., $m_A(x) \mid p(x)$).*

Proof. Suppose $m_A(x) \mid p(x)$, so $p(x) = q(x)m_A(x)$ for some polynomial $q(x)$. Substituting A:

$$p(A) = q(A)m_A(A) = q(A) \cdot 0 = 0.$$

Suppose $p(A) = 0$. By the polynomial division algorithm, there exist unique polynomials $q(x)$ and $r(x)$ such that:

$$p(x) = q(x)m_A(x) + r(x), \quad \text{where } r(x) = 0 \text{ or } \deg(r) < \deg(m_A).$$

Evaluating at A:

$$p(A) = q(A)m_A(A) + r(A) \implies 0 = q(A) \cdot 0 + r(A) \implies r(A) = 0.$$

If $r(x) \neq 0$, then $r(x)$ would be an annihilating polynomial for A with $\deg(r) < \deg(m_A)$, contradicting the minimality of $\deg(m_A)$. Thus, $r(x) = 0$, proving $m_A(x) \mid p(x)$. ■

Theorem 4.2.6 (Root Correspondence Theorem). *Let $A \in M_n(\mathbb{K})$. A scalar $\lambda \in \mathbb{K}$ is a root of the characteristic polynomial $p_A(x)$ if and only if λ is a root of the minimal polynomial $m_A(x)$.*

Consequently, $p_A(x)$ and $m_A(x)$ have the exact same distinct roots (eigenvalues), differing only in the multiplicities of those roots.

Proof. Let λ be a root of $p_A(x)$, so λ is an eigenvalue of A with non-zero eigenvector v ($Av = \lambda v$). Applying $m_A(A) = 0$ to v:

$$0 = m_A(A)v = m_A(\lambda)v.$$

Since $v \neq 0$, $m_A(\lambda) = 0$, so λ is a root of $m_A(x)$.

Conversely, let λ be a root of $m_A(x)$, so $m_A(x) = (x - \lambda)q(x)$ for some polynomial $q(x)$ with $\deg(q) < \deg(m_A)$. Since $m_A(x)$ is minimal, $q(A) \neq 0$. Choose a vector w such that $v = q(A)w \neq 0$. Then:

$$(A - \lambda I_n)v = (A - \lambda I_n)q(A)w = m_A(A)w = 0w = 0.$$

Thus, $Av = \lambda v$ with $v \neq 0$, showing λ is an eigenvalue of A, and hence a root of $p_A(x)$. ■

If the characteristic polynomial of $A \in M_n(\mathbb{K})$ factorizes into linear factors over $\mathbb{K}$ as:

$$p_A(x) = (x - \lambda_1)^{n_1}(x - \lambda_2)^{n_2} \cdots (x - \lambda_k)^{n_k}, \quad \text{where } \sum_{i=1}^{k} n_i = n,$$

then its minimal polynomial $m_A(x)$ has the factorization structure:

$$m_A(x) = (x - \lambda_1)^{m_1}(x - \lambda_2)^{m_2} \cdots (x - \lambda_k)^{m_k}, \quad \text{where } 1 \leq m_i \leq n_i \text{ for each } i.$$

The exponent m_i in $m_A(x)$ represents the size of the largest Jordan block associated with eigenvalue λ_i.

Example 4.2.7 (Computing Characteristic and Minimal Polynomials for 3×3 Matrices). Find the characteristic polynomial $p_A(x)$ and minimal polynomial $m_A(x)$ for the following matrices:

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Solution. For matrix $A = 3I_3$: The characteristic polynomial is:

$$p_A(x) = \det(xI_3 - A) = \det \begin{bmatrix} x-3 & 0 & 0 \\ 0 & x-3 & 0 \\ 0 & 0 & x-3 \end{bmatrix} = (x-3)^3.$$

By the Root Correspondence Theorem, the minimal polynomial must be of the form $m_A(x) = (x-3)^m$ for some $1 \leq m \leq 3$. Testing $m = 1$:

$$m(A) = A - 3I_3 = 3I_3 - 3I_3 = 0_{3\times3}.$$

Thus, the minimal polynomial is $m_A(x) = x - 3$.

For matrix B: The characteristic polynomial is:

$$p_B(x) = \det(xI_3 - B) = \det \begin{bmatrix} x-3 & -1 & 0 \\ 0 & x-3 & 0 \\ 0 & 0 & x-3 \end{bmatrix} = (x-3)^3.$$

The candidate minimal polynomials are $m_1(x) = x-3$, $m_2(x) = (x-3)^2$, and $m_3(x) = (x-3)^3$.

- Test $m_1(B) = B - 3I_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq 0$.
- Test $m_2(B) = (B - 3I_3)^2$:

$$(B - 3I_3)^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0.$$

Thus, the minimal polynomial for B is $m_B(x) = (x-3)^2$. ■

Example 4.2.8 (Determining Minimal Polynomial from Candidate Factors)**.** Let $A \in M_4(\mathbb{R})$ have characteristic polynomial $p_A(x) = (x-2)^3(x+1)$. Determine all possible candidate minimal polynomials for A, and find $m_A(x)$ if it is known that $\operatorname{rank}(A - 2I_4) = 2$.

Solution. Since $p_A(x) = (x-2)^3(x+1)$, the root -1 has multiplicity 1 in $p_A(x)$, so its exponent in $m_A(x)$ must be 1. The root 2 has multiplicity 3 in $p_A(x)$, so its exponent m in $m_A(x)$ satisfies $1 \le m \le 3$.

The possible candidate minimal polynomials are:

$$\begin{aligned} q_1(x) &= (x-2)(x+1), \\ q_2(x) &= (x-2)^2(x+1), \\ q_3(x) &= (x-2)^3(x+1). \end{aligned}$$

To determine the correct exponent m for $(x-2)$, recall that the nullity of $(A-2I_4)$ represents the number of Jordan blocks associated with $\lambda = 2$. By the Rank-Nullity Theorem:

$$\operatorname{nullity}(A - 2I_4) = 4 - \operatorname{rank}(A - 2I_4) = 4 - 2 = 2.$$

There are 2 linear independent eigenvectors (Jordan blocks) for $\lambda = 2$. Since the total algebraic multiplicity is 3, the sizes of these two Jordan blocks must be 2 and 1. The size of the largest Jordan block is 2.

Because the exponent of $(x - \lambda)$ in $m_A(x)$ equals the size of the largest Jordan block for λ, we have $m = 2$.

Thus, the minimal polynomial is $m_A(x) = (x-2)^2(x+1)$. ■

4.3 Eigenvalues and Eigenvectors as Invariants of Linear Transformations

When a linear operator $T : V \to V$ on a finite-dimensional vector space V is represented by a matrix, the matrix entries depend on the specific choice of basis for V. However, certain fundamental spectral quantities—namely **eigenvalues**, **eigenspaces**, **characteristic polynomials**, **trace**, and **determinant**—remain invariant under basis changes.

Definition 4.3.1 (Eigenvalue, Eigenvector, and Eigenspace)**.** Let V be a vector space over a field $\mathbb{K}$, and let $T : V \to V$ be a linear operator.

(i) A scalar $\lambda \in \mathbb{K}$ is called an **eigenvalue** of T if there exists a non-zero vector $v \in V \setminus \{0\}$ such that:

$$T(v) = \lambda v.$$

(ii) Any non-zero vector $v \in V \setminus \{0\}$ satisfying $T(v) = \lambda v$ is called an **eigenvector** of T corresponding to λ.

(iii) The set of all eigenvectors corresponding to λ, together with the zero vector, forms a subspace of V called the **eigenspace** of T at λ, denoted by $E_\lambda(T)$ or V_λ:

$$E_\lambda(T) = \ker(T - \lambda I) = \{v \in V \mid T(v) = \lambda v\}.$$

For a square matrix $A \in M_n(\mathbb{K})$, these definitions apply to the linear operator $T(x) = Ax$ on $\mathbb{K}^n$, giving $Ax = \lambda x$ and $E_\lambda(A) = \mathrm{null}(A - \lambda I_n)$.

Definition 4.3.2 (Algebraic and Geometric Multiplicity)**.** Let λ be an eigenvalue of $A \in M_n(\mathbb{K})$.

(i) The **algebraic multiplicity** $m_a(\lambda)$ is the multiplicity of λ as a root of the characteristic polynomial $p_A(x) = \det(xI_n - A)$.

(ii) The **geometric multiplicity** $m_g(\lambda)$ is the dimension of the eigenspace $E_\lambda(A)$:

$$m_g(\lambda) = \dim(E_\lambda(A)) = \mathrm{nullity}(A - \lambda I_n) = n - \mathrm{rank}(A - \lambda I_n).$$

Theorem 4.3.3 (Fundamental Multiplicity Inequality)**.** *For every eigenvalue λ of a matrix $A \in M_n(\mathbb{K})$, the geometric multiplicity is at least 1 and cannot exceed the algebraic multiplicity:*

$$1 \leq m_g(\lambda) \leq m_a(\lambda).$$

4.3.1 Basis Independence and Invariance Under Linear Transformations

Let V be an n-dimensional vector space over $\mathbb{K}$, and let $T : V \to V$ be a linear operator. If $\mathcal{B} = \{b_1, \ldots, b_n\}$ and $\mathcal{C} = \{c_1, \ldots, c_n\}$ are two ordered bases for V, the matrix representations $A = [T]_\mathcal{B}$ and $B = [T]_\mathcal{C}$ are related by a similarity transformation:

$$B = P^{-1}AP,$$

where $P = [I_V]_\mathcal{B}^\mathcal{C}$ is the non-singular change-of-basis matrix from $\mathcal{C}$ to $\mathcal{B}$.

Theorem 4.3.4 (Invariance of Eigenvalues under Change of Basis)**.** *Let $T : V \to V$ be a linear operator, and let $A = [T]_\mathcal{B}$ and $B = [T]_\mathcal{C}$ be matrix representations of T with respect to two different bases $\mathcal{B}$ and $\mathcal{C}$. Then:*

(i) A and B have the exact same characteristic polynomial: $p_A(\lambda) = p_B(\lambda)$.

(ii) A and B have the exact same eigenvalues with identical algebraic and geometric multiplicities.

(iii) $\mathrm{Tr}(A) = \mathrm{Tr}(B)$ *and* $\det(A) = \det(B)$.

Consequently, eigenvalues are intrinsic properties of the operator T, independent of the basis chosen to represent T.

Proof. Since $B = P^{-1}AP$ for a non-singular change-of-basis matrix P:

$$\begin{aligned} p_B(\lambda) &= \det(\lambda I_n - B) = \det(\lambda I_n - P^{-1}AP) \\ &= \det(P^{-1}(\lambda I_n - A)P) = \det(P^{-1})\det(\lambda I_n - A)\det(P) \\ &= \det(\lambda I_n - A) = p_A(\lambda). \end{aligned}$$

Since $p_A(\lambda) = p_B(\lambda)$, A and B share identical roots (eigenvalues) and algebraic multiplicities. Furthermore, since $\operatorname{rank}(P^{-1}MP) = \operatorname{rank}(M)$ for any matrix M:

$$m_g(\lambda)_B = n - \operatorname{rank}(B - \lambda I_n) = n - \operatorname{rank}(P^{-1}(A - \lambda I_n)P) = n - \operatorname{rank}(A - \lambda I_n) = m_g(\lambda)_A.$$

Thus, geometric multiplicities are also preserved. ■

Theorem 4.3.5 (Transformation of Eigenvector Coordinates). *Let $T : V \to V$ be a linear operator, $A = [T]_{\mathcal{B}}$, $B = [T]_{\mathcal{C}}$, and $B = P^{-1}AP$. If $x \in \mathbb{K}^n$ is an eigenvector of A corresponding to eigenvalue λ ($Ax = \lambda x$), then:*

$$y = P^{-1}x$$

is the eigenvector of B corresponding to the same eigenvalue λ ($By = \lambda y$).

Proof. Substitute $y = P^{-1}x$ into By:

$$By = (P^{-1}AP)(P^{-1}x) = P^{-1}A(PP^{-1})x = P^{-1}Ax = P^{-1}(\lambda x) = \lambda(P^{-1}x) = \lambda y.$$

Since P is non-singular and $x \neq 0$, $y = P^{-1}x \neq 0$. Thus, y is an eigenvector of B with eigenvalue λ. ■

4.3.2 Linear Independence of Eigenvectors

Theorem 4.3.6 (Linear Independence of Eigenvectors for Distinct Eigenvalues). *Let $T : V \to V$ be a linear operator. If $v_1, v_2, \ldots, v_k \in V$ are eigenvectors of T corresponding to **distinct** eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k \in \mathbb{K}$ (so $\lambda_i \neq \lambda_j$ for $i \neq j$), then the set $\{v_1, v_2, \ldots, v_k\}$ is linearly independent.*

Proof. We prove the theorem by induction on k, the number of eigenvectors.

Base Case ($k = 1$): Since v_1 is an eigenvector, $v_1 \neq 0$. The single-element set $\{v_1\}$ is linearly independent.

Inductive Hypothesis: Assume that any set of $k-1$ eigenvectors corresponding to distinct eigenvalues is linearly independent.

Inductive Step: Let $v_1, v_2, \ldots, v_k$ be eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_k$. Consider the vector relation:

$$c_1v_1 + c_2v_2 + \cdots + c_kv_k = 0. \tag{1}$$

Apply the linear operator T to both sides of equation (1):

$$\begin{aligned} T(c_1v_1 + c_2v_2 + \cdots + c_kv_k) &= T(0) \\ c_1T(v_1) + c_2T(v_2) + \cdots + c_kT(v_k) &= 0 \\ c_1\lambda_1v_1 + c_2\lambda_2v_2 + \cdots + c_k\lambda_kv_k &= 0. \end{aligned} \tag{2}$$

Multiply equation (1) by the scalar λ_k:

$$c_1\lambda_kv_1 + c_2\lambda_kv_2 + \cdots + c_k\lambda_kv_k = 0. \tag{3}$$

Subtract equation (3) from equation (2):

$$c_1(\lambda_1 - \lambda_k)v_1 + c_2(\lambda_2 - \lambda_k)v_2 + \cdots + c_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1} + 0v_k = 0.$$

This is a linear combination of the $k-1$ eigenvectors $\{v_1, \ldots, v_{k-1}\}$. By the inductive hypothesis, $\{v_1, \ldots, v_{k-1}\}$ is linearly independent, so every coefficient must be zero:

$$c_i(\lambda_i - \lambda_k) = 0 \quad \text{for all } 1 \leq i \leq k-1.$$

Since all eigenvalues are distinct, $\lambda_i - \lambda_k \neq 0$ for $i < k$, which forces $c_i = 0$ for $1 \leq i \leq k-1$.

Substituting $c_1 = c_2 = \cdots = c_{k-1} = 0$ back into equation (1) leaves $c_kv_k = 0$. Since $v_k \neq 0$, $c_k = 0$.

Thus, $c_1 = c_2 = \cdots = c_k = 0$, proving $\{v_1, v_2, \ldots, v_k\}$ is linearly independent. ■

Example 4.3.7. Find the eigenvalues, algebraic multiplicities, geometric multiplicities, and eigenspaces for the matrix:

$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \in M_3(\mathbb{R}).$$

Solution. First, compute the characteristic polynomial $p_A(\lambda) = \det(\lambda I_3 - A)$:

$$\begin{aligned} p_A(\lambda) &= \det \begin{bmatrix} \lambda - 4 & 0 & -1 \\ 2 & \lambda - 1 & 0 \\ 2 & 0 & \lambda - 1 \end{bmatrix} \\ &= (\lambda - 1) \det \begin{bmatrix} \lambda - 4 & -1 \\ 2 & \lambda - 1 \end{bmatrix} \\ &= (\lambda - 1)\left((\lambda - 4)(\lambda - 1) - (-2)\right) \\ &= (\lambda - 1)(\lambda^2 - 5\lambda + 6) = (\lambda - 1)(\lambda - 2)(\lambda - 3). \end{aligned}$$

The eigenvalues are $\lambda_1 = 1$, $\lambda_2 = 2$, and $\lambda_3 = 3$. Each eigenvalue has algebraic multiplicity $m_a = 1$. Since $1 \leq m_g \leq m_a = 1$, each eigenvalue also has geometric multiplicity $m_g = 1$.

We compute the eigenspace for each eigenvalue:

For $\lambda_1 = 1$: Solve $(A - 1I_3)x = 0$:

$$\begin{bmatrix} 3 & 0 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies x_1 = 0, \quad x_3 = 0, \quad x_2 \text{ is free.}$$

$$E_1(A) = \operatorname{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

For $\lambda_2 = 2$: Solve $(A - 2I_3)x = 0$:

$$\begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ -2 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies x_3 = -2x_1, \quad x_2 = -2x_1.$$

$$E_2(A) = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \right\}.$$

For $\lambda_3 = 3$: Solve $(A - 3I_3)x = 0$:

$$\begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies x_3 = -x_1, \quad x_2 = -x_1.$$

$$E_3(A) = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} \right\}.$$

Since $\lambda_1, \lambda_2, \lambda_3$ are distinct, the three basis vectors are linearly independent and form an eigenvector basis for $\mathbb{R}^3$. ■

Example 4.3.8. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Compute $B = P^{-1}AP$, verify that $\det(xI_2 - A) = \det(xI_2 - B)$, and show how the eigenvectors of A transform into eigenvectors of B.

Solution. First, compute P^{-1}. Since $\det(P) = (1)(-1) - (1)(1) = -2$:

$$P^{-1} = -\frac{1}{2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}.$$

Now compute $B = P^{-1}AP$:

$$AP = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 5 & 1 \end{bmatrix},$$

$$B = P^{-1}(AP) = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}\begin{bmatrix} 3 & -1 \\ 5 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -1 & -1 \end{bmatrix}.$$

Computing characteristic polynomials:

$$p_A(x) = \det\begin{bmatrix} x-1 & -2 \\ -3 & x-2 \end{bmatrix} = (x-1)(x-2) - 6 = x^2 - 3x - 4 = (x-4)(x+1),$$

$$p_B(x) = \det\begin{bmatrix} x-4 & 0 \\ 1 & x+1 \end{bmatrix} = (x-4)(x+1) = x^2 - 3x - 4.$$

Thus, $p_A(x) = p_B(x)$, and both matrices share eigenvalues $\lambda_1 = 4$ and $\lambda_2 = -1$.

For $\lambda_1 = 4$, an eigenvector x_1 of A satisfies $(A - 4I_2)x_1 = 0$:

$$\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies x_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

The corresponding eigenvector y_1 of B is given by $y_1 = P^{-1}x_1$:

$$y_1 = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5/2 \\ -1/2 \end{bmatrix}.$$

Verifying $By_1 = 4y_1$:

$$By_1 = \begin{bmatrix} 4 & 0 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} 5/2 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 10 \\ -2 \end{bmatrix} = 4\begin{bmatrix} 5/2 \\ -1/2 \end{bmatrix} = 4y_1.$$

This confirms that $y_1 = P^{-1}x_1$ is an eigenvector of B with eigenvalue $\lambda_1 = 4$. ■

4.4 The Cayley–Hamilton Theorem and Its Computational Applications

The **Cayley–Hamilton Theorem** is one of the foundational results in linear algebra. It asserts that every square matrix satisfies its own characteristic equation, converting higher-degree matrix power calculations and matrix inversions into manageable lower-degree polynomial evaluations.

Theorem 4.4.1 (Cayley–Hamilton Theorem). *Let $A \in M_n(\mathbb{K})$ be a square matrix over a field $\mathbb{K}$, and let $p_A(\lambda) = \det(\lambda I_n - A)$ be its characteristic polynomial, expanded as:*

$$p_A(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0.$$

Then the matrix A satisfies its own characteristic equation:

$$p_A(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1A + c_0I_n = 0_{n\times n}.$$

Proof. Let $C(\lambda) = \lambda I_n - A$ be the characteristic matrix of A. The classical adjugate (adjoint) matrix $B(\lambda) = \operatorname{adj}(\lambda I_n - A)$ is a matrix whose entries are cofactors of $\lambda I_n - A$, which are polynomials in λ of degree at most $n-1$.

Therefore, $B(\lambda)$ can be uniquely expanded as a matrix polynomial in λ with constant coefficient matrices $B_0, B_1, \ldots, B_{n-1} \in M_n(\mathbb{K})$:

$$B(\lambda) = B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \cdots + B_1\lambda + B_0.$$

By the fundamental property of the adjugate matrix ($M \cdot \operatorname{adj}(M) = \det(M)I_n$):

$$(\lambda I_n - A)B(\lambda) = \det(\lambda I_n - A)I_n = p_A(\lambda)I_n.$$

Substituting the expansion of $B(\lambda)$ and $p_A(\lambda)$:

$$(\lambda I_n - A)\left(B_{n-1}\lambda^{n-1} + B_{n-2}\lambda^{n-2} + \cdots + B_0\right) = (\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0)I_n.$$

Expanding the left-hand side:

$$B_{n-1}\lambda^n + (B_{n-2} - AB_{n-1})\lambda^{n-1} + (B_{n-3} - AB_{n-2})\lambda^{n-2} + \cdots + (B_0 - AB_1)\lambda - AB_0.$$

Equating coefficients of identical powers of λ on both sides:

$$\begin{aligned} B_{n-1} &= I_n, \\ B_{n-2} - AB_{n-1} &= c_{n-1}I_n, \\ B_{n-3} - AB_{n-2} &= c_{n-2}I_n, \\ &\vdots \\ B_0 - AB_1 &= c_1 I_n, \\ -AB_0 &= c_0 I_n. \end{aligned}$$

Left-multiply the first equation by A^n, the second by A^{n-1}, the third by A^{n-2}, ..., the k-th by A^{n-k+1}, and the last by $A^0 = I_n$:

$$\begin{aligned} A^n B_{n-1} &= A^n, \\ A^{n-1}B_{n-2} - A^n B_{n-1} &= c_{n-1}A^{n-1}, \\ A^{n-2}B_{n-3} - A^{n-1}B_{n-2} &= c_{n-2}A^{n-2}, \\ &\vdots \\ AB_0 - A^2 B_1 &= c_1 A, \\ -AB_0 &= c_0 I_n. \end{aligned}$$

Summing all left-hand sides yields a telescoping sum where every intermediate matrix term cancels out completely:

$$0_{n\times n} = A^n + c_{n-1}A^{n-1} + c_{n-2}A^{n-2} + \cdots + c_1 A + c_0 I_n = p_A(A).$$

Hence, $p_A(A) = 0_{n\times n}$. ■

The Cayley–Hamilton theorem provides powerful techniques for three primary computational tasks:

1. Matrix Inversion

If $A \in M_n(\mathbb{K})$ is non-singular, then $\det(A) \neq 0$, which implies $c_0 = (-1)^n \det(A) \neq 0$.
By the Cayley–Hamilton Theorem:

$$A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0 \implies -c_0 I_n = A\left(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I_n\right).$$

Multiplying by $-\frac{1}{c_0}$ gives an explicit formula for the inverse A^{-1}:

$$A^{-1} = -\frac{1}{c_0}\left(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_2 A + c_1 I_n\right).$$

2. Reduction of Higher Matrix Powers A^m ($m \geq n$)

Any matrix power A^m ($m \geq n$) can be expressed as a linear combination of $I_n, A, A^2, \ldots, A^{n-1}$.
By the polynomial division algorithm, divide x^m by the characteristic polynomial $p_A(x)$:

$$x^m = q(x)p_A(x) + r(x), \quad \text{where } \deg(r) < n.$$

Let $r(x) = \alpha_{n-1}x^{n-1} + \cdots + \alpha_1 x + \alpha_0$. Substituting A:

$$A^m = q(A)p_A(A) + r(A).$$

Since $p_A(A) = 0$, this simplifies to:

$$A^m = r(A) = \alpha_{n-1}A^{n-1} + \cdots + \alpha_1 A + \alpha_0 I_n.$$

3. Evaluation of Matrix Polynomials and Functions

For any high-degree polynomial $f(x)$, divide $f(x)$ by $p_A(x)$ to get $f(x) = q(x)p_A(x) + r(x)$, yielding $f(A) = r(A)$.

Example 4.4.2 (Verification and Inverse for a 2×2 Matrix)**.** Verify the Cayley–Hamilton Theorem for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, and use it to compute A^{-1} and A^4.

Solution. Compute the characteristic polynomial $p_A(\lambda) = \det(\lambda I_2 - A)$:

$$p_A(\lambda) = \det \begin{bmatrix} \lambda - 1 & -2 \\ -3 & \lambda - 4 \end{bmatrix} = (\lambda - 1)(\lambda - 4) - 6 = \lambda^2 - 5\lambda - 2.$$

To verify $p_A(A) = A^2 - 5A - 2I_2 = 0$:

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}.$$

$$A^2 - 5A - 2I_2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

This verifies the Cayley–Hamilton Theorem.
To compute A^{-1}, rearrange $A^2 - 5A - 2I_2 = 0 \implies 2I_2 = A^2 - 5A = A(A - 5I_2)$:

$$A^{-1} = \frac{1}{2}(A - 5I_2) = \frac{1}{2}\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}\right) = \frac{1}{2}\begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}.$$

To compute A^4, rearrange $A^2 = 5A + 2I_2$:

$$A^3 = A \cdot A^2 = A(5A + 2I_2) = 5A^2 + 2A = 5(5A + 2I_2) + 2A = 27A + 10I_2,$$
$$A^4 = A \cdot A^3 = A(27A + 10I_2) = 27A^2 + 10A = 27(5A + 2I_2) + 10A = 145A + 54I_2.$$

Substituting A and I_2:

$$A^4 = 145\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} + 54\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = \begin{bmatrix}145+54 & 290\\ 435 & 580+54\end{bmatrix} = \begin{bmatrix}199 & 290\\ 435 & 634\end{bmatrix}.$$

■

Example 4.4.3 (Verification and Inverse for a 3×3 Matrix). **Verify the Cayley–Hamilton Theorem for** $A = \begin{bmatrix}2 & -1 & 1\\ -1 & 2 & -1\\ 1 & -1 & 2\end{bmatrix}$, **and compute** A^{-1}.

Solution. Compute $p_A(\lambda) = \det(\lambda I_3 - A)$:

$$p_A(\lambda) = \lambda^3 - \mathrm{Tr}(A)\lambda^2 + c_1\lambda - \det(A).$$

- $\mathrm{Tr}(A) = 2 + 2 + 2 = 6$.
- $c_1 = \det\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix} + \det\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix} + \det\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix} = 3 + 3 + 3 = 9$.
- $\det(A) = 2(4-1) - (-1)(-2+1) + 1(1-2) = 6 - 1 - 1 = 4$.

Thus, $p_A(\lambda) = \lambda^3 - 6\lambda^2 + 9\lambda - 4$.

We compute powers of A:

$$A^2 = \begin{bmatrix}6 & -5 & 5\\ -5 & 6 & -5\\ 5 & -5 & 6\end{bmatrix}, \quad A^3 = \begin{bmatrix}22 & -21 & 21\\ -21 & 22 & -21\\ 21 & -21 & 22\end{bmatrix}.$$

Evaluating $p_A(A) = A^3 - 6A^2 + 9A - 4I_3$:

$$\begin{bmatrix}22 & -21 & 21\\ -21 & 22 & -21\\ 21 & -21 & 22\end{bmatrix} - \begin{bmatrix}36 & -30 & 30\\ -30 & 36 & -30\\ 30 & -30 & 36\end{bmatrix} + \begin{bmatrix}18 & -9 & 9\\ -9 & 18 & -9\\ 9 & -9 & 18\end{bmatrix} - \begin{bmatrix}4 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 4\end{bmatrix} = 0_{3\times 3}.$$

This verifies the theorem.

To compute A^{-1}, use $A^3 - 6A^2 + 9A - 4I_3 = 0 \implies 4I_3 = A(A^2 - 6A + 9I_3)$:

$$A^{-1} = \frac{1}{4}(A^2 - 6A + 9I_3).$$

$$A^2 - 6A + 9I_3 = \begin{bmatrix}6 & -5 & 5\\ -5 & 6 & -5\\ 5 & -5 & 6\end{bmatrix} - \begin{bmatrix}12 & -6 & 6\\ -6 & 12 & -6\\ 6 & -6 & 12\end{bmatrix} + \begin{bmatrix}9 & 0 & 0\\ 0 & 9 & 0\\ 0 & 0 & 9\end{bmatrix} = \begin{bmatrix}3 & 1 & -1\\ 1 & 3 & 1\\ -1 & 1 & 3\end{bmatrix}.$$

Thus:

$$A^{-1} = \frac{1}{4}\begin{bmatrix}3 & 1 & -1\\ 1 & 3 & 1\\ -1 & 1 & 3\end{bmatrix}.$$

■

Example 4.4.4 (High-Degree Polynomial Evaluation via Cayley–Hamilton). **Let** $A = \begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$. **Evaluate** $f(A) = A^5 - 4A^4 + 3A^3 + 2A^2 - 5A + 4I_2$.

Solution. Compute the characteristic polynomial of A:

$$p_A(\lambda) = \det \begin{bmatrix} \lambda - 2 & -1 \\ -1 & \lambda - 2 \end{bmatrix} = (\lambda - 2)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

By the Cayley–Hamilton Theorem, $A^2 - 4A + 3I_2 = 0$.

Perform polynomial division of $f(x) = x^5 - 4x^4 + 3x^3 + 2x^2 - 5x + 4$ by $p_A(x) = x^2 - 4x + 3$:

$$\begin{aligned} f(x) &= x^3(x^2 - 4x + 3) + 2(x^2 - 4x + 3) + 3x - 2 \\ &= (x^3 + 2)(x^2 - 4x + 3) + (3x - 2). \end{aligned}$$

Substituting A into $f(A)$:

$$f(A) = (A^3 + 2I_2)\underbrace{(A^2 - 4A + 3I_2)}_{=0} + (3A - 2I_2) = 3A - 2I_2.$$

Evaluating $3A - 2I_2$:

$$f(A) = 3\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 6-2 & 3 \\ 3 & 6-2 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}.$$

■

4.5 Inner Product Spaces on $\mathbb{R}^n$ and $\mathbb{C}^n$, Norms, and Orthogonality

To introduce geometric notions such as length, distance, angle, and orthogonality into $\mathbb{R}^n$ and $\mathbb{C}^n$, we equip these spaces of column vectors with an **inner product structure**.

Definition 4.5.1 (Inner Product on $\mathbb{K}^n$)**.** Let $\mathbb{K}^n$ denote the space of column vectors with entries in $\mathbb{K}$ (where $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$). An **inner product** on $\mathbb{K}^n$ is a function $\langle \cdot, \cdot \rangle : \mathbb{K}^n \times \mathbb{K}^n \to \mathbb{K}$ that assigns a scalar $\langle x, y \rangle \in \mathbb{K}$ to each pair of vectors $x, y \in \mathbb{K}^n$, satisfying the following axioms for all vectors $x, y, z \in \mathbb{K}^n$ and all scalars $\alpha, \beta \in \mathbb{K}$:

(i) **Conjugate Symmetry:**

$$\langle x, y \rangle = \overline{\langle y, x \rangle}.$$

(When $\mathbb{K} = \mathbb{R}$, this simplifies to standard symmetry: $\langle x, y \rangle = \langle y, x \rangle$).

(ii) **Linearity in the Second Argument:**

$$\langle x, \alpha y + \beta z \rangle = \alpha \langle x, y \rangle + \beta \langle x, z \rangle.$$

(iii) **Conjugate Linearity in the First Argument:**

$$\langle \alpha x + \beta y, z \rangle = \bar{\alpha} \langle x, z \rangle + \bar{\beta} \langle y, z \rangle.$$

(iv) **Positive Definiteness:**

$$\langle x, x \rangle \geq 0 \quad \text{for all } x \in \mathbb{K}^n, \quad \text{and} \quad \langle x, x \rangle = 0 \iff x = 0.$$

The space $\mathbb{K}^n$ equipped with an inner product $\langle \cdot, \cdot \rangle$ is called an **inner product space**.

Example 4.5.2 (Standard Euclidean Inner Product)**.** On $\mathbb{R}^n$, the standard inner product (dot product) of $x = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}^T$ and $y = \begin{bmatrix} y_1 & \dots & y_n \end{bmatrix}^T$ is:

$$\langle x, y \rangle = x^T y = \sum_{i=1}^{n} x_i y_i.$$

For $x = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $y = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^3$:

$$\langle x, y\rangle = (1)(4) + (2)(-1) + (-3)(2) = 4 - 2 - 6 = -4.$$

Example 4.5.3 (Standard Complex Inner Product). On $\mathbb{C}^n$, the standard complex inner product of $x = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}^T$ and $y = \begin{bmatrix} y_1 & \dots & y_n \end{bmatrix}^T$ is:

$$\langle x, y\rangle = x^* y = \sum_{i=1}^{n} \bar{x}_i y_i.$$

For $x = \begin{bmatrix} 1+i \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 3i \\ 1-i \end{bmatrix}$ in $\mathbb{C}^2$:

$$\begin{aligned} \langle x, y\rangle &= \overline{(1+i)}(3i) + \overline{(2)}(1-i) \\ &= (1-i)(3i) + 2(1-i) \\ &= (3i - 3i^2) + (2 - 2i) = (3i + 3) + 2 - 2i = 5 + i. \end{aligned}$$

Example 4.5.4 (Real Matrix-Weighted Inner Product). Let $A \in M_n(\mathbb{R})$ be a real symmetric, positive-definite matrix (so $A^T = A$ and $x^T A x > 0$ for all $x \neq 0$). The function defined by:

$$\langle x, y\rangle_A = x^T A y$$

is a valid inner product on $\mathbb{R}^n$.

For example, choosing the diagonal matrix $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ gives the weighted inner product on $\mathbb{R}^2$:

$$\langle x, y\rangle_A = 2x_1y_1 + 3x_2y_2.$$

For $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$:

$$\langle x, y\rangle_A = 2(1)(4) + 3(2)(1) = 8 + 6 = 14.$$

Example 4.5.5 (Frobenius Inner Product on Real Matrices). For real $m \times n$ matrices $A, B \in M_{m\times n}(\mathbb{R})$, the **Frobenius inner product** is defined using the trace operator:

$$\langle A, B\rangle_F = \mathrm{Tr}(A^T B) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} b_{ij}.$$

For $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 4 & 1 \end{bmatrix}$ in $M_2(\mathbb{R})$:

$$\langle A, B\rangle_F = 1(2) + 2(-1) + 0(4) + 3(1) = 2 - 2 + 0 + 3 = 3.$$

Example 4.5.6 (Frobenius Inner Product on Complex Matrices). For complex $m \times n$ matrices $A, B \in M_{m\times n}(\mathbb{C})$, the **complex Frobenius inner product** is defined by:

$$\langle A, B\rangle_F = \mathrm{Tr}(A^* B) = \sum_{i=1}^{m}\sum_{j=1}^{n} \bar{a}_{ij} b_{ij}.$$

For $A = \begin{bmatrix} i & 1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2i & 0 \\ 1 & 1+i \end{bmatrix}$ in $M_2(\mathbb{C})$:

$$\begin{aligned}\langle A, B\rangle_F &= \bar{i}(2i) + \bar{1}(0) + \bar{0}(1) + \bar{2}(1+i) \\ &= (-i)(2i) + 0 + 0 + 2(1+i) \\ &= -2i^2 + 2 + 2i = 2 + 2 + 2i = 4 + 2i.\end{aligned}$$

Example 4.5.7. Let $C[a, b]$ be the space of continuous real-valued functions defined on the closed interval $[a, b]$. The standard inner product on $C[a, b]$ is given by integration:

$$\langle f, g\rangle = \int_a^b f(x)g(x)\,dx.$$

For $f(x) = x$ and $g(x) = x^2$ on the interval $[0, 1]$:

$$\langle f, g\rangle = \int_0^1 (x)(x^2)\,dx = \int_0^1 x^3\,dx = \left[\frac{x^4}{4}\right]_0^1 = \frac{1}{4}.$$

Example 4.5.8. For complex-valued continuous functions $f, g : [a, b] \to \mathbb{C}$, the inner product is given by:

$$\langle f, g\rangle = \int_a^b \overline{f(x)}g(x)\,dx.$$

For $f(x) = e^{ix}$ and $g(x) = e^{2ix}$ on the interval $[0, 2\pi]$:

$$\begin{aligned}\langle f, g\rangle &= \int_0^{2\pi} \overline{e^{ix}}e^{2ix}\,dx = \int_0^{2\pi} e^{-ix}e^{2ix}\,dx \\ &= \int_0^{2\pi} e^{ix}\,dx = \left[\frac{e^{ix}}{i}\right]_0^{2\pi} = \frac{e^{2\pi i} - e^0}{i} = \frac{1-1}{i} = 0.\end{aligned}$$

Thus, e^{ix} and e^{2ix} are orthogonal in this function space.

4.5.1 Standard Inner Products on $\mathbb{R}^n$ and $\mathbb{C}^n$

Definition 4.5.9 (Real Space $\mathbb{R}^n$)**.** The **standard inner product** (or Euclidean dot product) on $\mathbb{R}^n$ is a function $\langle\cdot,\cdot\rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ defined for vectors $x = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}^T$ and $y = \begin{bmatrix} y_1 & \dots & y_n \end{bmatrix}^T$ by:

$$\langle x, y\rangle = x^T y = \sum_{i=1}^n x_i y_i = x_1y_1 + x_2y_2 + \cdots + x_ny_n.$$

Definition 4.5.10 (Complex Space $\mathbb{C}^n$)**.** The **standard inner product** on $\mathbb{C}^n$ is a function $\langle\cdot,\cdot\rangle : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ defined for vectors $x = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}^T$ and $y = \begin{bmatrix} y_1 & \dots & y_n \end{bmatrix}^T$ by:

$$\langle x, y\rangle = x^* y = (\bar{x})^T y = \sum_{i=1}^n \bar{x}_i y_i = \bar{x}_1y_1 + \bar{x}_2y_2 + \cdots + \bar{x}_ny_n.$$

4.5.2 Induced Norms and Fundamental Inequalities

Definition 4.5.11 (Induced Vector Norm)**.** The **norm** (or length) of a vector $x \in \mathbb{K}^n$ induced by the standard inner product is the non-negative real number $\|x\|$ defined by:

$$\|x\| = \sqrt{\langle x, x\rangle} = \left(\sum_{i=1}^n |x_i|^2\right)^{1/2}.$$

A vector $u \in \mathbb{K}^n$ with $\|u\| = 1$ is called a **unit vector** (or **normalized vector**).

Theorem 4.5.12 (Cauchy–Schwarz Inequality). *For any two vectors $x, y \in \mathbb{K}^n$:*

$$|\langle x, y\rangle| \leq \|x\|\|y\|,$$

with equality holding if and only if x and y are linearly dependent.

Proof. If $y = 0$, then $\langle x, 0\rangle = 0$ and $\|x\|\|0\| = 0$, so equality holds trivially.
Assume $y \neq 0$, so $\|y\| > 0$. For any scalar $t \in \mathbb{K}$, positive definiteness implies:

$$0 \leq \|x - ty\|^2 = \langle x - ty, x - ty\rangle = \langle x, x\rangle - \bar{t}\langle y, x\rangle - t\langle x, y\rangle + |t|^2\langle y, y\rangle.$$

Choose the specific scalar $t = \frac{\langle y,x\rangle}{\langle y,y\rangle} = \frac{\overline{\langle x,y\rangle}}{\|y\|^2}$. Substituting t into the inequality:

$$\begin{aligned} 0 &\leq \|x\|^2 - \frac{\langle x, y\rangle}{\|y\|^2}\langle y, x\rangle - \frac{\overline{\langle x, y\rangle}}{\|y\|^2}\langle x, y\rangle + \frac{|\langle x, y\rangle|^2}{\|y\|^4}\|y\|^2 \\ &= \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2} - \frac{|\langle x, y\rangle|^2}{\|y\|^2} + \frac{|\langle x, y\rangle|^2}{\|y\|^2} \\ &= \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2}. \end{aligned}$$

Rearranging terms yields $|\langle x, y\rangle|^2 \leq \|x\|^2\|y\|^2$. Taking the positive square root gives:

$$|\langle x, y\rangle| \leq \|x\|\|y\|.$$

Equality holds if and only if $\|x - ty\|^2 = 0 \iff x - ty = 0 \iff x = ty$, meaning x and y are linearly dependent. ∎

Theorem 4.5.13 (Properties of the Vector Norm). *For all $x, y \in \mathbb{K}^n$ and $\alpha \in \mathbb{K}$:*

(i) $\|x\| \geq 0$, *and* $\|x\| = 0 \iff x = 0$.

(ii) ***Absolute Homogeneity:*** $\|\alpha x\| = |\alpha|\|x\|$.

(iii) ***Triangle Inequality:*** $\|x + y\| \leq \|x\| + \|y\|$.

Proof. We prove property (iii) using the Cauchy–Schwarz Inequality:

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y\rangle = \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle \\ &= \|x\|^2 + \langle x, y\rangle + \overline{\langle x, y\rangle} + \|y\|^2 \\ &= \|x\|^2 + 2\,\mathrm{Re}(\langle x, y\rangle) + \|y\|^2 \\ &\leq \|x\|^2 + 2|\langle x, y\rangle| + \|y\|^2 \quad (\text{since } \mathrm{Re}(z) \leq |z|) \\ &\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \quad (\text{by Cauchy–Schwarz}) \\ &= (\|x\| + \|y\|)^2. \end{aligned}$$

Taking square roots yields $\|x + y\| \leq \|x\| + \|y\|$. ∎

4.5.3 Orthogonality, Pythagorean Theorem, and Parallelogram Law

Definition 4.5.14 (Orthogonality). Two vectors $x, y \in \mathbb{K}^n$ are said to be **orthogonal** (denoted $x \perp y$) if their inner product is zero:

$$x \perp y \iff \langle x, y\rangle = 0.$$

Theorem 4.5.15 (Pythagorean Theorem in $\mathbb{K}^n$). *If $x, y \in \mathbb{K}^n$ are orthogonal vectors ($x \perp y$), then:*

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2.$$

Proof. Expanding $\|x+y\|^2$:

$$\|x+y\|^2 = \langle x+y, x+y\rangle = \|x\|^2 + \langle x,y\rangle + \langle y,x\rangle + \|y\|^2.$$

Since $x \perp y$, $\langle x,y\rangle = 0$ and $\langle y,x\rangle = \overline{\langle x,y\rangle} = 0$. Thus:

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2.$$

■

Theorem 4.5.16 (Parallelogram Law). *For any two vectors $x, y \in \mathbb{K}^n$:*

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

Proof. Expand both norm square terms:

$$\begin{aligned}\|x+y\|^2 &= \|x\|^2 + \langle x,y\rangle + \langle y,x\rangle + \|y\|^2,\\ \|x-y\|^2 &= \|x\|^2 - \langle x,y\rangle - \langle y,x\rangle + \|y\|^2.\end{aligned}$$

Adding the two equations cancels the cross-terms $\langle x,y\rangle$ and $\langle y,x\rangle$:

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

■

Example 4.5.17 (Computing Complex Inner Products and Norms). Let $x = \begin{bmatrix} 1+i \\ 2 \\ -i \end{bmatrix}$ and $y = \begin{bmatrix} 2i \\ 1-i \\ 3 \end{bmatrix}$ in $\mathbb{C}^3$. Compute $\langle x,y\rangle$, $\|x\|$, $\|y\|$, and verify $\langle y,x\rangle = \overline{\langle x,y\rangle}$.

Solution. Using the standard complex inner product $\langle x,y\rangle = \sum_{j=1}^{3} \bar{x}_j y_j$:

$$\begin{aligned}\langle x,y\rangle &= \overline{(1+i)}(2i) + \overline{(2)}(1-i) + \overline{(-i)}(3)\\ &= (1-i)(2i) + 2(1-i) + (i)(3)\\ &= (2i - 2i^2) + (2-2i) + 3i = (2i+2) + 2 - 2i + 3i = 4 + 3i.\end{aligned}$$

Now compute $\langle y,x\rangle = \sum_{j=1}^{3} \bar{y}_j x_j$:

$$\begin{aligned}\langle y,x\rangle &= \overline{(2i)}(1+i) + \overline{(1-i)}(2) + \overline{(3)}(-i)\\ &= (-2i)(1+i) + (1+i)(2) + 3(-i)\\ &= (-2i - 2i^2) + 2 + 2i - 3i = (-2i+2) + 2 - i = 4 - 3i.\end{aligned}$$

Notice that $\overline{\langle x,y\rangle} = \overline{4+3i} = 4 - 3i = \langle y,x\rangle$, confirming conjugate symmetry.

Now compute the norms:

$$\|x\| = \sqrt{|1+i|^2 + |2|^2 + |-i|^2} = \sqrt{(1^2+1^2) + 4 + 1^2} = \sqrt{2+4+1} = \sqrt{7}.$$

$$\|y\| = \sqrt{|2i|^2 + |1-i|^2 + |3|^2} = \sqrt{2^2 + (1^2 + (-1)^2) + 9} = \sqrt{4+2+9} = \sqrt{15}.$$

■

Example 4.5.18 (Verification of the Cauchy–Schwarz Inequality). Using $x = \begin{bmatrix} 1+i \\ 2 \\ -i \end{bmatrix}$ and $y = \begin{bmatrix} 2i \\ 1-i \\ 3 \end{bmatrix}$ from the previous example, verify the Cauchy–Schwarz Inequality $|\langle x, y\rangle| \leq \|x\|\|y\|$.

Solution. From the previous calculations:

$$\langle x, y\rangle = 4 + 3i \implies |\langle x, y\rangle| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5.$$

$$\|x\| = \sqrt{7}, \quad \|y\| = \sqrt{15} \implies \|x\|\|y\| = \sqrt{7 \times 15} = \sqrt{105} \approx 10.246.$$

Comparing the values:

$$5 \leq \sqrt{105} \implies |\langle x, y\rangle| \leq \|x\|\|y\|.$$

This verifies the Cauchy–Schwarz Inequality. ■

Example 4.5.19 (Verification of the Parallelogram Law). Let $x = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ in $\mathbb{R}^2$. Verify the Parallelogram Law $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$.

Solution. Compute the norms of x and y:

$$\|x\|^2 = 3^2 + 1^2 = 10 \implies 2\|x\|^2 = 20,$$

$$\|y\|^2 = 1^2 + 2^2 = 5 \implies 2\|y\|^2 = 10.$$

Thus, the right-hand side is $2\|x\|^2 + 2\|y\|^2 = 20 + 10 = 30$.

Now compute $x + y$ and $x - y$:

$$x + y = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \implies \|x + y\|^2 = 4^2 + 3^2 = 25,$$

$$x - y = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \implies \|x - y\|^2 = 2^2 + (-1)^2 = 5.$$

The left-hand side is $\|x + y\|^2 + \|x - y\|^2 = 25 + 5 = 30$.

Since $30 = 30$, the Parallelogram Law holds. ■

4.6 Orthonormal Sets and Geometric Interpretation of Length and Angles

Orthonormal sets represent the ideal coordinate systems in $\mathbb{R}^n$ and $\mathbb{C}^n$. They dramatically simplify vector decompositions, projection calculations, and geometric analysis of length and angles.

4.6.1 Orthogonal and Orthonormal Sets

Definition 4.6.1 (Orthogonal and Orthonormal Sets). Let $S = \{u_1, u_2, \ldots, u_k\}$ be a set of column vectors in $\mathbb{K}^n$ ($\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$).

(i) S is called an **orthogonal set** if every pair of distinct vectors in S is orthogonal:

$$\langle u_i, u_j\rangle = 0 \quad \text{for all } i \neq j.$$

(ii) S is called an **orthonormal set** if it is an orthogonal set and every vector in S is a unit vector ($\|u_i\| = 1$). Using the Kronecker delta δ_{ij}:

$$\langle u_i, u_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Theorem 4.6.2 (Linear Independence of Orthogonal Sets). *Every orthogonal set of non-zero vectors $S = \{u_1, u_2, \ldots, u_k\} \subset \mathbb{K}^n \setminus \{0\}$ is linearly independent.*

Proof. Suppose there exist scalars $c_1, c_2, \ldots, c_k \in \mathbb{K}$ such that:

$$c_1 u_1 + c_2 u_2 + \cdots + c_k u_k = 0.$$

To isolate the coefficient c_i, take the inner product of both sides with the i-th vector u_i:

$$\left\langle u_i, \sum_{j=1}^{k} c_j u_j \right\rangle = \langle u_i, 0 \rangle \implies \sum_{j=1}^{k} c_j \langle u_i, u_j \rangle = 0.$$

Since $\langle u_i, u_j \rangle = 0$ for all $j \neq i$, the sum collapses to a single term:

$$c_i \langle u_i, u_i \rangle = 0 \implies c_i \|u_i\|^2 = 0.$$

Since $u_i \neq 0$, $\|u_i\|^2 > 0$, forcing $c_i = 0$. Since this holds for every $i = 1, 2, \ldots, k$, S is linearly independent. ∎

4.6.2 Orthonormal Bases and Fourier Expansion

Definition 4.6.3 (Orthonormal Basis). An **orthonormal basis** (ONB) for $\mathbb{K}^n$ is a basis $\mathcal{U} = \{u_1, u_2, \ldots, u_n\}$ that is an orthonormal set.

Theorem 4.6.4 (Fourier Expansion Theorem). *Let $\mathcal{U} = \{u_1, u_2, \ldots, u_n\}$ be an orthonormal basis for $\mathbb{K}^n$. Then every vector $x \in \mathbb{K}^n$ can be expressed uniquely as:*

$$x = \sum_{i=1}^{n} \langle u_i, x \rangle u_i = \langle u_1, x \rangle u_1 + \langle u_2, x \rangle u_2 + \cdots + \langle u_n, x \rangle u_n.$$

*The scalar coefficients $c_i = \langle u_i, x \rangle$ are called the **Fourier coefficients** of x with respect to $\mathcal{U}$.*

Proof. Since $\mathcal{U}$ is a basis, there exist unique scalars $c_1, \ldots, c_n$ such that $x = \sum_{j=1}^{n} c_j u_j$. Taking the inner product of u_i with x:

$$\langle u_i, x \rangle = \left\langle u_i, \sum_{j=1}^{n} c_j u_j \right\rangle = \sum_{j=1}^{n} c_j \langle u_i, u_j \rangle = \sum_{j=1}^{n} c_j \delta_{ij} = c_i.$$

Substituting $c_i = \langle u_i, x \rangle$ back yields the Fourier expansion formula. ∎

Theorem 4.6.5 (Parseval's Identity). *Let $\{u_1, u_2, \ldots, u_n\}$ be an orthonormal basis for $\mathbb{K}^n$. For any vector $x \in \mathbb{K}^n$, the norm square of x is the sum of the absolute squares of its Fourier coefficients:*

$$\|x\|^2 = \sum_{i=1}^{n} |\langle u_i, x \rangle|^2.$$

Proof. Substitute the Fourier expansion $x = \sum_{i=1}^{n} c_i u_i$ into $\|x\|^2 = \langle x, x \rangle$:

$$\|x\|^2 = \left\langle \sum_{i=1}^{n} c_i u_i, \sum_{j=1}^{n} c_j u_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{n} \bar{c}_i c_j \langle u_i, u_j \rangle = \sum_{i=1}^{n} \bar{c}_i c_i (1) = \sum_{i=1}^{n} |c_i|^2.$$

Since $c_i = \langle u_i, x \rangle$, $\|x\|^2 = \sum_{i=1}^{n} |\langle u_i, x \rangle|^2$. ∎

4.6.3 The Gram–Schmidt Orthogonalization Process

The **Gram–Schmidt process** is an algorithmic procedure that transforms any linearly independent set of column vectors into an orthonormal set spanning the exact same set of linear combinations.

Theorem 4.6.6 (Gram–Schmidt Orthogonalization Theorem)**.** *Let $\{v_1, v_2, \dots, v_k\}$ be a linearly independent set of vectors in $\mathbb{K}^n$. Then there exists an orthonormal set $\{u_1, u_2, \dots, u_k\}$ such that for every $1 \le m \le k$:*

$$\operatorname{span}\{u_1, u_2, \dots, u_m\} = \operatorname{span}\{v_1, v_2, \dots, v_m\}.$$

Construction of the Algorithm. We construct an orthogonal set $\{w_1, w_2, \dots, w_k\}$ sequentially, and then normalize each vector to obtain $u_i = \frac{w_i}{\|w_i\|}$:

First Vector: Set $w_1 = v_1$, and normalize to get $u_1 = \frac{w_1}{\|w_1\|}$.

Second Vector: Subtract the orthogonal projection of v_2 onto u_1:

$$w_2 = v_2 - \langle u_1, v_2 \rangle u_1.$$

Then normalize to get $u_2 = \frac{w_2}{\|w_2\|}$.

General m-th Vector: Having constructed orthonormal vectors $\{u_1, \dots, u_{m-1}\}$, subtract the orthogonal projections of v_m onto each previous vector:

$$w_m = v_m - \sum_{j=1}^{m-1} \langle u_j, v_m \rangle u_j.$$

Because $v_m \notin \operatorname{span}\{v_1, \dots, v_{m-1}\}$, $w_m \neq 0$. Normalize to obtain $u_m = \frac{w_m}{\|w_m\|}$.

Inductively, $\{u_1, u_2, \dots, u_k\}$ forms the required orthonormal set. ■

4.6.4 Geometric Interpretation of Length, Angle, and Projection

Standard inner products provide a precise bridge between vector algebra in $\mathbb{K}^n$ and classical geometry:

1. **Length (Norm):** $\|x\| = \sqrt{\langle x, x \rangle}$ represents the Euclidean distance from the origin to the point x.

2. **Distance Between Vectors:** The distance $d(x, y)$ between two vectors $x, y \in \mathbb{K}^n$ is defined by:

$$d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle}.$$

3. **Angle in $\mathbb{R}^n$:** For two non-zero vectors $x, y \in \mathbb{R}^n$, the Cauchy–Schwarz Inequality guarantees that $-1 \le \frac{\langle x,y \rangle}{\|x\|\|y\|} \le 1$. The unique **angle** $\theta \in [0, \pi]$ between x and y is defined by:

$$\cos\theta = \frac{\langle x, y \rangle}{\|x\|\|y\|} \implies \theta = \arccos\left(\frac{\langle x, y \rangle}{\|x\|\|y\|}\right).$$

4. **Orthogonal Projection:** The orthogonal projection of a vector x onto a non-zero vector u is given by:

$$\operatorname{proj}_u(x) = \frac{\langle u, x \rangle}{\|u\|^2} u.$$

The vector $x - \operatorname{proj}_u(x)$ is perpendicular to u, decomposing x into orthogonal components parallel and perpendicular to u.

Example 4.6.7 (Fourier Expansion Relative to an Orthonormal Basis)**.** Show that

$$\mathcal{U} = \left\{ u_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix}, u_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\\0\end{bmatrix}, u_3 = \begin{bmatrix}0\\0\\1\end{bmatrix} \right\}$$

is an orthonormal basis for $\mathbb{R}^3$, and express $x = \begin{bmatrix}3\\1\\4\end{bmatrix}$ using Fourier coefficients.

Solution. We check orthonormality by computing inner products $\langle u_i, u_j \rangle$:

$$\langle u_1, u_1 \rangle = \frac{1}{2}(1+1+0) = 1, \quad \langle u_2, u_2 \rangle = \frac{1}{2}(1+1+0) = 1, \quad \langle u_3, u_3 \rangle = 1.$$

$$\langle u_1, u_2 \rangle = \frac{1}{2}(1-1+0) = 0, \quad \langle u_1, u_3 \rangle = 0, \quad \langle u_2, u_3 \rangle = 0.$$

Thus, $\mathcal{U}$ is an orthonormal set of 3 vectors in $\mathbb{R}^3$, making it an orthonormal basis.

Now compute the Fourier coefficients $c_i = \langle u_i, x \rangle$:

$$\begin{aligned} c_1 &= \langle u_1, x \rangle = \frac{1}{\sqrt{2}}(1(3) + 1(1) + 0(4)) = \frac{4}{\sqrt{2}} = 2\sqrt{2}, \\ c_2 &= \langle u_2, x \rangle = \frac{1}{\sqrt{2}}(1(3) - 1(1) + 0(4)) = \frac{2}{\sqrt{2}} = \sqrt{2}, \\ c_3 &= \langle u_3, x \rangle = 0(3) + 0(1) + 1(4) = 4. \end{aligned}$$

Applying the Fourier expansion theorem:

$$x = 2\sqrt{2}u_1 + \sqrt{2}u_2 + 4u_3.$$

Verifying Parseval's identity: $\|x\|^2 = 3^2 + 1^2 + 4^2 = 26$, and $c_1^2 + c_2^2 + c_3^2 = (2\sqrt{2})^2 + (\sqrt{2})^2 + 4^2 = 8 + 2 + 16 = 26$. ■

Example 4.6.8 (Gram–Schmidt Orthogonalization Process)**.** Apply the Gram–Schmidt process to transform the linearly independent set $\{v_1, v_2\} \subset \mathbb{R}^3$ into an orthonormal set $\{u_1, u_2\}$, where:

$$v_1 = \begin{bmatrix}1\\1\\0\end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix}1 & 0 & 1\end{bmatrix}^T.$$

Solution. Construct the first orthogonal vector $w_1 = v_1$:

$$w_1 = \begin{bmatrix}1\\1\\0\end{bmatrix} \implies \|w_1\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}.$$

Normalize w_1 to get u_1:

$$u_1 = \frac{w_1}{\|w_1\|} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix}.$$

Construct the second orthogonal vector $w_2 = v_2 - \langle u_1, v_2 \rangle u_1$: First, compute the inner product:

$$\langle u_1, v_2 \rangle = \frac{1}{\sqrt{2}}(1(1) + 1(0) + 0(1)) = \frac{1}{\sqrt{2}}.$$

Then compute w_2:

$$w_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}.$$

Compute the norm of w_2:

$$\|w_2\| = \sqrt{(1/2)^2 + (-1/2)^2 + 1^2} = \sqrt{1/4 + 1/4 + 1} = \sqrt{3/2} = \frac{\sqrt{6}}{2}.$$

Normalize w_2 to get u_2:

$$u_2 = \frac{w_2}{\|w_2\|} = \frac{2}{\sqrt{6}} \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}.$$

The resulting orthonormal set is $\{u_1, u_2\}$. ■

Example 4.6.9 (Distance, Angle, and Projection Calculations)**.** Let $x = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}$ in $\mathbb{R}^3$. Calculate the distance $d(x, y)$, the angle θ between x and y, and the orthogonal projection $\text{proj}_y(x)$.

Solution. First, compute the individual norms and inner product:

$$\|x\| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3,$$
$$\|y\| = \sqrt{0^2 + 3^2 + 4^2} = \sqrt{25} = 5,$$
$$\langle x, y \rangle = 1(0) + 2(3) + 2(4) = 0 + 6 + 8 = 14.$$

To compute the distance $d(x, y) = \|x - y\|$:

$$x - y = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} \implies d(x, y) = \sqrt{1^2 + (-1)^2 + (-2)^2} = \sqrt{1 + 1 + 4} = \sqrt{6}.$$

To compute the angle θ:

$$\cos\theta = \frac{\langle x, y \rangle}{\|x\|\|y\|} = \frac{14}{3 \times 5} = \frac{14}{15} \implies \theta = \arccos\left(\frac{14}{15}\right) \approx 0.367 \text{ radians } (21.04^\circ).$$

To compute the orthogonal projection $\text{proj}_y(x)$:

$$\text{proj}_y(x) = \frac{\langle y, x \rangle}{\|y\|^2} y = \frac{14}{25} \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 42/25 \\ 56/25 \end{bmatrix}.$$

■

Exercise 4.6.10. Solve the following exercise.

1. Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \in M_2(\mathbb{R})$ and $p(x) = x^3 - 3x^2 + 2x - 1$. Evaluate the matrix polynomial $p(A)$.

2. Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \in M_3(\mathbb{R})$. Show that A is nilpotent, state its index of nilpotency k, and evaluate $(I_3 - A)^{-1}$ and e^A as finite matrix polynomials.

3. Prove that if $A \in M_n(\mathbb{K})$ is an idempotent matrix ($A^2 = A$), then every eigenvalue of A is either 0 or 1.

4. Find the characteristic polynomial $p_A(\lambda)$ and the minimal polynomial $m_A(x)$ for the matrix:
$$A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

5. Compute the characteristic polynomial $p_A(x)$ and minimal polynomial $m_A(x)$ for:
$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

6. A matrix $A \in M_5(\mathbb{R})$ has characteristic polynomial $p_A(x) = (x-2)^3(x-5)^2$.
   (a) List all possible candidate minimal polynomials $m_A(x)$ for A.
   (b) If it is given that $\operatorname{rank}(A - 2I_5) = 3$ and $\operatorname{rank}(A - 5I_5) = 4$, uniquely determine $m_A(x)$.

7. Find the eigenvalues, algebraic multiplicities, geometric multiplicities, and a basis for each eigenspace of:
$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

8. Let $A, B \in M_n(\mathbb{K})$ be similar matrices ($B = P^{-1}AP$). Prove that A and B have the exact same characteristic polynomial, trace, determinant, and eigenvalues.

9. Prove that if $v_1, v_2, \ldots, v_k \in \mathbb{K}^n$ are eigenvectors of a matrix A corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$, then $\{v_1, v_2, \ldots, v_k\}$ is linearly independent.

10. Let $A \in M_n(\mathbb{K})$ be an invertible matrix. If λ is an eigenvalue of A with corresponding eigenvector x, prove that $\lambda \neq 0$ and that λ^{-1} is an eigenvalue of A^{-1} with the exact same eigenvector x.

11. Verify the Cayley–Hamilton Theorem for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Use the theorem to compute A^{-1} and A^4.

12. Verify the Cayley–Hamilton Theorem for the matrix:
$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix},$$
and use it to find A^{-1}.

13. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Using the Cayley–Hamilton Theorem, evaluate the high-degree matrix polynomial:
$$f(A) = A^5 - 2A^4 - 3A^3 + 2A^2 - 3A + 5I_2.$$

14. Given $x = \begin{bmatrix} 2+i \\ 3 \\ -2i \end{bmatrix}$ and $y = \begin{bmatrix} 1-i \\ i \\ 2 \end{bmatrix}$ in $\mathbb{C}^3$:
    (a) Compute the complex inner product $\langle x, y \rangle = x^* y$.
    (b) Compute the norms $\|x\|$ and $\|y\|$.
    (c) Verify the Cauchy–Schwarz Inequality $|\langle x, y \rangle| \leq \|x\| \|y\|$.
15. Prove the Parallelogram Law for vectors in $\mathbb{K}^n$:
$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$
16. Prove the Pythagorean Theorem in $\mathbb{K}^n$: if $x, y \in \mathbb{K}^n$ are orthogonal ($x \perp y$), then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.
17. Show that $S = \left\{ u_1 = \frac{1}{3} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, u_2 = \frac{1}{3} \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}, u_3 = \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} \right\}$ is an orthonormal basis for $\mathbb{R}^3$.
    Find the Fourier coefficients of $x = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}$ relative to S.
18. Apply the Gram–Schmidt process to transform the linearly independent set $\{v_1, v_2, v_3\} \subset \mathbb{R}^3$ into an orthonormal set $\{u_1, u_2, u_3\}$, where:
$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$
19. For the vectors $x = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ in $\mathbb{R}^3$:
    (a) Compute the distance $d(x, y) = \|x - y\|$.
    (b) Find the angle θ between x and y.
    (c) Compute the orthogonal projection $\mathrm{proj}_y(x)$.
20. Prove Parseval's Identity: if $\{u_1, u_2, \ldots, u_n\}$ is an orthonormal basis for $\mathbb{K}^n$, then for any vector $x \in \mathbb{K}^n$:
$$\|x\|^2 = \sum_{i=1}^{n} |\langle u_i, x \rangle|^2.$$

Concluding Note to Students

Congratulations!

"Mathematics is not about numbers, equations, or computations: it is about understanding."

— **William Paul Thurston**

Dear Students,

Congratulations on successfully completing the course on **Theory of Matrices (MMT122J)**! Reaching this milestone reflects your dedication, perseverance, and growing mathematical maturity.

Throughout this book, you have mastered the essential tools of linear algebra—transitioning from matrix algebra, structural symmetries, and elementary row operations (Units I–II) to linear independence, basis, rank, trace, spectral theory, the Cayley–Hamilton theorem, and inner product spaces (Units III–IV).

Matrix theory is the universal language of modern science, technology, engineering, and advanced mathematics. The conceptual clarity, abstract reasoning, and analytical problem-solving skills you have cultivated here will serve as an enduring foundation for your future studies and competitive examinations such as **CSIR NET-JRF, GATE, SET, IIT-JAM, and Ph.D. entrance tests**.

Keep exploring, stay curious, and carry this logical confidence forward into every domain of your academic journey. I wish you every success and fulfillment in all your future endeavors!

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